CONTROL AND OPTIMIZATION OF DISTRIBUTED ENERGY RESOURCES USING DYNAMIC ALGORITHMS

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Abstract

As concerns for the environment and energy independence leads a transition towards a power grid that depends increasingly on energy from renewable resources like solar and wind, the integration and intelligent control of distributed energy resources (DER) including photovoltaic (PV) arrays, controllable loads, energy storage (ES) and the batteries in plug-in electric vehicles (EVs) will be critical to realizing a power grid that can handle both the variability and unpredictability of renewable energy sources as well as increasing system complexity. In addition to providing added system reliability, DERs acting in coordination can be leveraged to address supply-demand imbalances through Demand Response (DR) and/or price signals on the electric power grid by enabling continuous bidirectional load balancing. Intelligent control and integration has the capability to reduce or shift demand peaks and improve grid efficiency by offsetting the need for spinning reserves and peaking power plants.

In this dissertation, the use of dynamic and distributed algorithms that can handle the higher penetration of renewable resources in a more open and transparent energy market is explored. Specifically, we look at solving the power scheduling problem using Model Predictive Control (MPC) to ensure a dynamic response and the Alternating Direction Method of Multipliers (ADMM) to distribute the optimization problem when apropos. MPC allows the DERs to be adaptive and robust while ADMM encourages each DER to cooperate to achieve system-level goals while still operating and functioning independently. This enables policies and incentives to co-develop and work with optimization and control technologies to ensure a smooth transition to an infrastructure that can run on renewable energy resources and a more distributed grid. Climate change and ecological concerns coupled with economic concerns over fossil fuel prices are addressed not only through the integration of cleaner energy sources, but also by providing a way to encourage participation throughout the grid. Since the forthcoming grid will require integrating renewable energy at all different

levels, we look specifically at an example of generation in a grid-connected PV system with storage as well as a distributed microgrid with onsite PV generation and demand response capabilities. We present simulation results that demonstrate the ability of the algorithms to respond dynamically to external price signals and provide benefits to the grid while respecting and maintaining the functional requirements of the local resources. Each example uses real data taken from measurements at generation and demand sites. While we focus on two of the more popular solutions currently being explored, the platforms using the algorithms are application agnostic in the sense that they can include a range of DERs with varying objectives. Since the algorithms used are both flexible and scalable, devices can be easily integrated as more come online. The platforms can also be implemented in remote areas and developing countries. In addition to demonstrating that such platforms can be implemented dynamically in real-time, the algorithms can also be used in models and simulations as a design tool to inexpensively develop future systems with more generation from renewable resources which can still operate efficiently and reliably.

In the first example, battery storage is used for shifting and smoothing the power generated by the PV array. MPC allows the storage units to automatically and dynamically adapt to changes within and external to the system. This makes the system both flexible and robust to measurement inaccuracy and physical model imprecision, accounting for the impacts of system variability as well as changes in the operating environment more readily and directly than a statistical model. Model parameters and forecasts are readily updated over the planning horizon with the availability of additional information.

With the microgrid, we specifically consider a commercial site equipped with a PV array, EV charge stations, stationary battery, and curtailable load. The microgrid can participate in the wholesale market on the power grid or operate in an islanded state. As with the grid-connected PV system, the microgrid operates within an MPC framework to ensure the microgrid dynamically adapts and is robust to changes in the operating environment. Instead of being centrally controlled, however, both optimization and control is distributed among the individual DERs using an iterative decentralized method known as the ADMM. Each DER operates independently and autonomously using its own local operating conditions and communicates minimally through the exchange of simple messages. The messages are automatically determined and represent accurate cost signals that align the incentives of each DER in the system

to attain the responses necessary to meet system-wide as well as individual objectives. In addition to providing a way for the microgrid to rapidly reach consensus in a coupled and constrained system, ADMM will also keep the control problem tractable and maintain system reliability as more and more DERs simultaneously come online.

The simulations from each example demonstrate that even with limited or incomplete information, the algorithms perform close to optimal. While complex models and prediction methods can be used, we show that they are not required to achieve good results. Additionally, no assumptions need to be made about the distributions of the data and this is particularly advantageous when having to deal with real data for which such assumptions do not always hold. In both examples, we are also able to optimize the power schedules in such a way as to provide more flexible and responsive systems as seen by the power grid. Each DER cooperates to ensure the systems behave in a more predictable manner even while maintaining their independence. This allows more intermittent renewable energy resources to be integrated onto the grid without requiring more operational reserves or upgrade in line capacities to handle the variable and unpredictable generation. Finally, ADMM combined with MPC demonstrates a symbiotic relationship since the warm start from MPC dramatically speeds up computation time in ADMM while distributing the optimization problem with ADMM allows coupled optimization problems to be solved more efficiently. This is especially true with recent advances in optimization which allow inexpensive embedded processors to efficiently execute computations in tens to hundreds of microsecond time scales.

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Chapter 1

Introduction

Over the last several decades, the generation and delivery of energy through the electric grid has become increasingly challenging. The infrastructure we currently have was originally built for economies of scale of generation and did not anticipate and account for the growing and increasingly diverse demand that resulted as more loads became electrified. There was little attention given to making the system more flexible and responsive on the distribution side of the grid, both in terms of incentive mechanisms and enabling technologies. As a result, dynamic balancing of generation and load had to be carried out primarily from the generation side and operating reserves had to be procured in order to prevent brownouts and blackouts. Deregulation of the electricity industry in many industrialized nations also led to underfunded and aging infrastructures that failed to keep up with other modernizing sectors as utility companies focused mostly on recouping costs as quickly as possible in order to maximize profit. Consequently, the electric grid has become increasingly inefficient and unreliable over time [DRT+12, oE03], lacking the communications infrastructure that has been integrated into other sectors of technology. With no open market system on the retail side and no dynamic prices for consumers to respond to, there is additionally a misalignment of economic incentives. The information asymmetry means that growing generation and operation costs are spread out across all rate payers so that there is little impetus for most consumers to participate in the energy market. In addition to these problems, there is a growing awareness of the unsustainability and environmental impact of a grid that continues to run primarily on fossil fuels. The combustion processes for generating power are typically only 45% efficient and only 60% at best [Adm12] while creating pollution that both impacts the climate and

environment. This accounting also does not consider the externalized costs associated with environmental damage and health impacts from burning fossil fuels for energy.

As our awareness of these issues has grown, a range of solutions has been pursued to try and improve the current power system. Two solutions that have gained broad support due to their potential to enable significant transformations are the deployment of renewable energy resources and demand side management (DSM). Electricity generated from renewable resources like solar and wind can bring significant benefits to the power grid with its inexhaustible supply and abundance. It is estimated that there is enough solar energy hitting the earth in 1 hour to power the entire globe for a year [TLC06]. With the concern over climate change and human impact on the planet increasing as our population continues to grow and non-renewable resources being rapidly exhausted, a shift away from fossil fuel based power generation will result in fewer carbon emissions, less damage to the environment and an opportunity to build a more sustainable infrastructure. A less often cited but equally important benefit of switching to renewable energy is its inherent ability to decentralize our power grid through distributed generation (DG). Since these resources are abundant and some form of renewable energy exists locally in almost any geographical location, transitioning to a power system that runs on renewable energy being generated throughout the system will both diversify and democratize the supply of electricity by making the power system more resilient to points of technical failure on the grid as well as market monopolization of generation resources and security threats.

At the same time, we need to create a system that incentivizes and enables consumers on the demand side to participate in energy management so that local resources can be used to help the grid operate more efficiently and resiliently. This is the basic idea behind DSM. The energy management responsibility is now shared with the consumer by directly relating kilowatt-hour price to the actual cost of consumption through an open market. It enables consumers to takes advantage of local resources such as collocated renewable energy sources and become more energy independent. The consumer becomes an active participant in an internet of energy and is able to take a greater role in deciding how the grid is developed and operated. The energy management problem is also made easier by distributing the required control efforts. Both DG and DSM by individuals, businesses and communities can act as alternatives or supplement centralized power. This novel approach to thinking about

our energy infrastructure essentially envisions an electricity network where it is possible to connect, distribute and share resources spatially and temporally, making the grid more flexible and resilient. Proximity to the end user will additionally result in fewer transmission losses and a reduced transmission load, leading to greater efficiency and reduced environmental impact. Since resources are used more efficiently, the need to overbuild and expand transmission and distribution (T&D) lines and address siting and permitting problems that often accompany these upgrades is reduced as well.

DR is one of the more functional and beneficial forms of DSM on the power grid and is arguably an integral part of the transition to a more participatory and flexible system. Specifically, DR emphasizes dynamic response to the conditions on the grid by shifting demand to off-peak times when generation resources are available and there is less congestion and stress on the T&D system. Its current role is often less well defined, however, since utilities have not traditionally been able to rely on it for significant amounts of load reduction. But as loads on the distribution side become increasingly electrified and embedded controls are made more affordable and ubiquitous, their ability to implement DR will not only be inevitable but also necessary. Without DR, the influx of electric loads could overload a grid not designed to handle the sudden demand and this could potentially result in catastrophic failure. Equipping the loads with DR capabilities and coupling them with DG and renewable energy generation hence not only ensures the grid is still operational when a single transmission line or a central power plant fails, but empowers consumers to enact preventative measures in such contingencies. In case of severe failure, systems with DR and DG can operate off-grid and unburden or even aid in the recovery of the grid. The line between DR and energy efficiency also begins to blur in a more dynamic system as shifting and reducing energy will all be carried out simultaneously in the most cost-effective allocation by the same control systems. Coordination is needed because energy efficiency and DR schedule the same energy resources on the demand side and hence both affect the amount and type of generation that needs to be procured [GRLS10].

As we strive to realize these solutions however, we still face significant hurdles. The biggest challenges facing renewable energy and a more distributed and decentralized grid infrastructure with DSM are the inherent intermittency of renewable resources as well as the shortage of enabling technologies and incentives that would

encourage participation on the demand side of the grid. In the case of intermittency, the output of a PV array will vary with the available sunlight and the output of a wind farm will depend on wind speeds. The only renewable resources that have consistent output are hydroelectric and geothermal power and these are geographically limited resources. Since most renewable power sources in developed countries are connected to the electricity grid, the system operators must handle the associated intermittency while trying to meet their primary objectives of keeping the system reliable and the operation costs low. Load intermittency is currently managed on the existing electrical power system by varying the output of fossil fuel power plants acting as spinning reserves or in extreme and prolonged drops in output, firing up a fast-responding peaking power plant. As both solar and wind penetration increases, however, the rapid ramp rate from full power to partial or minimal power output leads to greater unpredictability in the power availability on the grid. Hence while intermittency may be negligible when renewable energy supplies only a small proportion of the power on the grid, the spinning reserves and peaking power plants already used to manage the grid and handle current intermittency problems may prove insufficient to maintain power quality and system reliability as the proportion of power generated from renewable resources increases. Simply increasing the output of these fossil-fuel based power plants also does not provide a long term solution that leads us away from nonrenewable resources towards a more sustainable power infrastructure since it means that a higher penetration of renewable resources will be inherently coupled to a need for more reserve capacity on the grid. This means that integration of renewable energy may reduce the efficiency of the power system and add climate and environmental costs if not done intelligently. Securing more operating reserves to handle intermittency will also cause electricity rates to go up and detract from the need to build a more efficient and flexible infrastructure by deferring the resources needed to modernize the technology on the grid to make it more adaptive and responsive.

This leads to the other major hurdle facing a more resilient and sustainable grid. Intelligent integration of renewable energy sources by itself will not result in a more flexible and robust system. Creating an incentive structure and providing a platform for participation throughout the distribution side of the grid is critical to achieving both a more responsive and resilient system that can depend primarily on renewable sources of energy. One reason for this is because in the U.S. as well as in other developed countries, the current centralized infrastructure is one of the main bottlenecks

with integrating renewable energy. In order to generate renewable energy on a large scale centrally, vast amounts of space are generally required and so generation must often be located in remote areas and transported to high demand areas. Although this provides economies of scale and makes managing the intermittency and operational costs of the resource more tractable in some cases, it could prove to be inefficient if not impossible in other cases if a grid connection is not readily available. This means that the amount of renewable energy that can be integrated into the grid centrally is limited. It is generally accepted that in order for the amount of renewable energy to significantly increase on the grid, the infrastructure needs to be decentralized such that it can take advantage of the distributed nature of renewable resources wherever it is located. This is particularly the case if it is already collocated with the load at the distribution limits of the grid and can be intelligently managed locally without having to significantly upgrade the grid infrastructure to accommodate the additional generation or increase land and resource use [BKP11, Ass13, PDH$^+$]. Especially with power from PV, the imperative for large scale centralized generation is being called into question as the enabling technologies for a highly distributed grid topology are rapidly becoming more available and affordable. The other reason is that the cost of electricity strongly depends on generation and demand throughout the day and hence is not static. This must be relayed to consumers if the demand side is to be incentivized to respond to changing conditions on the grid and act as a buffer to intermittent generation. By creating an interconnected system that aligns the interests of all participants and ensures an equitable market environment, demand side responsiveness increases and helps tackle the problem of demand inflexibility to grid conditions. The result is better resource utilization, an efficient transmission and distribution system for electricity, and a more flexible and participatory infrastructure with fewer single points of failure. Developing a more interconnected grid over large geographical areas not only allows encourages participation in energy management throughout the system, but also allows renewable resources from a variety of climates to be combined to reduce overall variability. The last benefit should be emphasized since renewable energy by itself cannot solve the energy problem if it is not accompanied by a cultural shift in the way we view and use energy. An inclusive approach for operating the grid enables more efficient resource use without compromising power quality, stability and reliability. Many policy-makers in countries like the U.S., Canada, the UK and Germany are now beginning to recognize the ability

of DG and DSM to play this role and are starting to introduce incentives to bring relevant technologies into the mainstream. There are policies in place now which streamline the interconnection process as well as selling power back onto the grid. While this is a promising start in the right direction, much work still needs to be done to ensure more widespread participation through social and economic incentives coupled with easily deployed technology platforms.

The current challenges that need to be addressed no longer lie so much in the technology, which is available at increasingly lower prices and rapidly becoming economically feasible [WMB05, PD11]. The greatest barrier to implementation and integration is the absence of an open and transparent platform that addresses the dynamic energy management problem by allowing each energy resource to participate in an efficient power exchange system. Collectively, we refer to all distributed generation, storage and loads as Distributed Energy Resources (DERs). There is currently almost no incentive for a DER to respond to the variable costs of power delivery and offer quantifiable system benefits including handling the intermittency of renewable energy generation. Development and deployment critically depend on this since technology manufacturers cannot shoulder the risk of a technology without a guaranteed market but system operators are unwilling to provide that market until they can be assured of reliable and quantifiable benefits from each DER. An open platform would simultaneously ensure a clear route to market for dynamically controlled DERs to provide system-wide benefits while ensuring that the benefits in terms of load deferment potential, reduced operating costs, reliability of resource and carbon savings could be quantified and verified. Benefits and costs on a per-device basis could be determined in order to incentivize service providers. Even in rural and remote areas where renewable energy is already well-suited and where small PV systems are already providing electricity to millions of households [REN13b], a coordinating infrastructure that can intelligently schedule DERs on-site could allow for better resource utilization and hence lower both initial and operational costs while improving flexibility and reliability.

In all scenarios, an open platform for energy exchange that would address implementation and integration barriers requires two key enablers: a transparent framework for all DERs to interact on would have to be made accessible and each DER would also have to be equipped with the ability to respond dynamically to changing

grid conditions. This would offer a way for each DER to independently meet objectives and constraints while operating in an open market. DER owners would in this way be able to influence the development and operation of the grid. For example, if controllable DERs can show themselves to be a more cost effective strategy for scheduling power because they can provide a relatively less expensive way to buffer at times of imbalance or intermittency, some of the savings that result from this effect can be diverted towards incentives for incorporation of controllable DERs into the grid and away from unnecessary upgrades to the system or procurement of more operating reserves.

By encouraging a diversity of DERs that are located at all levels of the grid and dispersed geographically, the intermittency of renewable resources could also be managed in real-time autonomously. This would directly handle the difficult problem of finding a resource that can be cycled with the time constant of PV intermittency. The rapid response of many DERs makes them well positioned to smooth out the high-frequency component of PV intermittency and resources with sufficient capacity to be partially ramped up and down can be controlled without having a significant effect on the owner of the resource [GBH11, GRLS10, U. 06, HWB13]. Each participant on the grid would also be incentivized to behave rationally and better utilize local energy resources or even participate in the generation of power to sell back to the utility when it makes sense economically. This approach of meeting the supply demand imbalance locally would hence also automatically and immediately mitigate inefficiencies in transmission and distribution.

The other key issue that would be addressed is whether the aggregated behavior of controllable DERs can be consistently relied upon or possibly have a negative and harmful impact on the stability of the power grid. Using responsive electronics has the potential to convert the grid into a dispersed and networked system that responds in real-time to a broad range of system states, but actually deploying decentralized control must ensure that load balancing is still feasible and system reliability is not compromised. Even though complex systems utilizing self-regulation through emergence have been generally demonstrated to be more resilient and flexible than simpler top-down command and control systems [OPMO13, WGM$^+$13, Eto07], there is still some uncertainty as to whether or not the combined affect of many DERs can always be relied upon. Dynamic control algorithms that can coordinate DERs in an open

market system resolve these problems by optimizing and controlling aggregated behavior through a power transaction market where the utility function of each resource is quantified and accounted for in the operation of the system. Each resource still retains local autonomy but the dynamic market ensures that an optimal solution for minimizing the total cost to the system is found by having all participants compromise and reach a consensus without violating system constraints.

In the research for this dissertation, the primary goal is to show that dynamic algorithms can be used to schedule DERs to handle the higher penetration of renewable resources while bringing about a more open and transparent energy market. Within an open platform, these algorithms would form the intelligence layer above the communications and security layers installed in the hardware. We adapt and use algorithms developed for power scheduling [BPC+11, PB13, PB14, KCLB14] to schedule the energy resources, allowing them to respond to the intermittency of renewable resources as well as fluctuating prices. It is important to stress that each resource can act autonomously since a consumer cannot be expected to be actively engaged with energy management at all times. DERs can be aggregated at any level and work within the current infrastructure with their own internal market even while the entire grid is still transitioning to a more dynamic infrastructure. Externalized costs could also be represented through these algorithms in order to facilitate convergence to a market equilibrium that reflects a more realistic clearing price. Our work looks at both centralized generation in a grid-tied PV storage system and a distributed microgrid with on-site PV generation and DR capabilities since the future grid will require a diversity of solutions for utilizing renewable energy in order to realize the range of benefits available with different levels of integration. Climate change and ecological concerns coupled with economic concerns over fossil fuel prices post-peak oil are addressed not only through the integration of cleaner energy sources, but also by providing a way to participate throughout the system via a distributed and decentralized infrastructure. The latter capability should be emphasized since it directly addresses the concern over whether consumers will act in response to market signals. Other work has shown that when given choices and the opportunity to manage their own energy profiles, many consumers elect to do nothing or lose motivation to participate after an initial period of time [PKMB+10, MIT11, Com13a, Par12, EQ09, All09, PGK+09]. The reasons cited often have to do with the high opportunity costs resulting in little economic benefit to gain. With the dynamic algorithms we propose, accurate cost

signals can be automatically integrated and each participating resource operates autonomously based on external conditions and consumer preferences. The amount of active participation required is entirely up to the consumer and privacy is maintained. Additionally, the ability of the supplier or other large scale participant to take advantage of the customer and monopolize the system through information asymmetry is mitigated by decentralizing control and enabling all participants to respond. This also recognizes and promotes the idea that technologies cannot work in isolation and must be considered as part of a system since they can behave symbiotically with complimentary benefits that influence system dynamics. Such benefits include responding to demand peaks, intermittency from solar generation and extended contingencies due to outage events. The result is a more constant power profile and a reduced dependence on operating reserves for support, as well as increased system flexibility and responsiveness. This reduces the cost of managing the electricity grid through increased system stability while simultaneously reducing carbon emissions through more efficient distribution of power. In addition to demonstrating load balancing and rapid response times capable of lowering demand forecasts, DER functionality is maintained without compromising system reliability.

Our approach using dynamic and autonomous algorithms also keeps the control problem tractable as system complexity rises with increasing levels of participation. The large amount of data from forecasts and other measurements that are now being made throughout the grid can be naturally and readily integrated. This enables realization of long term environmental, social and economic goals without requiring a costly system overhaul by utilizing the current resources more efficiently. Policies and incentives are allowed to co-develop and work with the technologies to ensure a smooth transition to an infrastructure that can run on renewable resources. There is no attempt to pick a winning technology and skew the market in any direction. Each DER or aggregate of DERs simply determine their own local costs with an agreed upon metric to price electrons so that no shared costs of generating and distributing electricity are externalized. For example, a dynamic cost could factor in environmental costs as well as operational costs of the power delivery system and each DER would then have to account for this in its own operation. The incentives of each stakeholder in the system are in this way automatically aligned through the optimization algorithms to attain the responses necessary to meet both system-wide and individual objectives [Kin13]. This approach is also dynamic in that it constantly

adapts to changing operating conditions and is robust to measurement inaccuracy and physical model imprecision. The impacts of system variability are more readily and directly account for than they would be in a statistical model. Risk associated with operating the system is consequently reduced and the system can effectively self-heal. While we focus on two of the more popular solutions currently being explored, the algorithms are also application agnostic in the sense that they can include a range of DERs with varying objectives. Since the algorithms used are both flexible and scalable, devices can be easily integrated as more come online and could be applied to a range of scenarios including in remote areas and developing countries.

In addition to demonstrating that such algorithms can be implemented dynamically in real-time, we show that they can also be used in models and simulations as a design tool to inexpensively develop future systems which can still operate efficiently and reliably with more generation from renewable resources. This allows for optimization of the range of mechanisms that can be used to manage the intermittency problem, including the use of collocated energy storage systems, dynamic prices representing system-wide conditions and curtailment of output from renewable energy generation. The algorithms will be able to determine avoided costs and tradeoffs in a complex operating environment. It can also be used to find the optimal locations to upgrade the electricity network but we focus on the other methods since upgrading the grid often entails political and legal issues that are outside the scope of our current work. Focusing on efficiency and conservation solutions like DR and strategically placed storage, especially as the enabling technologies become more accessible and affordable, is also a more effective way to reduce emissions and climate change and can minimize overbuilding of the grid and power generators [oE13b, AES, Eto07, Far, GRLS10, PD11]. The modeling platform would help provide the further investigation and required analysis needed to determine the effectiveness of these mechanisms acting both individually and cooperatively while offering greater confidence in terms of knowing how much resource is available.

1.1 Dissertation contributions

This dissertation models and simulates real world systems using an integrated dynamic optimization and control framework through which DERs can be scheduled and controlled. Specifically, we look at a grid-tied PV storage system operated by

the Public Service Company of New Mexico (PNM) and a distributed microgrid with on-site PV generation, battery storage, curtailable load and a fleet of DR-enabled EVs. In both cases, real measurement data is used to optimize and control the systems. The primary contributions of this research includes:

- Using MPC with both centralized and distributed optimization algorithms to carry out dynamic simulations based on real systems. Data and measurements from the system are integrated into both the models and predictions. Realistic objectives and constraints are also specified, including smoothing of renewable energy generation, minimizing real-time energy costs, consumer utility, and DER cycling and curtailment costs. While DER control and scheduling is a field that is currently rapidly developing and previous work has shown how algorithms can work at a highly theoretical level, this work attempts to model and simulate real systems in a complete framework to demonstrate that the algorithms used to control and schedule the DERs can work well with actual data in real world conditions.

- Demonstrating that a dynamic controller using MPC can perform near-optimal with limited access to information. Using convex models and simple prediction methods, results come very close to prescient scenarios and this suggests that MPC can be used in embedded controllers to schedule DERs in a way that handles dynamic conditions on the power grid and allows for greater integration of renewable energy resources while maintaining system reliability. DERs scheduled in this way can act as flexible and reliable resources on the grid while operating independently. Increasing responsiveness in the distribution system in this way also has significant implications for reducing costs associated with system upgrades and procuring more operating reserves.

- Demonstrating the synergies that result when a distributed algorithm like ADMM is combined with MPC. Previous work has looked at how distributed methods can be used to schedule DERs or looked at how MPC can be used for dynamic control of DERs, but there hasn't been an integrated approach looking at how a dynamic MPC framework can be used to execute the schedules determined by the distributed algorithms in a robust and dynamic manner with imperfect and incomplete information. Our work shows that this is not only possible, but it performs very well in cases where DER behavior is coupled due to capacity

limited lines and a cooperative approach between the resources is required. This allows various key objectives ranging from integrating renewable generation to DSM to be easily combined. The relationship between MPC and ADMM is symbiotic in the sense that the warm start from MPC can dramatically speed up computation time during the ADMM iterations while distributing the optimization problem with ADMM allows coupled optimization problems to be solved more efficiently in the MPC framework.

1.2 Outline

The remainder of this thesis is divided into several chapters:

In chapter 2, we provide background on the current power system and the problems it faces. We also look in detail at the obstacles facing a future grid with more renewable resources deployed and discuss the state of the technology used to address these hurdles using DERs. This helps inform our choice of dynamic algorithms as well as determines our modeling objectives and constraints when simulating the real world examples.

We then describe the methods we use to simulate and model our systems in chapter 3. This includes the MPC algorithm that enables dynamic and robust control and the distributed ADMM method used to achieve market consensus. In addition to the operating framework and distributed optimization method, we also describe the locally implemented prediction methods we use to help schedule DERs.

Chapter 4 looks at the example of centralized generation with the grid-tied PV storage system. In this chapter, we first describe the model we use and then describe the numerical results obtained running simulations with data taken from PNM's Prosperity Energy Project. Since it only makes sense to use distributed algorithms in complex operating environments when there is coupling between resources either through prices or constraints, this example focuses on the benefits of using an adaptive framework for integrating renewable resources onto the grid.

In Chapter 5, we look at the second example which explores a distribution-level microgrid with onsite PV generation, partially curtailable load and a battery storage unit for backup. In this example, we again describe the model used for the system and then describe the numerical results. The data was taken from both measurements at a commercial building and a transportation study looking at regional driver behavior.

Scenarios are considered where many DERs have to operate cooperatively due to coupling constraints and so in this example, ADMM is used in an MPC framework to demonstrate how power flow can be managed within the microgrid in a distributed manner.

Finally, we conclude our results in Chapter 6 and make recommendations for future work.

Chapter 2

Background

2.1 Load balancing and the electricity grid

Since the inception of the power grid in the 1870s, the primary role of the system operator responsible for coordinating and controlling the grid has been to keep the cost of power delivery low while maintaining stability and reliability across the lines. These objectives have led to a grid topology oriented around the economies of scale that come with conventional generation technologies like large coal- and gas-fired power plants which can produce up to several gigawatts of power reliably. With relatively predictable demand levels, this infrastructure made the most economic sense since plants could be scheduled well in advance to meet the anticipated load. As electricity demand increased and regular spikes in demand began to form, however, daily peaks in a more unpredictable load had to be addressed and accounted for. Higher, more diverse and more geographically distributed demand meant that the technological problems and cost bottlenecks were no longer so much in procuring sufficient power generation but rather in the dynamics of power generation and delivery. By the end of the 20th century, demand growth was outpacing generation and the economies of scale that had been the primary motivation for centralized power were being undermined by shortfalls in both generation and the capacity of T&D at peak times throughout the grid. Since the grid was built out in a cobbled together network of power lines to meet new demand and lacked a comprehensive shared vision for system structure or function, inefficient power delivery and a fragmented and inconsistent regulatory regime became the main drivers for both power costs and power reliability problems [MIT11, WS13]. Coupled with aging and deteriorating equipment that has not been

upgraded significantly since the 1960s due to a chronic lack of investment, today's grid is leading rapidly to an overpriced and vulnerable power delivery system with corridor constraints resulting in brownouts and occasional blackouts [GA11, Ami11]. The current centralized infrastructure has also led to increasingly higher environmental and energy costs as the sustainability of a grid primarily powered by fossil fuels is called into question.

To better understand what inefficiencies and problems the grid currently faces, it is instructive to have a basic understanding of how the grid works. In order to maintain stability and keep the supply and demand in balance on the electric power grid, the amount of electricity being generated must be equal to the amount being consumed at any instant in time since there is little buffering or storage capability on the T&D system. The grid frequency is a system-wide indicator of overall power imbalance and fluctuates continuously in response to changing demand and the practical impossibility of controlling generation instantaneously to track all changes in demand. System frequency will fall if demand exceeds supply because generators will start to slow down slightly and will rise if supply exceeds demand and generators speed up. Most electric systems today use alternating current (AC) with a nominal frequency of 50 or 60Hz. The system operator is required to keep the frequency within ± 0.5Hz of the nominal value in order to maintain grid stability and the mains frequency normally remains within ± 0.2Hz. Since there is little control or flexibility on the demand side, the rate at which this frequency changes is primarily dependent on the system's total inertia and the aggregate response of the generators' control systems.

In the U.S., the current power system consists of interconnected areas with tie-line constraints and the responsibility of the system operators in each area is to meet their own load while maintaining scheduled tie-line transactions. In order to accomplish this, they depend on generators with a supplementary control loop referred to as the Automatic Governor Control (AGC) that addresses not just local load imbalances but also keeps control of active power across the tie-lines and prevents steady state frequency error from increasing indefinitely [KNL96]. With no load control, the natural inertia of the governors automatically tries to adjust their speed to respond to changes in frequency. But the response is relatively slow and the load increase results in a droop effect in speed-governor control that prevents frequency from ever fully recovering. AGC takes effect and adjusts the generator output to

eventually restore frequency after several minutes [KNL96, BTB04, TDL05] This generator control action is determined by estimating the Area Control Error (ACE) from the tie-line flow and system frequency along with parameters that represent the natural responsiveness of the system. Although it keeps the control error manageable, the relatively slow response time can result in the AGC transient response degrading for large control areas due to errors that result in estimating the ACE from parameters that are nonlinear. Higher frequency swings caused by electromechanical oscillations can additionally become undamped in a power system that is stability limited with heavier loading and this results in unstable operation [TDL05]. In a deregulated power system, the concept of a purely vertically integrated utility also ceases to exist since generators are no longer required to have secondary control and provide ancillary services. But the objective of restoring frequency and net interchanges for each control area remains and this makes managing the system far more complex and challenging [BMT04]. This means that the resiliency of the grid to load imbalances is now lacking and often inadequate.

An important part of load balancing and keeping power costs low is the ability of the system operator to predict and meet load at any given time. Peak load times will vary by region due to industry mix and local climate but are typically highest in the late afternoons during warm seasons and in the mornings and evenings during cold seasons. Although changes in the everyday demand pattern can be predicted fairly accurately to within $1 - 2\%$, there is always uncertainty associated with random variability [Ove12]. This not only results in frequency instability, but can cause voltage fluctuations due to local power imbalances. While demand is only expected to gradually trend upwards over the near future, the load factor is expected to decrease as a consequence of increasing peak demand [MIT11]. This trend is due in part to the increase in residential and commercial loads and the relative decline in market share of industrial loads. The load duration curve is only expected to get steeper, especially as the transportation sector transitions off fossil fuels and more EVs plug into the grid without incentives to encourage smart charging. This means that the potential for power imbalances will increase if not managed carefully. Generation contingencies such as sudden large generation losses occurring due to generator faults or loss of transmission lines may also occur and must similarly be accounted for. The sudden reduction in generation results in sharp decreases in system frequency and voltage and without preventative action, the integrity of the electricity grid is compromised.

Because load on the grid is currently lacking in flexibility, load serving entities must be able to provide for both base load and peak load and have a peak reserve margin with a capacity that is $10\% - 18\%$ above their projected peak power demand [CF12]. This is typically achieved by purchasing the bulk of the anticipated power through contracts or on the day-ahead market and then providing for shortfalls to real-time demand with additional purchases on the spot market or through operating reserves such as spinning reserves and peaking power plants.

Spinning reserves are the extra generating capacity that can be called on at extremely short notice as a primary response mechanism for frequency control and balancing of supply and demand. It is provided by having certain generators operate below maximum output in order to create a buffer of spare capacity and by this means a safety margin to protect against faults. These spinning reserve generators must be compensated for lost generating opportunity and for stress to the governor control systems and steam valves that control power output. Already synchronized to the power grid, the generators then operate by frequency response and alter their output of active power on a second-to-second basis to correct imbalances in supply and demand, thus keeping the frequency around the nominal value. Spinning reserves have the fastest response and must be able to last up to 1 hour to qualify as operating reserves. Their rapid response time makes them the highest valued operating reserve [CF12]. In contrast, peaking power plants typically run natural gas generators with fast start-up times and low capital costs but are both inefficient and costly to run since they are specifically designed to operate during infrequent peak demand scenarios and not to produce power efficiently to meet baseloads. Peaking power plants are typically needed for only a few hundred hours per year and must be able to respond within a 10 minute timeframe [CAI07, CF12]. Since the system operator is only able to schedule generation capacity in advance to meet the anticipated base load, unanticipated load imbalances must be removed through requests to spinning reserves and peaking power plants if DR is not available. These generators operating as reserves cost the power utilities and taxpayers millions of dollars annually because they constitute power that must be purchased in addition to the power that utilities must purchase in order to meet everyday demand [HDJ+13]. They are particularly expensive, however, since they are purchased as capacity and not as energy on the ancillary services market to provide the power needed during the infrequent peak demand events. It also makes generation far less efficient since generation equipment

is designed to operate most efficiently at full power and not when partially loaded. The kind of generation used for rapid response also typically runs on fossil fuels and the extra fuel burned to keep the generator spinning will produce a significant environmental burden and increased greenhouse gas (GHG) emissions [DEKM10]. Hence the poor utilization of resources combined with the necessary redundancy in the grid to ensure power reliability has led to an infrastructure that is now both inefficient and expensive to operate and maintain. The requirement for the power system to be equipped to meet peak load with a reserve margin can lower capacity utilization such that more than 20% of generation capacity as well as transmission and distribution capacity gets used less than 12% of the time [MIT11]. The costs of building, maintaining and operating this idle capacity is passed onto consumers as higher electricity rates.

Having reserves ready as described above to handle a growing and fluctuating load has been the traditional approach to energy management on the grid. This has not only resulted in large inefficiencies throughout the system both in terms of generation and power delivery, but is also unsustainable since it requires building out more generation along with T&D capabilities than would have otherwise been necessary to handle under-frequency events. New large scale generation plants are becoming increasingly harder to cite and the grid upgrades required for more power to be transmitted is growing more costly. Deregulation has further exacerbated the problem by removing any incentive for the T&D infrastructure to adapt and modernize services. While the intention of deregulation was to break up the traditional vertically-integrated monopolies that operated the grid and introduce a competitive wholesale market to reduce overall system costs and inefficiencies, it lead to a more complex operating environment where long-term planning for infrastructure upgrades took a back seat to other economic considerations. This included the ability to recapture capital costs through immediate scheduling of power plants and grid equipment which were now seen primarily as dispatchable assets in the investment analysis. In many cases, this meant that there was no buffer to provide backup in case of generation outages and power is delivered continually over long distances throughout the power grid at near capacity. The result is fluctuating power flows that can significantly impact system reliability and lead to uncontrolled cascading effects throughout the grid [PSS14, SRTF10].

2.1.1 Current trends

Improvements in power grid dynamics are possible if the system operator has additional support from connected energy resources throughout both the transmission and distribution systems that can respond to system imbalances and provide energy reserves or other ancillary services when needed. By improving dynamic stability in the system in this way, the grid can operate more efficiently and the amount of real power that can be transferred over the transmission system is increased since the increased ability of the system to absorb and handle large unexpected power swings allows the grid to operate nearer to capacity without worrying about contingency events that could potentially lead to blackouts and brownouts. More generators are also allowed to operate on baseline loading since they are not regulating as much and this means they operate more efficiently.

Hence to address the problems facing the current aging power infrastructure, a number of efforts are focusing on locating smaller generation units closer to sites of demand and enabling DERs to respond locally instead of trying to meet increasing demand through more centralized generation [SFEB, RLL+10, U. 07]. Embedded intelligence installed at the distribution side of the network would increase energy efficiency and reduce both operational and environmental costs throughout the entire grid by coordinating DERs across the network and allowing them to adapt dynamically to real-time grid conditions [DRT+12, FMXY12, Voj08]. DERs employed in this way could help alleviate bottlenecks in transmission in addition to increasing the flexibility and resiliency of the system. They enable demand to be balanced with generation efficiently in a distributed fashion to meet both base load and load peaks. Even small amounts of local generation and load adjustment from the distribution side can significantly reduce the need for spinning reserves and peaking power plants by eliminating just a fraction of the demand that occurs during peak periods [Com06].This not only reduces cost to the consumers but also eliminates the environmental cost of reserve generators and can significantly extend the life of power equipment. By removing congestion and preventing any one supplier from monopolizing the market, distributed resources also encourage fairer electricity prices [U. 06, Kin13]. Another benefit of distributed generation is that it circumvents the problem of local opposition to siting new lines and the significant cost of building out the lines. Many have consequently argued that investments in the future grid should go towards having it operate more like an open and transparent internet, allowing more equal access to

producers and consumers at all levels [MIT11, Kin13, WMB05]. In an interconnected and open system still facing capacity constraints, utilities should start to view DERs not as competition or as an unwelcome disruption but rather as a resource that can be utilized to keep the grid running more flexibly and reliably. As we move towards a cleaner and more sustainable grid that relies increasingly on intermittent and variable generation from wind turbines and solar cells, system operators now have to also figure out how to handle sharp fluctuations and unpredictable behavior on the generation side in addition to maintaining system reliability and minimizing cost. But the increased flexibility and resiliency offered by DERs responding locally can allow utilities to more easily increase and integrate intermittent renewable energy resources by allowing for distributed generation across more diverse geographical regions and balancing them locally with controllable resources that can buffer the intermittency.

While it has been recognized that the aging infrastructure of the grid requires significant investment to maintain functionality and reliability, one U.S. Department of Energy (DOE) study estimated that upgrading the grid with embedded intelligence, sensing and communication capabilities would save between 46 and 117 billion dollars over the next two decades [KKMC+03]. This "smarter" infrastructure would be able to better predict and anticipate the demands of the system, and to increase consumer participation through either dynamic pricing or DR strategies. Dynamic control and optimization of grid operations and DERs would also allow for easier integration of renewable energy and increased participation of DERs throughout the system. The DOE emphasized the need for this real-time control and response to be automated and properly incentivized to remove any barriers to entry into the power market [Ame]. To this end, funding has been allocated to develop more robust energy control technologies for DERs throughout the grid to enable the automated and distributed management of both electricity and information [JW12, Bet12]. These upgrades create a more flexible and resilient system with minimal overbuilding, especially as the enabling technologies for a distributed approach become more accessible and affordable [IA09, Voj08]. Of particular interest is the ability to manage bidirectional energy flows in a more meshed distribution system since this would allow DERs to operate to their full capabilities without worrying about the safety and reliability issues associated with reverse power flows. Bidirectional information flow between the DERs and the utilities is also important since it enables better utilization of the resources and mitigates the need for peaking power plants, spinning reserves and

avoids the extreme brownouts and blackouts scenarios. It enables the network to self-heal and ensures a more reliable supply of electricity with reduced vulnerability to single points of failure.

2.2 Electricity market

The current power transmission and distribution systems in most industrialized nations operate on a competitive bulk power market. In the U.S., load serving entities must purchase $60 - 95\%$ of their electricity on this deregulated, wholesale market and make up for real-time shortfalls through the spot market or through operating reserves [U. 06, Ove12]. Generators competing on the wholesale market bid a kilowatt-hour price and the system operator arranges these offers in ascending order in a bid stack and dispatches enough generation to meet the anticipated load from consumer bids. Each generator dispatched is then compensated by the price the last generator bids. Additional parameters such as generator start-up times, actual line congestion and reliability concerns can lead to wholesale prices that swing wildly in the course of a day. These changes occur at hourly intervals at each node in the system where predictions are made about constraints that could cause congestion and bind transmission in order to calculate the nodal price or locational marginal price (LMP) for a kilowatt-hour of electricity. The average retail price of electricity in the U.S. is around \$100/MWh whereas the wholesale price can go from almost nothing to near \$1000/MWh [MIT11]. While such extreme swings are not common, the daily price is bound to be affected by the generator and load mix on the grid. For example, inflexible base-load generators that are costly to shut down or start-up will bid at a price at or below zero in order to ensure they are dispatched. Generation capacity can also be pulled out of the mix and used as operating reserves instead, lowering the clearing price that each generator receives. Prices can additionally be manipulated for less legitimate reasons such as deliberately withholding power to maximize profit during peak demand or forcing physical constraints to increase congestion pricing [Ove12, NGNT04, Hog12]. Hence while the LMP accurately reflects the supply-demand equilibrium price in theory, economic dispatch in practice can often artificially change the clearing price.

In the retail market, the situation is very different. While some states and European countries allow consumers to purchase from competing retail suppliers with the

utility operating the delivery system and being compensated for distribution services, the majority of states have little or no retail competition and a retailer will have a regional monopoly. These retailers or load serving entities dispatch the electricity purchased on the wholesale market at a higher price to ensure they can cover all their costs for both power purchases and contingencies. The consumer is also often not exposed to real-time prices that reflect the variable wholesale price. Instead, prices are based on average costs sufficient to recover costs of generation and distribution and make returns on investments. Prices will typically have at most 2 or 3 price levels in a given day to reflect average demand during different parts of the day as shown in Fig.2.1. If the system is run inefficiently and is constantly facing demand peaks that result in dispatch of expensive forms of generation, the additional costs are simply added to the kilowatt-hour rates and distributed to all ratepayers equally. These problems are only further exacerbated by the fact that consumers who do not face real-time prices have no incentive to reduce or shift demand during peak times. One analysis suggests that even on a time-of-use (TOU) rate schedule with a few different price levels, off-peak rates are still almost three times higher than the marginal cost of electricity [Fri11]. The general consensus is that improving grid efficiency and keeping rates low will require providing the correct incentives to both generators and consumers by offering dynamic rates that reflect the time-varying costs of supplying power [MIT11, Kin13, Bor09]. The market needs a coherent policy that incorporates GHG emissions, DERs, and renewable and distributed generation to give incentives for investment in a more adaptive and reliable grid.

Recognizing the shortcoming of the retail electricity market, some wholesale markets are now beginning to open up to end-users and allow for more communication between generators and consumers. Providing a market signal that rewards energy efficiency and flexibility in consumption at the distribution level enables DERs with storage capacity and/or flexible schedules to use their resources more efficiently and participate in operating the system more effectively [MIT11, GRLS10, Bor09]. This alignment of interests on the demand-side is critical and integral to enabling active participation of consumers since it ensures that the benefits of responding at the distribution level are shared between the system operator and the end-user rather than just the former [Kin13, U. 06, Hel09, Det06]. While large commercial and industrial sectors are starting to move in this direction, dynamic pricing combined with technology to automate response to price changes at all levels of the grid is still a long

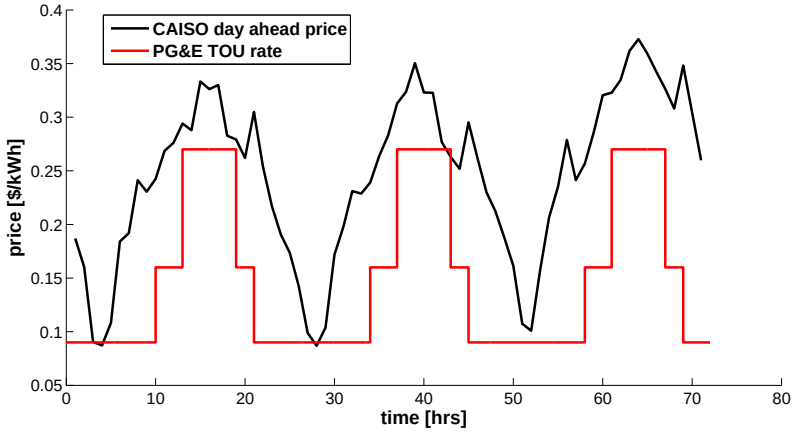

Figure 2.1: CAISO wholesale day-ahead price compared to PG&E's time-of-use (TOU) rate structure. While the price peaks align, the wholesale prices are much more volatile and dependent on grid operating conditions.

way from being realized and a platform to realize such a framework is greatly needed [All09, PGK+09, MIT11, Fri11]. Once the demand-side becomes more empowered to respond to grid conditions, the system-wide benefits of a competitive electricity market can then be reallocated to consumers through both lower electricity prices and increased reliability [Jos08, IA09]. By directly accessing and responding to market conditions, consumers can also cut out overhead costs and reduce the market power of electricity suppliers when production runs short [Bor09]. It provides a response mechanism that decentralizes control and increases bi-directional transactions on the grid while simultaneously load balancing. To implement such a system, the final clearing cost would additionally include access fees and congestion costs to address the problem of paying for access to T&D wires. Since operational costs would be inherently represented, independent companies providing T&D services would be able to maintain and operate the system as well as have the resources and incentives to upgrade the system and provide a market for DG and other DERs [MIT11].

2.3 Renewable energy

Renewable energy has been growing exponentially on the grid over the last decade, constituting the largest fraction of newly installed generation capacity worldwide since 2008 and already providing more than 19% of electricity generation worldwide [Adm13b, REN13b, Age12]. Although the majority of this energy currently comes

from hydroelectricity, the share coming from renewable sources like solar and wind is increasing rapidly. Wind power has gone up at an annual rate of 20% with a global installed capacity of 238 000 MW at the end of 2011 [LWH12, REN13b]. The growth of photovoltaics has been even more impressive, surpassing wind in 2004 as the world's fastest growing energy source. Since 2007, PV capacity has more than doubled every two years with a 63% annual average growth rate and the worldwide capacity at the end of 2011 was 67 000 MW [Age10, REN13b]. This has been due primarily to the sharp reduction in the manufacturing and installation costs of PV. These capital costs make up almost the entire levelized cost of electricity (LCOE) since PV systems do not use fuel and are extremely low maintenance. Over the last two decades, the price has dropped by more than an order of magnitude [Lac13]. The International Energy Agency (IEA) predicts that solar power generation may provide most of the world's electricity by 2060 [Age11].

The main benefits of generating electricity from renewable energy sources like solar and wind are that the energy is renewable, available locally and has a minimal environmental footprint with little or no pollution. Recognizing these advantages, over a hundred countries around the world are currently rolling out policies and programs to encourage rapid development of technologies for renewable power generation [REN13b]. A wide range of policies also exist at the state and local levels. For example, many states in the U.S. have already required utilities to meet renewable portfolio standard (RPS) targets to include a certain percentage of renewable energy in their portfolios by a specified date and have also provided tax incentives to encourage development and use of renewable energy technologies [CtIRE, Leo13]. Other countries throughout Europe and Asia have similar programs and incentives in place. While strong public support for renewable energy has put pressure on utilities to integrate more resources into the grid even before technologies have reached grid parity, the costs are already rapidly coming down for all forms of renewable energy and the costs of fossil fuels are heading in the opposite direction [AM]. As the need to quantify externalities through mechanisms like carbon taxes and cap and trade accompany the technological progress that is rapidly decreasing the cost of installing renewable energy, the price gap between renewable energy generation and conventional generation will soon vanish with grid parity already within reach in some regions [Kro13b, EBE+11, Fri11]. This means that integrating renewable energy even without incentives now appears to be a smart long term economic investment for both

sustainability and energy security [Age13b].

In developing countries as well as in rural and remote areas with no power infrastructure, using renewable resources to directly meet load can be particularly beneficial since transmission and distribution of energy generated from centralized power plants can be both difficult and expensive [REN13b, MBl11]. The geographically diverse and decentralized nature of renewable energy resources means that DG using solar and wind will be well suited in these areas, potentially becoming the way the approximately 1.4 billion people around the world without access to grid electricity [EMS$^+$, Age11] can leapfrog a traditional grid infrastructure dependent on centralized power from fossil fuel resources. Not only will this have significant environmental benefits, but it will also avoid resource monopolies, exposure and vulnerability to global commodity market swings, and possible mismanagement of resources due to weak, unstable or corrupt institutions.

However, despite the significant leaps renewable energy technologies have made and the many benefits they give rise to, both solar and wind still only contribute a very small fraction of the world's energy and both stronger policies and better technologies are needed to encourage widespread adoption and integration into the grid. Many engineers and scientists are already advancing plans to transition the world to an energy system that is supplied primarily with renewable resources but studies suggest that while this transition is both possible and affordable, it is necessary to rally the required policy and industry support to set up the regulations and incentives for a renewable energy infrastructure to really take hold [JD, REN13b]. In addition to bringing down the relative cost of renewable energy, the problem of intermittency must also be addressed in order for renewable energy resources to be truly sustainable and widely adopted as an alternative to fossil fuels. With most wind and solar generation systems, the capacity factor is only in the range of $10 - 40\%$ [Adm13a]. In addition to this, the weather dependence of these resources means that they have higher ramp rates and variability in output relative to traditional sources of generation. This means that as penetration increases, the ability of the system operator to balance load and supply will be significantly hindered as intermittency causes voltages to fluctuate and the stability and reliability of the grid becomes compromised [SHC$^+$12]. The existing electrical power system already incorporates significant load intermittency which is managed through generator dispatch but this is at significant cost and is already reaching its limit with the aging power infrastructure. Many studies have

estimated that the variability in supply combined with the variability of demand could begin to pose a significant challenge using existing centralized dispatch mechanisms and lead to serious increases in operating reserve requirements and other protection mechanisms as penetration increases beyond 30% [RP10, BI04, THPB08, DEKM10, WNRN08]. Hence while the variability and unpredictability of wind and solar pose no issues at very low levels of penetration, increasing penetration means that operating and maintaining system reliability will require a change in operating strategies in order to provide greater flexibility and resilience as well as prevent retail costs from shooting upwards [Cor09, Ele10]. Currently, there are few system-wide market incentives to encourage technologies that enable greater flexibility in the power system even though this will be critical to integrating more intermittent renewable energy sources onto the grid and meeting the renewable energy targets that many states and regions have set for themselves.

Also, while in some situations renewable energy can help to meet peak demand and reduce the need for grid upgrades or expensive operating reserves, generation and demand peaks are frequently misaligned. As penetration of renewable resources increases, more buffering capacity will be needed to ensure load can be met at all times to handle the misalignment as well as the inconstancy and unpredictability of solar and wind. Currently, as with load variability and intermittency, the only meaningful option to handle this issue is to dispatch operating reserves including spinning reserves and peaking power plants [PKMB+10]. Hence instead of reducing the need for transmission reinforcements and generator dispatch, increasing penetration of renewable energy resource could potentially make the system less efficient if not integrated carefully [CF, MNPG+]. While storage and DR have been considered viable options, the former is still too expensive for large scale deployment and the latter is only available in small amounts and has yet to demonstrate that it can be consistently relied upon when needed. Finding ways to connect local energy resources in a way that allows them to transparently and openly exchange power in an intelligent manner will be critical to ensuring renewable energy can be ramped up without negating the economical and environmental benefits they help bring about [Kin13, WMB05]. For example, strategically placing just small amounts of storage at locations where congestion is expected could go a long way in terms of providing the power system with both flexibility and reliability. One could imagine placing storage units at the ends

of a congested corridor to alleviate congestion without having to upgrade the system. Studies have shown that small amounts of storage could significantly help in the transition towards a power grid that is dependent entirely on intermittent renewable resources [DEKM10, BST+13].

In addition to the practical problems of integrating variable and intermittent generation resources, an unanticipated fallout of the subsidy structure for renewable energy is that it has lead to solar and wind undercutting the margins required for the utilities to operate the power grid [Kin13, MIT11, Sho13]. Renewable energy sources have very low or zero operating costs and the subsidies mean that they can bid a negative price while still receiving positive compensation using the excess power they produce above what they deliver to the utilities through purchase power agreements (PPA). This means that their bids are accepted first on the open market and they are used whenever available. While this also reduces the market clearing price, increasing the penetration of these renewable energy resources in the generation mix may result in more conventional generation plants running inefficiently at lower output either because they are priced out of the bid stack or because they are required to help provide the larger capacity margin as operating reserves in order to assure the same level of system reliability with more intermittent renewable sources online [CF12, MIT11]. Hence the introduction of more renewable energy onto the grid through the current market and subsidy structure can and has resulted in higher costs both in terms of price for energy and environmental impact. This problem has recently become much more prominent as a consequence of the rapidly falling technology prices and a maturing industry in which companies in the renewable energy space are beginning to consolidate and grow in order to maintain profitability. Combined with the net metering problem on the retail side described in the following section on distributed generation, this has led to a power market that will become progressively unsustainable as penetration of renewable energy resources increases and system operators have to absorb unexpected risks assuming no measures are taken. Generation from renewable resources is unarguably necessary for a sustainable energy infrastructure, but subsidies must be phased out and replaced with a market where energy is priced dynamically to reflect the actual costs of operating the system as well as all the environmental and social costs of utilizing different energy sources.

2.4 Distributed generation

DG from renewable resources like solar and wind arose out of concerns over externalized environmental and health costs of conventional centralized generation, the deteriorating equipment and constrained capacity of the delivery system for bulk power, the increasingly distributed nature of loads, and the increasing price for energy. Prior to the surge in renewable energy, DG was often not practical or economically accessible and was deployed only in very niche circumstances. Utilities were not keen on DG because its financial benefits tended to be site-specific and required a large initial effort to realize [U. 07, Ass13]. While the LCOE of DG from renewable resources is still relatively greater than conventional generation from centralized power plants, the scalability and rapidly declining costs of PV and wind means that DG can now be deployed throughout the distribution system close to loads and start to compete against the economy of scale from large centralized generators [SFEB, RLL+10]. With DG becoming economically feasible, the other potential advantages of deployment are significant. This includes reducing demand peaks and increasing system resiliency. Long term benefits such as reductions in land citing costs and improvements in infrastructure resilience would also be realized [U. 07, Ass13]. Additionally, since DG units now commonly produce their power from renewable resources like solar and wind, local generation has become both more sustainable and beneficial for the environment.

The total installed worldwide capacity for DG systems is expected to grow rapidly over the next 5 years and in the U.S. alone there are about 12 million DG units installed with a total capacity of about 200GW [Adm13c, Lac13, Mee13, U. 07]. Additionally, solar is currently the most rapidly growing generation sector and a significant fraction of this has been due to customer-sited PV generation [Adm11, Age12]. Current installations of DG are predominantly based on solar and other renewable resources, and provide a viable way to diversify energy sources and allow for more meshed power transmission on the electricity grid. They are used to either supplant or supplement centralized power in order to meet local demand efficiently. In some parts of the country, municipalities and other local governing bodies have created regional power purchasing pools to aggregate energy resources like PV power and battery electric storage [BH02, CtIRE]. This approach allows for lower relative costs since it reduces the amount of capital needed for grid connection equipment and exploits economy of scale. It also empowers local communities by enabling them to solicit competitive electricity suppliers and use municipal financial instruments to

help fund the development of the renewable energy resources. There is additionally the possibility for local communities to participate in energy management and offer ancillary services to the grid.

Currently, there are policies and incentives both at the state and federal level to promote renewable energy and DG. The majority of states also employ some form of net metering which compensates end users generating energy at the retail rate rather than at the wholesale price [CtIRE]. This introduces another problem, however, since the fixed cost of operating the distribution system is what constitutes the main difference between these rates and this effective subsidy to DG is typically recovered by passing the cost onto all ratepayers. In other words, using DG to offset retail costs can make it difficult to recoup the costs of operating and maintaining the grid and force utilities to pass on those costs disproportionately to rate payers who do not have the capacity or capability to install DG on-site. In addition to this price discrepancy, DG at higher levels of penetration can go beyond reducing load and result in reversed power flow from the substation into the transmission grid if the power flows are not carefully managed. The power system is not currently equipped to handle such flow patterns and this could lead to high-voltage swings that could potentially damage both customer and utility equipment [MIT11]. Integration could also result in power transients that impact power quality, voltage stability, and protection equipment located upstream [RLL+10, Ass13]. All of this could potentially affect system reliability and control, especially as DG penetration continues to grow and a larger percentage of DG comes from intermittent renewable resources like solar and wind. In an already convoluted and complex system, energy management and load balancing can consequently become even more challenging, especially during contingency situations. Since rates are regulated and can only be increased if approved by the regulatory bodies, utilities do not have adequate incentives to make the necessary investments to accommodate the influx of power from DG [Kin13]. Hence while the value of DG for increasing grid flexibility, efficiency and resiliency are evident, the current way the market is run does not provide the necessary incentives to develop and increase the amount of DG [MIT11]. New methods to operate and control the grid will be critical to enabling significant penetration of DG as well as renewable energy and this can only be done cost-effectively if an open and transparent incentive structure that recognizes both the benefits and costs of DG on the system is implemented. Again, as with renewable energy more generally, this means integrating

the cost of operating and maintaining the grid as well as accounting for the negative externalities associated with conventional use of fossil fuels. This would not only help the system operators keep the grid operating, but would also enable the power system to run more efficiently with lower demand peaks and consequently lower rates for all rate payers over the long term [Kin13].

2.5 Demand response

As noted earlier, demand levels on the grid can change rapidly and unexpectedly, particularly in the residential and commercial sectors. The resulting supply-demand imbalance can threaten the integrity of the grid over very large areas within seconds. As a subset of DSM, DR addresses supply-demand imbalance on the electric power grid by controlling the amount of load connected to the grid at any one given time [PD11]. When a deregulated market was introduced to reduce overall system costs and inefficiencies, a fallout of this was that demand inflexibility resulted in volatility in electricity prices and issues in maintaining system reliability. DR addresses this tradeoff between system reliability and efficient generation of power by giving demand the flexibility to deal with load imbalances. Although DR had previously been used to some extent over controllable loads, it was not widely relied upon since there was not a substantial amount of load online that could be controlled [GRLS10, Com13a, MRLG10]. A large part of the DR that was available was also dependent on voluntary changes in service level, reducing or shifting electricity usage to less critical times when demand was lower. These payments for reduction often imposed a transaction cost by requiring agreement on base levels of use, contracts, notification, and explicit compensation, and hence limited participation to large customers or groups. DR candidates had to have necessary metering and communication capability for notification, bidding and confirmation processes and even then their response time of several minutes meant their response was geared more towards addressing longer-term energy shortages than short-term capacity shortages. And even among the customers who could afford to grant such control to the utility, many were only willing to grant control over part of their loads. So although direct control was more valuable to the utility since it increased confidence that loads would be reduced when needed on short notice, it did not really play a major role in the control strategy of the system operator. [NPC05] The situation we are left with today is that while

load shedding and DR are currently available to some extent, they are generally only accessible through dispatches made via traditional telecommunications lines and this process is neither autonomous nor dynamic. When such load response strategies are fully automated, however, they can provide a valuable reliability and power quality resource by enabling continuous bidirectional load balancing [All09, PGK+09]. This capability also empowers the demand side to moderate supplier control in a bid-based market and displace generation by increasing demand elasticity in response to system conditions as shown in Fig. 2.2. During the California electricity crisis in 2001, for example, it is estimated that if consumers could have reduced demand by 5% during the peak hours, the market would have been far more resilient to the intentional withholding of supply and enabled a 50% price reduction [Age03, Bor09]. Despite this potential, the DR potential in the U.S. is currently only 6% of peak demand and although there is the potential for another 10% over the next 5 years, the rate of DR increase has been stagnating over the last few years due to both lack of an incentive structure and capability to quantify benefits [Gro09, Com13b]. With the trend moving away from large power stations towards more DG and meshed power transmission, this ability to intelligently integrate DR enabled technologies and on-site DG to create a transaction-based collaboration on increasingly complex and intermittent networks becomes increasingly important. This is particularly critical with high penetrations of renewable energy sources where intermittency becomes more and more problematic. Fig. 2.4 and Fig. 2.3 show how DR is able to address the problems of supply variability and unpredictability by making the grid more flexible through load shifting and curtailment respectively. Since load is located relatively close to DG, it can also help absorb generation directly and both reduce line losses from T&D and prevent transient problems that may result from DG [SFEB, RLL+10, CF, DB06].

At the distribution level, DR can be integrated into any plug load to enable it to provide important services for the electricity grid such as smoothing peaks in consumer demand and continuous balancing of demand and generation. Implementation involves using power electronic interfaces to respond to signals from the grid by reducing and/or shifting load to a less critical time [CSK13, KCKT08, BLR+10, AES, PD11]. It can be enabled in any load which needs electrical energy to function but to which the delivery of is not time critical. Loads with inherent storage capacities like EVs and refrigerators are particularly good candidates for DR as they allow the excess output from local generation to be absorbed in situations when the grid cannot.

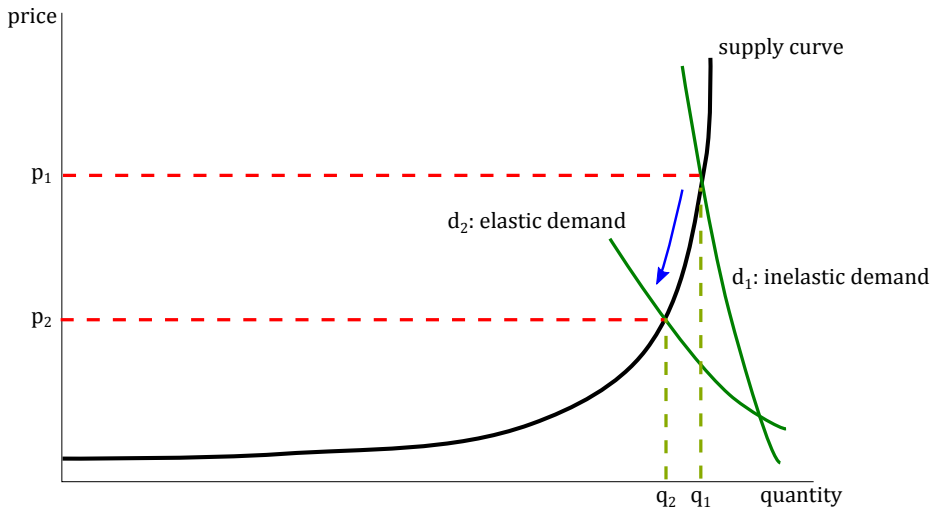

Figure 2.2: Effect of DR on price elasticity. When demand goes from being inelastic (d_1) to becoming more elastic (d_2), it can reduce the required power and consequently the price in a strained electricity market. *Adapted from [Age03]*

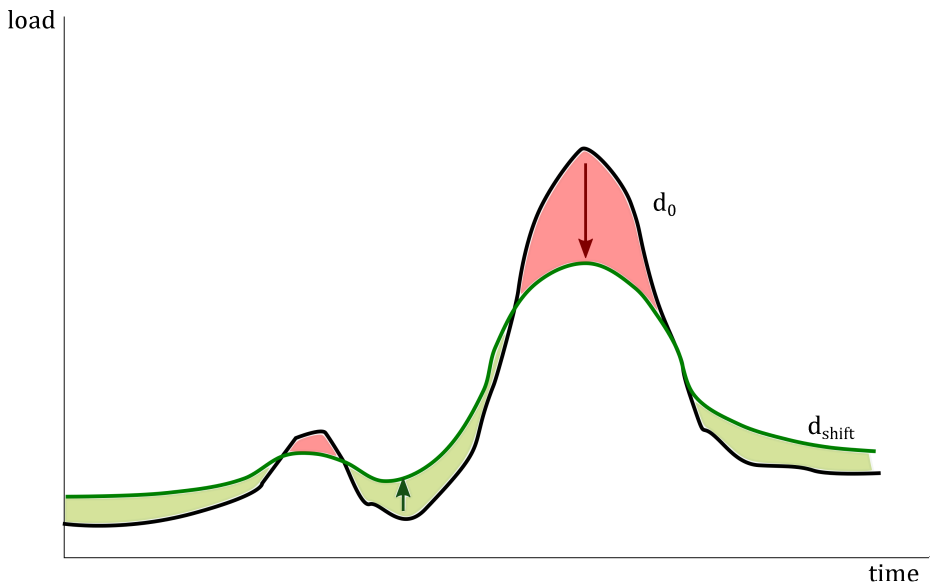

Figure 2.3: Load shifting moves electrical demand from peak times to off-peak times. The load does not actually reduce net consumption but instead recoups lost energy at the off-peak times.

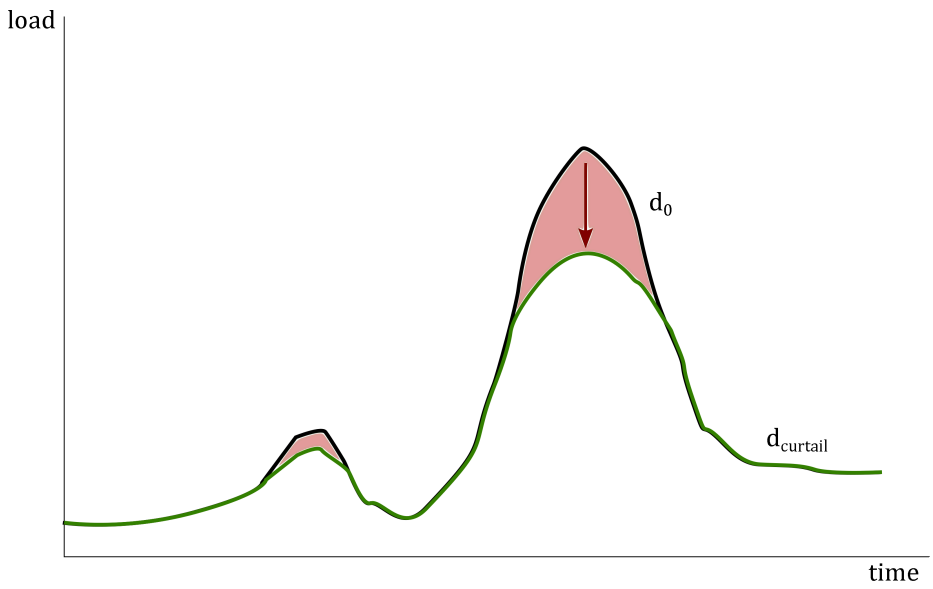

Figure 2.4: Load curtailment or load shedding simply reduces load at times of high demand when the grid is strained. The loads give up some convenience in order to allow the system to continue operating.

This reduces the need for curtailment of generation and firms up power from DG by providing power during shortfalls. Any appliance load that operates through a duty cycle such as industrial or domestic air conditioners, immersion heaters, heat pumps and refrigeration could also be used to provide a grid balancing service by timing their duty cycles in response to system load imbalances. Adaptations of DR may additionally be implemented in other appliances like dishwashers and washing machines where an intensive heating part of their program could be delayed until the load on the power grid is decreased. DR can hence be integrated into the load as part of the operation and control mechanism, modifying the control strategy of the resource and providing an extremely simple and cost-effective way of helping to manage the power grid. Individual loads receive signals reflecting the state of the power grid while monitoring their own states in order to switch on or off at optimal moments [PD11]. This can help smooth overall system load and offset spikes in peak-load or spikes due to generation loss on the electric grid. Communication technologies have made DR dispatchable on time scales of a few minutes as operating reserves, regulation or other ancillary services [BLR+10, PD11]. This means that they can help to maintain grid stability while allowing the grid to operate closer to its physical limits. In addition to helping load balance, the controller will also ensure that at no point does the load

stray out of its acceptable operating range in order to maintain its functionality and remain unnoticeable to the end user. This is a crucial point since the benefit of DR to the individual is often insufficient to offset potential inconveniences and productivity losses.

In contrast to the relatively sluggish response of AGC mechanisms, DR enabled loads can react very quickly to changes in power flow. While baseload can generally only be reduced slightly or deferred for a finite period of time, the fast-acting controllers should be able to handle most of the short-term smoothing and allow the spinning reserve generators to change their output much more gradually if at all [Kir07, Kir03, Eto07, PD11, CH11]. Leveling load curves with DR will therefore provide considerable benefit in terms of wear and tear on a generator since it will allow the generators to operate more efficiently and with a steadier output [REN13a, PKMB$^+$10]. Although a single device constitutes only a very small load, deploying an aggregate of devices able to respond at the distribution level also has the potential to save significant amounts of energy since it would avoid the need to utilize the most expensive and inefficient operating reserves required to provide rapid response to sudden load peaks [WMSS13, U. 06, MIT11, CH11]. The benefit of increasing the load factor is particularly obvious in regions with variable climates. Avoiding the need for large capital expenditures to build the actual peaking power plants and transmission lines to serve the heightened demand peaks that only occur a few hours per year would potentially lead to a $10-20\%$ reduction in wholesale prices [Off04, BJR02].

As with renewable subsidies, there has been an effort from the federal government to incentivize some amount of DR. In 2011, the Federal Energy Regulatory Commission (FERC) issued Order 745 to encourage the deployment of DR in the U.S. by providing compensation to economic DR participating in wholesale power markets [Com11]. While well-intentioned, the order has resulted in controversy with energy economists asserting that incentives cannot be correctly priced in this way without encountering unintended consequences. California along with other affected parties have filed suit against FERC to challenging the legality of Order 745 [Eis13]. Hence incentive-based DR using the structured dynamic pricing schemes proposed for DG will be critical to integrating DR into the system. This can significantly reduce the price of energy often at lower costs than supply-side resources without incurring unintended consequences [U. 06, MIT11, RI10, AES, Kin13]. Using an open market

system would also avoid the tragedy of the commons by ensuring both system-wide benefits and costs are redistributed to all participants on the grid equitably. Applied to the demand side, loads would be able to tap into their resources to both minimize harm to the grid and maximize the value they can contribute. As the amount of DR increases, the loads would in turn help to level demand and reduce overall electricity prices for all ratepayers.

The aforementioned savings come solely from the reduced requirement for spinning reserves and peaking plants and does not include the additional benefits that will arise from allowing for renewable energy integration onto the grid, improving transmission line utilization and gaining breathing time to schedule reserves and help the grid recover from frequency excursions or avoid blackouts. The ability to smooth peaks in demand will also lead to energy conservation and increase avoided costs from highly capital-intensive investments in generation and T&D. These avoided construction costs can increase the value of DR by multiples of 5 to 20 [NPC05]. DR could additionally aid in black start recovery after a power outage. Generally black starts are made more difficult because of the large number of loads attempting to draw power simultaneously at start up when voltages are low. This causes huge overloads that trip local breakers and delay full system recovery. Responsive loads could mitigate this by shedding load quickly or islanding while the running generators spin up during generation loss, then switching back in cooperatively to bring frequency and voltage back to their nominal values when generation capacity increases. This capability decreases the likelihood of peak demands tripping off severe blackouts and brownouts and hence ensures system reliability.

The extent to which DR enabled loads could displace traditional forms of ancillary services for both operation and regulation will depend on the extent to which the aggregated response can reliably provide stability and security to the power grid at times of sudden increase in demand or loss of generation, as well as during times of fluctuating and intermittent wind or solar power. While pilot projects have already demonstrated the potential of DR to realize many of the above-mentioned benefits, few customers have yet to adopt fully automated DR due to insufficient evidence showing methods that can be reliably and successfully deployed and hence lack of policy and market incentives to cover cost of implementation [PKMB+10, Com13a, Par12, EQ09]. Technologies that can improve distributions systems by making demand more responsive to real-time costs need to be incentivized [MIT11, Kin13, Fri13]. There

also needs to be a merging between DR and energy efficiency as the line between them begins to blur in a more dynamic system and shifting and reducing energy is carried out simultaneously in the most cost-effective allocation by the same control systems. Coordination is needed because energy efficiency and DR schedule the same energy resources and recruit the same customers [GRLS10]. Low system interoperability also means that systems cannot seamlessly integrate with neighboring players and more freely exchange information to assist in managing supply and demand. There remains a need to intelligently control and integrate DR technologies and on-site DG to allow customers to manage their energy resources reliably and flexibly in response to system conditions [PD11]. This should be done autonomously without requiring dispatch in order to ensure immediate response while maintaining resource autonomy. Control architectures that help to firm up renewable resources and can exploit the potential of DERs to provide grid support additionally require modernized system design as well as enabling hardware and software.

2.6 Microgrids

As the grid grows increasingly coupled at the interconnection level, centralized control becomes near impossible especially as system operators try to address the challenge of efficiently integrating renewable energy [MIT11]. Integration of and cooperation between DERs will be critical to creating a network that can handle both the variability and unpredictability of renewable energy sources as well as increasing system complexity. Particularly advantageous is the fact that many DERs do not require time critical power. While these DERs do add complexity, intelligent aggregation and control of their power schedules can potentially serve as a considerable system reliability and stability resource while simultaneously providing a means for greater power system flexibility [Kir07, KLI+08, KIHD08]. Autonomous control action with DERs, however, has typically meant some degree of uncertainty in the availability of response due to both lack of direct control and the diversity in consumer utility [CH11, Eto07]. One of the major challenges that needs to be addressed is whether load balancing and power management can operate automatically and be distributed across the system in a way that avoids exacerbating line congestion [TDL05]. The biggest concern is lack of control over power flow. While system operators cannot control where electricity goes, they can control how much comes from where and

load balancing has simply meant scheduling enough generation while respecting the constraints of the system by not overloading the transmission and distribution lines. This traditionally requires analyzing the historical usage of electricity on a minute-by-minute basis and projecting forward demand growth on the existing grid. With an influx of generators and loads connected at geographically dispersed locations to the power grid, however, the flow of electricity along the path of least resistance to the end user is no longer easily determined or controlled, especially with the ad-hoc nature of the U.S. grid [MIT11]. Although population diversity among the DERs over a large enough area will mean that their net response in general should be relatively smooth, this is inadequate to a system operator who needs to manage local power flows within constraints throughout the grid and maintain a high degree of power system reliability at all times. In order to merit support for DER response through demand-side policies, it is necessary to ensure that DERs can deliver balancing services on a cost-effective basis without resulting in the potential for any negative impacts that could completely negate or eclipse the benefits of integration. This includes preventing an aggregate of DERs from becoming synchronized after long or consecutive contingencies and causing an undesired recovery effect in which a rebound peak results from too many controllable loads attempt to come back online at once [Ham07, RLL$^+$10, LT12]. Not doing so would potentially delay recovery and put additional strain on an already taxed grid, causing huge overloads that can trip local breakers and possibly accelerating degradation of the lifetime and functionality of the DER. Such impacts must be accounted for and balanced against the potential positives of using DERs to improve grid operation.

To address these issues, many research efforts are now considering an infrastructure that is hierarchical in nature, with DERs aggregated and intelligently controlled through microgrids [ZD, DH05, KIHD08, Las11, LSK, LR, SFEB, Ili07]. A microgrid is a localized grouping of DERs including electricity generation resources, energy storage units, and various loads [HAIM07]. Generation and loads in the microgrid can be residential, community-owned, industrial and/or commercial and are normally interconnected at low voltage. The entire network is then connected to the main grid at the point of common coupling (PCC). Power transmission is typically in the range of 3kW to 10MW. The microgrid provides a natural way of managing increasing system complexity since aggregation of resources able to respond to changes on the grid will result in a more resilient system through consolidation into small balancing

areas able to operate autonomously [PL]. When connected to the grid, the microgrid masks the inherent complexity by behaving as if it were a single entity that can intelligently respond to incentives and constraints set by the system operator [HAIM07, Las11]. Using advanced algorithms to dynamically control an aggregation of DERs in a microgrid also means that resources can be pooled together. While some of the load in a microgrid will be inflexible high-priority load, the on-site demand coming from DR enabled loads capable of power scheduling alongside on-site generation and storage will give the microgrid the capability to shape the microgrid power profile and reduce or shift demand peaks. This allows for participation in both the ancillary and capacity markets. Control within the microgrid can be centralized, decentralized or a hybrid of both depending on the hierarchical nature of the system [TH, ZD, DH05, SFEB, CH11]. This new model of energy management and control allows the diversity of resources within a microgrid to provide highly reliable electric power, additionally avoiding technical problems that may arise from the increasing penetration and interconnection of DG and renewable energy. Microgrids also typically generate power locally from renewable resources and so in this case they would directly deal with intermittency issues. Dynamically controlling the local DERs and actively balancing demand and generation would effectively be used to help meet more stringent system requirements for interconnection resulting from the increased variability and unpredictability of diverse generation from renewable energy. In addition to using the DERs to help buffer local intermittent generation and allow for more renewable resources to be added to the power grid, microgrids directly taking advantage of local renewable generation and manage local resources more efficiently [Las07, PL, LP]. The overall effect is better maintenance of distribution level voltages, fewer T&D losses, and more effective utilization of generators. This will also allow for lower starting reserve requirements and enable deferment of construction for both transmission and generation, resulting in reduced operation costs [Boa12, MYK$^+$13].

The microgrid takes DG and DR one step further with its potential ability to disconnect or "island" from the main grid and function autonomously when necessary [MYK$^+$13, KIL05, HAIM07, Las11, SFEB, PLMM06]. Islanding is the scenario in which a microgrid disconnects itself when power from the grid is no longer available and continues to obtain power from on-site DG. This is particularly advantageous during power cuts since it enables the microgrid to still utilize DG in contingency situations (*e.g.*when a generator fails on the grid) where it can be most useful. There

is still concern, however, that reversed power flows could harm utility workers fixing damaged lines. The other equally important concern is that power output from the generators may begin to drift from tolerance levels if inverters do not have a grid signal to synchronize to. This may lead to power quality problems such as abnormal voltage and frequency excursions as well as out-of-phase reclosing and interference with the restoration of normal grid conditions [SFEB, PLMM06, YWM+05, SMH08]. Consequently, all DG is traditionally required to disconnect as soon as there is a problem on the grid and this significantly reduces DG's inherent ability to increase grid flexibility and resiliency. With microgrids, the ability to island and demonstrate dynamic flexibility and responsiveness to external conditions can avoid these issues without exacerbating contingencies and instead increase system resiliency.

In order for a microgrid to operate both as a grid-connected system and in an islanded state, some form of storage is required to either provide energy arbitrage or replace the stabilizing reserve offered by the power system [PLMM06, Mee13, oE13b]. The storage helps dynamically balance supply and demand, maintaining system voltage and ensuring power is delivered when and where it is needed. This becomes especially important when using local generation coming from renewable energy resources like PV and wind to meet demand from autonomous devices as well as vehicles. While the upfront capital investments required in a microgrid are often greater than traditional grid infrastructure costs due to storage and other operational infrastructure required to run the network, energy markets have come to recognize that microgrids with DG are able to address the problems associated with large scale centralized generation by offering increased flexibility, resiliency, and environmental benefit. The transfer of microgrid technologies to power grids in developing and transition economies will also be of interest since it will help reduce difficulties with supply-side investment and also help maintain an adequate supply-demand balance, which is a common issue for these economies. Development of the technology may therefore additionally generate an export market for technology and expertise on dynamic control methods, and help countries meet clean development objectives through construction of more efficient electric power systems [Com10, SS10]. Unfortunately, there is still no way to capture many of these advantages using traditional utility accounting methods and until such benefits can be quantified, the development and deployment of microgrids and locally controlled DERs will continue to be hampered. An incentive structure needs to be put in place to recognize that microgrids can help

to dynamically balance the supply and demand for electric power if integrated and controlled intelligently. This means that in addition to incorporating all externalities, prices should also be dynamic and represent real-time system states. Not only would this address some of the integration problems associated with distributed resources, it would also leverage and aggregate the efforts of the resources to help improve the efficiency, flexibility, reliability, economics, and sustainability of both the production and distribution of electricity.

2.6.1 Implementation

To implement and operate a microgrid, each DER must be capable of bidirectional communication. However, the bandwidth required has been shown to be relatively manageable [SFEB, BLR+10, PD11, CH11]. Systems with bidirectional communication technology and embedded processing have also become increasingly inexpensive and pervasive, enabling power systems that can run autonomously all the way from generation to distribution [SFEB]. Signals can be sent on separate lines or through the power lines themselves at network communication speeds that now make complex architectures and control systems with large numbers of DERs viable. These signals can be in the form of voltages, frequency and/or external signals from the system operator. Through these communication lines, advanced metering interfaces (AMI) enable real-time prices to be communicated to consumers and usage information to be transmitted back. This makes it possible to provide real-time incentives to DERs throughout the grid to reduce system peaks and conserve energy, resulting in a more efficient use of grid assets which in turn lowers energy rates. Real-time prices are in this way a form of incentive based control for influencing active power flow using bidirectional energy management interfaces. After the initial round of AMI deployments, utilities now have more experience with the technology and widespread deployment is only a matter of time as the technology matures [MIT11, Bor09]. It is estimated that there is currently already a 30% penetration of AMI meters in the U.S. and that more than half the households in the country will have an AMI meter in operation by mid-decade [fEE12, Com13b]. The compute power and CPU speed of embedded controllers in DERs has similarly been improving significantly, making it possible to deploy computationally intensive applications in real-time. Software built on top of the hardware would be able to integrate dynamic pricing and allow consumers to connect DERs with independent and distinct utility functions. Each

DER would automatically respond to the prices and maintain a more level load with a higher load factor while respecting the requirements of the consumer [U. 06, BJR02]. This distributed intelligence with advanced controls could provide a highly reliable aggregated response while allowing DERs to retain autonomy and privacy. In order to achieve this, however, hundreds of thousands of devices simultaneously online must be able to operate in real-time in a coordinated and cooperative manner. Optimization at the distribution level will have to go beyond addressing traditional problems like loss minimization and power quality and consider the foreseeable transition to a more dynamic power system.

In a typical microgrid, there will minimally be an electric load, local generation and some form of storage. The load will be some combination of a fixed load and a controllable load equipped for DSM. Details for some more common examples of these DERs is provided below.

Photovoltaics

The size of a PV system will depend on its intended purpose. When the system is grid-connected, its use can range from locally balancing load to generation through a larger power station, or some combination of the two. This means the size of the system can go from a few kWs to several hundred MWs. In off-grid scenarios, the total amount of generation must be at least equal to the total load. If solar is the only source of power generation in the system, some form of storage will be required to ensure load can be balanced at all times. The main advantages that PV systems have over both conventional and other renewable energy sources is that they are extremely low maintenance and can be readily integrated into building rooftops or land that has already been developed. In a grid-connected system, a synchronizing inverter converts the direct current (DC) output of the array into an alternating current (AC) output at 50 or 60 Hz depending on the grid frequency. It can also be used to provide power directly to a microgrid or a local off-grid system where energy can be shared by residential and/or commercial loads. The DC voltage of the inverter is typically kept as high as permitted in order to limit ohmic losses [Eng]. PV inverters also have additional functions such as maximum power point tracking and anti-islanding protection.

Each panel in a PV system has a conversion efficiency of around $10-20\%$ and will generate around $100-400$W. The panels are strung together to match the inverter

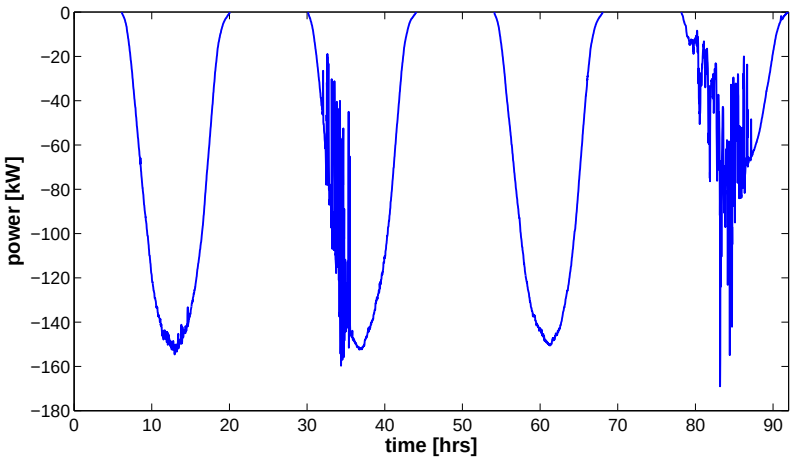

Figure 2.5: Power output of a PV array in Northern California in August. Capacity factor is lower and output becomes more volatile on overcast days.

voltage [BSN11]. In regions like the U.S. and Europe where the sun will typically shine for about 5 hours per day, a panel will produce around 1kWh/m^2 daily. Power production typically peaks at around local noon each day and an array will have a capacity factor in the $10-25\%$ range [Adm13a, AM11, Age10]. As shown in Fig. 2.5, this capacity factor will be lower on overcast days and power generation will become more variable and unpredictable.

Electric storage

While the cost of storage is still relatively high compared to other energy resources, innovation in storage technologies is rapidly driving down costs and making it a viable and practicable solution in many scenarios. This includes curtailing and shifting load peaks for a higher load factor, providing backup during voltage sags and power interruptions for improved power quality, buffering intermittent renewable energy sources, dynamically balancing multiple sources of generation and load, and running off-grid systems. With renewable energy in particular, storage turns the intermittent supply of energy into a dispatchable resource that can not only integrate more easily but can help reduce peak demand. This helps to lower the cost of operating and maintaining the power grid [Par13, HPM09, TOFD, MSC$^+$08]. Unanticipated contingencies can also be handled without relying on operating reserves that use fossil fuels since storage units can respond to dynamic energy prices and provide energy arbitrage.

In addition to holding power when it is available and discharging it when needed,

battery storage maintains unique advantages over other forms of storage. A typical battery is capable of rapid response and is also able to scale. This means that the characteristics of the battery system are independent of the total capacity that the individual units are aggregated into. As with other methods of storage, the cost of battery storage is still an obstacle. However, strategically placing just small amounts of storage can decouple the availability of generation resources from online loads to allow for more flexibility with DERs and would have significant impact on developing a more dynamic and efficient power system by freeing up congestion [DEKM10, BST+13, Mee13]. These benefits are felt at all levels from the generator to the system operator to the consumer.

Currently, there is significant interest in investing in more storage throughout the power system due to the influx of renewable energy [DEKM10, Boa12, THPB08, HPM07, BI04, LS05] but the development of the relevant technologies is still in its infancy stages. Where to install storage, how to size the storage units and how best to utilize its capacities are still all open research questions that need to be addressed [oE13b, MPKM+12, CDC+11, CGW12, OCB08, HD12, DJH+13]. A rule of thumb is to determine how much storage is needed for the purpose it was designed, whether this be smoothing, energy arbitrage or operating a system off-grid. System now online range from several hundred kilowatt-hours to several hundred megawatt-hours with the ability to charge and discharge rapidly at rates ranging from $1 - 100$ megawatts [CGNP13, oE13b].

Plug-in electric vehicles

About 94% of the U.S. transportation system still runs on petroleum, with more than 40% of this being imported from other countries [Adm11, Ner12]. The primary advantage of EVs is that they do not depend on gasoline for locomotion. This means that the vehicles provide a way to reduce fossil fuel dependence in the transportation sector while eliminating GHG emissions and pollutants when operating. For example, 40% of GHG emissions stem from the transportation sector in California and these emissions could potentially be reduced by 75% through vehicle electrification [Fri11]. The other advantage of EVs is their higher efficiencies. Conventional internal combustion engine (ICE) vehicles typically have efficiencies below 20% with most of the energy from gasoline being wasted as heat [Cen]. In contrast, EVs can have plug-to-wheel efficiencies around 60% [oE]. The EVs do not consume energy while at rest or

coasting and energy from braking can also be recaptured through regenerative braking. With fewer mechanical parts, EVs also have lower operating and maintenance costs and have been shown to match or exceed the life cycle miles of comparable ICE engine vehicles [Age13a].

Considering the environmental footprint of EVs from a well-to-wheels assessment, however, the net impact of the vehicles will depend heavily on the source of energy the batteries charge from since the pollution is essentially being shifted from the tailpipe to the location of the generation source. An EV charging from electricity generated from coal will have a very different footprint when compared to one that charges using electricity generated from renewable energy sources and this could potentially result in higher GHG emissions relative to a conventional vehicle [Ste12a, PDM07, Zeh13]. Nevertheless, the potential of EVs to move the transportation sector off fossil fuels and lead us towards a more sustainable and resilient energy infrastructure is significant [Age13a, Ste12b, LSA11]. Because electricity can be produced from renewable energy, EVs provides the option of charging with minimal pollution and emissions. This can be further reduced relative to a conventional ICE vehicle without changing the generation source simply by smart charging at off-peak times [FTCG13, vVBK$^+$, PDM07]. As both renewable energy and battery technology continues to advance at a rapid rate, the total environmental impact and total cost of operation (TCO) of the vehicle are anticipated to reduce to well below that of the conventional ICE vehicle [Ste12b, LSA11, vVBK$^+$]. Currently, the charging cost of an EV is around $3-4$ cents per mile assuming a national average rate of 11 cents per kWh of electricity and hence is already significantly less than the cost per mile of a conventional vehicle which can range from $8-20$ cents per kWh when gasoline is \$3.00 per gallon [U. b, U. a]. The finite supply of oil and the increase in global demand means that these gas prices can only be expected to trend upwards in the future and the relative cost of EVs will decline as a direct consequence [Age13a]. In terms of the vehicle cost, over the last 4 years the cost of the battery has come down by more than half and further cost reductions are expected through advances in battery technology and economies of scale [Age13a, KK13]. Some studies hence show that the TCO is already competitive with ICE vehicles [HB12, Ins13, CF12]. And while there is still debate over whether the amount saved from lower mileage costs is sufficient to offset the initial higher price tag of the vehicle, there is agreement that the price tag will continue to fall. Recognizing this potential, a 2013 DOE report laid out a road map with key objectives

that would make EVs competitive over a 10 year horizon [U. 13]. The issues addressed are primarily related to the battery. In addition to their cost, the limited capacity of the batteries continues to be a source of range anxiety among drivers. Battery capacity can range from $15 - 60$ kWh for EVs and hybrid EVs, and a vehicle can travel about 3 miles per kilowatt-hour [Gar, AD11, oE]. The critical goals would hence be to extend the range of the batteries to about 280 miles and reduce battery costs from their current price tag of U.S.$\$500/kWh$ to U.S.$\$125/kWh$ within a decade.

There are also many efforts to fund and develop a better public and private recharging infrastructure. While the home and workplace are expected to be the dominant locations for charging, more charge stations placed strategically in locations where there are no other charging alternatives or where high traffic is expected can go a long way in terms of helping to address range anxiety since most driving is local over distances less than 38 miles per day [SMN+09, MM09]. For home charging, a standard outlet can be used for Level 1 charging to charge an EV at $1-3$kW. Because such outlets are only equipped to handle peak loads for short durations, the charge rate is maintained below the capacity limit of the line in order to avoid overloading the circuit. An EV will typically requires $6-8$ hours to fully charge on Level 1. Level 2 charging using dedicated circuits allows vehicles to charge at around $7 - 24$kW in about $2-3$ hours. Charge stations capable of Level 2 charging have controls and protection in place that enable them to intelligently manage vehicle charging time along with the demands of other on-site loads to prevent tripping. These higher-voltage charging systems offer an additional flexibility advantage since they need less of the time a vehicle is typically plugged in for to fully recharge.

While transforming the transportation sector, the other area where EVs will simultaneously have a large impact is on the operation of the power system. EVs constitute a sizable load on the grid with the amount of energy required to charge the battery typically on the scale of what a home consumes daily [She14]. Increasing penetration of electric and hybrid electric vehicles especially in concentrated areas could potentially result in enormous demand peaks on the grid, overloading the distribution lines and leading to local brownouts or blackouts if the vehicles all charge simultaneously [TMA+09, SGKMF08, GKMD10, SPR09, PSJ+, RMM+09, HT, DSB10, PFRC+11, KCAY11, LSA11]. While this is obviously a cause for concern with utilities operating the power system, there is also general recognition that charging the vehicles intelligently could help the grid run more flexibly and efficiently [RFK12, GRSZ]. Although

storage can similarly improve efficiency and decrease energy losses on the grid, it is currently still a very costly solution. Using the energy storage inherent in EVs would address two problems with one solution by moving the vehicles off oil while simultaneously using their batteries for smart charging. EV chargers are able to exchange power almost instantaneously, making them much faster than calling up or even shutting down generation. A typical vehicle is also parked an average of $90 - 95\%$ of the time and needs only a few hours to fully charge [MKD, MBK+09, PKGE, Ste12b, KUH+08]. This flexible charging schedule along with rapid response capabilities and sizable capacity makes EVs well-positioned to take on the role of load balancing and shaping [BLR+10]. One of the prime examples of where EVs can already play a role is in load-leveling. Currently the power system is equipped to meet both base loads and peak loads and if there is a large difference between the two, the low load factor would mean a greater dependence on operating reserves that are continually over-generating during off-peak hours or only firing up inefficiently during peak hours. EVs could address this problem by simply shifting their charging times to absorb off-peak excess generation, thus leveling the load and increasing the load factor [SKMEW07, SV11, DS06, GRSZ]. Since the vehicles are already distributed, they could also be potentially managed so they would not produce the same congestion that dispatching additional generation might cause in a contingency. The EVs hence help create a system that can take advantage of rapid changes in the system while improving grid stability. They also enable the grid to better adapt as renewable energy becomes a larger part of the supply mix since their inherent storage capabilities natural compliments intermittent renewable energy generation by providing a high capacity energy cache that can stabilize the intermittency. Hence a symbiotic relationship between the vehicles and the renewable energy resources is possible [Ste12b, PLSAMDS, KUH+08, KT05b, TB12].

In addition to smart charging, recent work has looked to see if the rapid response of the batteries can be utilized to provide both capacity and energy-based ancillary services and give the vehicles added value [KT05a, DEKM10, oE13b, KUH+08, TK07, SES11, CTD+09]. Compared to the power system, EV fleets will have significantly higher power capacity with far less utilization since the vehicle needs stored energy at fairly predictable times and locations. In contrast, the system operator may need power almost immediately at unexpected times but does not care where that power comes from. While the EVs may not be able to provide kilowatt-hours

as inexpensively as generators on the power grid, their flexible schedules and underutilized capacity makes them prime candidates for providing operating reserves [KT05b]. This capability to provide ancillary services to the grid is generally referred to as Vehicle-to-Grid (V2G), In V2G, electricity is bidirectionally transferred between the EV and the grid to enable the vehicle battery to provide energy services when needed. Response to dynamic prices is one way the vehicles could be incentivized to interact with the grid through V2G. Vehicle owners would be able to set the parameters under which they are willing to sell energy, including their arrival and departure time as well as their desired state of charge for the battery. As a rapidly responding power source that is plugged in for the majority of the day and can be aggregated with other DERs, each EV could be offered some remuneration simply for being available without having to go through additional charge and discharge cycles. This could help the utilities to more efficiently manage demand peaks and operate the grid, reducing the need for more costly operating reserves like peak power plants and spinning reserves [WSE, DEKM10, KUH$^+$08, LLC13, CH10, TB12]. The aggregate of EVs essentially serves as a distributed battery storage system to buffer power during generation-load imbalances. In contingency situations, the EV batteries could also provide emergency backup power locally to prevent complete blackouts from occurring [Kro13a, KT05b, LK].

The potential value to the utilities of V2G has been estimated to be on the scale of several hundred to thousands of dollars per vehicle in the U.S. annually [TG, KT05a, DLRGNS]. However, the communications capability and the impact on battery lifetime still present sizable obstacles to bidirectional energy transfers from EVs [Ste12b, CNHD11, DEKM10, PWA10, TB12]. Lifetime is roughly a function of the energy throughput and given the high costs of the battery, any additional cycling is still currently too costly to warrant participation in V2G although some studies suggest that economic feasibility is possible [DEKM10, LPMC10, SES11, PAW, HH13, KT05a, KUH$^+$08]. Charging and discharging also currently has poor net efficiency since it requires converting the power continuously between DC and AC. These hurdles will likely come down as both battery and infrastructure technology progresses. For example, batteries with larger capacities in the future would greatly help to avoid the stresses associated with deep cycling which plays a critical role in battery lifetime degradation. Technology is also being developed that enable batteries to survive deeper cycles. In the meantime, the technology for smart charging is already available

now. Vehicles can absorb excess generation to provide partial smoothing without having to discharge and hence no additional impact on battery life. With both V2G and smart charging, implementing dynamic prices will be instrumental to incentivizing EV owners to recharging during off-peak times [The13, LLC13]. Presently, however, there is still no widely adopted incentive structure in place and the penetration of EVs is hence projected to be slow at the national level under current policies [Adm13b]. Developing the algorithms and platform for smart charging now and V2G in the future will be critical to both motivating the required infrastructure and enabling the full potential of EVs so that they can adapt quickly to dynamic grid conditions. This will ensure the vehicles can participate in a way that avoids adverse impacts and instead benefits the grid down the road.

2.7 Controls and optimization

Traditionally, control of DERs has been for the most part independent of and insulated from the optimization and control of power flows carried out by the system operator. With the exception of a few large loads signed up in DR programs, the system operator is limited to control of its own electrical equipment and the dispatch of generators through centralized control methods [COC13, WMSZT07]. However, with the rapid development of communication network infrastructure, bidirectional communication interfaces and embedded processing, DERs are now capable of being dynamically controlled either remotely or locally while accounting for system-wide costs and objectives [SFEB]. Embedded technologies also automate this process, enabling a far greater range of DERs both in terms of type and size that can be efficiently scheduled and controlled to benefit the consumer [IA09, Voj08]. This is at the same time of great benefit to the system operator since the optimization problem it needs to solve centrally in order to dispatch generation units and manage equipment continues to become increasingly complex and intractable with reliability and ability to scale being detrimentally affected [WMSZT07]. Distributing the energy management problem and operation of the grid significantly reduces the control effort of the system operator while allowing the grid to become more flexible and responsive.

The maturing of enabling technologies and the need to rethink grid operation has ignited considerable interest in solving the problems of energy management and power flow scheduling using localized, distributed and dynamic optimization methods

[KCLB14, KSK09, BKP11, IA09, Voj08]. Decreasing costs have made it much more cost effective to use optimization to solve these problems on embedded hardware. This is especially true with recent advances in convex optimization which allow problems to be solved in the millisecond and microsecond time scales ([WB10, MWB10]). Optimization algorithms can also be run offline on similar time scales prior to online implementation to reduce development time and ensure robust performance before deploying algorithms and policies online. This is critical with power systems since poor implementation can have significant negative consequences.

While there is an abundance of literature that has looked at the response of DERs at various locations on the grid, most work has focused on looking at how the DERs can be aggregated and centrally controlled in relatively predictable or unrealistically constrained environments [YT12, LCL11, TH, PSM10, CN09, CTD+09]. Some have looked specifically at integration of renewable energy and DG [HD12, AAAO, NAY11, OH11, TBBH10] while others have focused on demand side management [JJSP12, NP, OPJ+12, LCL11, ALP11, CCTH11, CKJCJHP12, DF12, GTL12, PSM10, SL11, SMQ, MQSX12]. There has also been much interest in control and optimization of microgrids [Las11, CN09, MSNAP, BCC11, CWS07, CDC+11, MPG+11, TH, YW11]. In all representations of the energy management and power scheduling problems, the most commonly adopted optimization approaches have ranged from variants of dynamic programming techniques to various artificial intelligence (AI) methods to model based predictive control algorithms.

With dynamic programming, an optimal solution can be found for the energy problem specified and stochastic methods can be used to make it more resilient to uncertainty. For relatively well-defined problems, this means that it can perform very well as an optimization tool [RBBP11, RI11, KP11, THAS13, HD12, ASDK14] However, unless approximations are made, the set of real world scenarios that it can actually model and solve is still fairly limited to well-ordered problems. With problems that have many states, the *curse of dimensionality* can lead to run times that grow exponentially [Bel03]. Assumptions also need to be made about the input data structures and probability distributions on unknown variables that may not accurately reflect real conditions at all times. These limitations diminish its usefulness in dynamically changing environments such as those found on the power grid.

AI is an umbrella term that encompasses methods such as neural networks, particle swarm optimization and machine learning. There is a great deal of interest in

using these methods in power systems due to their ability to handle and aggregate a large range of information for any number of scenarios [Kal07, PSM10, AAAO, MSNAP, BCC11, CWS07, CDC+11, MPG+11, YW11, SZPB12]. However, they can be challenging to deploy in dynamic systems since there is little flexibility even when adequately trained and assumptions about other agents in the system might need to be made. Such methods can fail to use common sense based on knowledge of a given system to help adapt to unexpected scenarios and changing environments. Tuning and automation is also difficult since you are effectively working with no transparency in a black box that will produce a solution regardless of whether it makes sense [Kes13, KJY+12].

Model based predictive control methods exploit analytical models of known system dynamics, objectives and constraints to help dynamically operate systems. The control algorithms have been used in a wide range of systems, including those in the power sector and energy management domain [AMT+12, HGL11, JQBS11, JJSP12, NHDSH, YKT+11, NP, OPJ+12, CK, HBP+12, SMQ, MQSX12, ATHS12, TBBH10]. The strengths of these methods when used in power systems include the ability to rapidly adapt and be flexible in dynamic environments, the ability to handle a wide range of inputs and constraints, and ease of implementation on-line through tuning of physically meaningful parameters. While predictive control does not guarantee optimality or robustness, it has a long track record of excellent performance in real-time systems [HW10, QB, MRRS00]. These techniques can run into issues, however, if not implemented correctly and this could result in poor results and even stability issues. Online optimization can also be computationally expensive for large and more complex systems.

Finally, there has been a recent ground swell of work looking at how distributed algorithms can be used to optimize the power grid and allow it to operate more efficiently [AC, GG, GG12, THAS13, BGSvdSJW13, YGG13, LCL11, ALP11, CCTH11, GTL11, GTL12, MCH13, CKJCJHP12, WCTT12, ATHS12]. The advantages of distributed and decentralized methods include scalability and ability to handle large complex problems where variables are coupled [WMB05]. This not only makes the problems more tractable, but allows them to be solved much more rapidly as the computational abilities of local resources and their ability to work in parallel are leveraged [BPC+11, PB13, PB14, KCLB14]. Another benefit in the case of power systems is that it allows resources to maintain their independence from the system

operator. These approaches, however, will not always converge as easily as centralized methods when the control points are uncoupled and the optimization problem is relatively simple. They must also be carefully tuned when deployed in any scenario to ensure that they do converge and perform well. Currently, research has mainly focused on showing how such methods could work at a highly theoretical level with simple examples and have not demonstrated that they would be able to work well in practical scenarios where the randomness and unpredictability of real data would need to be handled along with the uncertainties associated with a dynamic power grid. In the examples above, assumptions are often made about DER behavior that restrict their ability to include new objectives and constraints as well as adapt to uncertainty. Incomplete or imperfect input data is not uncommon in real systems, including the power grid, and this critically needs to be accounted for if such algorithms are to ever be deployed and implemented.

It should be noted that none of the methods mentioned are mutually exclusive. Hybrid approaches can be employed and some will be more appropriate than others depending on the scenario. Regardless, the advent of enabling technology has meant that these methods are now at the stage where they can help transition the power grid from traditional control methods to more dynamic, flexible and resilient control techniques.

Chapter 3

Method

In this chapter, we describe the algorithms we use to dynamically solve the power scheduling problem in order to control the DERs in our simulations. When considering real power flow in systems, the scheduling problem can be generally represented as [KCLB14],

$$\begin{array}{ll} \text{minimize} & f(p) = \sum_{i=1}^{N} f_i(p_i) \\ \text{subject to} & \sum_{i=1}^{N} p_i = 0 \end{array} \tag{3.1}$$

where the variable $p \in \mathbf{R}^{N \times T}$ denotes the power schedules for the N DERs over T time steps. f represents the total system cost and consists of the sum of all objectives in the system ranging from the system operator's goal of maintaining power quality and efficiency to each consumer's desire to minimize price while maintaining functionality of their loads. These individual objectives are represented by f_i. The constraint $\sum_{i=1}^{N} p_i = 0$ describes the need for power to balance and couples all the power schedules together in the system. Writing the problem in this way allows each DER to specify objectives and constraints independently, requiring only the physical constraint of power balance. This means that it can be extended to schedule a wide range of DERs regardless of whether they generate or consume power. Solving the scheduling problem will produce a fixed plan that minimizes the total cost over a desired time horizon for the given objectives and constraints. This plan helps allocate resources and balance power by providing power set points for each controllable DER at each time step. However, solving for these schedules assumes no uncertainty and also does not provide a control strategy. Additionally, specifying the problem in this way does not guarantee that the problem is tractable and hence that it can be solved

in real-time online.

In the following sections, we begin by describing how MPC can be used to solve the power scheduling problem and provide an adaptive control method that accounts for the uncertainty. We then describe how ADMM can be used to represent the power scheduling problem in a distributed form to distribute the computational effort when the problem becomes more complex and coupled. We also explain how the MPC framework can be used to envelop the ADMM algorithm and allow it to robustly operate and control the DERs in more complex and coupled systems. Finally, we describe the algorithms we use to make predictions of unknown variables that can then be passed on as inputs to the MPC formulation.

3.1 Model predictive control

MPC is a control policy that can be used to dynamically and autonomously control energy resources at all levels of the grid. Its ability to handle uncertainty in real-time allows the power scheduling problem to be solved robustly in a changing and unpredictable environment. Within the MPC algorithm, the optimization problem is solved dynamically at each time step to determine an action policy for each DER and enable scheduling of the controllable devices over a finite time horizon. In order to solve the optimization problem, predictions of unknown variables such as local load and generation are used. Although a formal statistical or stochastic model can be used to represent these unknown variables when making predictions in MPC, it is not needed for the policy to work and the controller can often perform very robustly even when predictions are poor [WB09]. Precise modeling of uncertainty in the system such as the dynamics of imbalances and disturbances is also not required since MPC is based on system models and does not place any restrictions on external inputs or dynamics [HGL11].

The versatility and flexibility of MPC when handling uncertainty and external inputs means that when compared to dynamic programming methods, it makes fewer assumptions and does not require probabilistic determination of all possible outcomes when scheduling resources. This makes it fast to implement and able to adjust to uncertainties. It can be implemented in a fully automated fashion and enables continuous and dynamic response to changing grid conditions. In contrast with non-model based AI techniques, MPC can also often perform better in situations where there

is some prior knowledge about the physical system since the model can incorporate this knowledge and provide an analytical solution to the problem. MPC directly integrates objectives, dynamics and constraints without having to learn and adjust controller parameters via a trial and error process, and this makes it particularly well suited for rapid online optimization of a locally controlled system which is dependent on and particular to a changing operating environment [Bem06]. It can be tuned easily through adjustment of physically meaningful parameters and this means that it can be used to reliably optimize real-time applications in an informed way [ABS07, AGANZ01]. Another advantage of the ability to directly integrate a wide range of inputs is that the system can operate closer to constraints and prevent variables from exceeding limits that could result in inefficient operation, damage to the physical system, and possible instability resulting from runaway feedback. MPC additionally accounts for nonlinearities due to DER saturation and rate limits, and can readily incorporate ramping and cycling constraints that will become increasingly prevalent in future grid scenarios. Costs in the optimization problem can include but are not limited to individual DER objectives, ramp rates and dynamic market prices based on system imbalances. When the objective and constraints in the system are convex, the power scheduling problem can be solved very efficiently and MPC often outperforms traditional control methods [MRRS00, MWB11]. While the open loop algorithm is a heuristic approach that does not guarantee optimality or stability, it has been used practically for decades in various industries for real-time controls [QB, MHL, GPM89]. In the majority of cases, systems are stable when the horizon is picked properly and the problem is tuned correctly.

To implement MPC, the power scheduling problem 3.1 is first rewritten as [MWB11],

$$\begin{array}{ll} \text{minimize} & \sum_{\tau=t}^{t+T_{\text{MPC}}} f(\hat{p}(\tau|t)) + \alpha_{\text{prev}}||\hat{p} - \hat{p}_{\text{prev}}||_2^2 \\ \text{subject to} & \sum_{i=1}^{N} \hat{p}_i(\tau|t) = 0 \qquad\qquad \tau = t, \ldots, t + T_{\text{MPC}} \end{array} \tag{3.2}$$

where the variables are again the DER power schedules $\hat{p} \in \mathbf{R}^{N \times T_{\text{MPC}}}$. The $\hat{}$ operator indicates an estimated state or quantity at time step τ and $\tau|t$ denotes an estimate of a quantity at time τ based on information available at the current time t. T_{MPC} is the MPC time horizon over which the DERs are scheduled. It gets around the fact that a simple closed-form expression generally does not exist for constrained optimization problems over an infinite horizon by approximating the specified performance

measures over the finite horizon [Lab10]. To implement MPC, we first solve the optimization problem at the current time step to determine conditional power schedules for each DER over the MPC time horizon. The controllers in the system use the information along with predictions of unknown quantities available to them locally at that time to jointly minimize their objective functions over the time horizon subject to individual constraints. $\hat{p}(t)^\star, \ldots, \hat{p}(t+T_{\mathrm{MPC}})^\star$ forms the solution to the power scheduling problem over the time horizon T_{MPC} and this updates the control policy using $p(t) = \hat{p}(t)^\star$. Each controller then executes this first step of the schedule and idles until the next time step at which point the entire optimization process is repeated in order to incorporate changes in operating environment, as well as new state measurements and external information that may have subsequently become available. Since convergence is on the order of milliseconds and microseconds [MWB11, MB12], actions and schedules are determined well before the next time step at which point the system responds to the updated state of the power grid (generally on the order of minutes). The MPC algorithm also takes advantage of the fact that consecutive control problems are often very similar to each other. Optimization is initialized with a 'warm start' using the previously computed solution, allowing the problem to be solved efficiently [MWB10]. The process of acquiring new information, making predictions and re-solving the problem is repeated at every time step thereafter. This iterative process using MPC provides an effective feedback mechanism that ensures the solution is robust to measurement errors, missing information and inaccurate forecasts, allowing the controllers to adjust their schedules in response to external disturbances that were unknown at the original time the schedules were computed. Compensating for prediction errors due to mismatch between the model and the actual system ensures that the control policy dynamically adjusts and is self-correcting as new information arrives and changes in the operating environment occur. MPC is thus well suited for use in dynamic operation of power systems, especially when there is uncertainty in the system [Mac02, Bem06, MWB11]. Computation time is also saved since it is often unnecessary to iterate any optimization problem to complete convergence with each MPC iteration constantly updating and self-correcting control actions based on new data and predictions [MWB10]. Another benefit is that everything can be implemented locally and autonomously since perfect information is not required to perform well.

In order to avoid oscillations between MPC iterations, we have also added a regularization term to the objective function in (3.2) which we specify as $\alpha_{\text{prev}}||\hat{p}-\hat{p}_{\text{prev}}||_2^2$, where $\hat{p}_{\text{prev}} \in \mathbf{R}^{N\times T_{\text{MPC}}}$ is the solution for the previous iteration and α^{prev} is the damping weight for oscillations between iterations. This prevents the solution at the current time step from diverging too far from the previous time step and ensures that smoothness across the MPC prediction horizon does not come at a cost to smoothness between the current time step and the previous time steps. In this way, the weighted regularization term accounts for stability issues. Individual objectives can also include terminal costs to represents potential longer term costs outside the finite time horizon. Algorithm 1 outlines the iterative process used by the MPC method.

Algorithm 1 Iterative optimization using MPC.

Initialize initial DER states $q_i(1) \quad \forall i = 1, \ldots, n$
for $\tau = 1 \rightarrow T$ **do**
  Predict unknown variables using updated historical power profiles available at time t;
  Solve optimization problem over MPC time horizon $[t, t + T_{\text{MPC}}]$;
  $q_i(t+1) \leftarrow \hat{q}_i(t+1)$;
end for

One of the disadvantages of MPC is that it requires online optimization and this can be problematic with more complex systems that require a large number of model coefficients to describe a response. The ability to perform online optimization computations quickly become significantly hampered. However, the flexibility of MPC can get around this by employing different optimization methods within the algorithm to deal with the more complex scenarios. This includes distributed methods like ADMM which break the problem up into several smaller problems that can be solved much more efficiently in parallel. For non-convex problems, heuristic methods such as sequential convex programming and convex relaxation can also be used to obtain good local solutions [Can04, KCLB14, GSC09, HPR02]. To speed up MPC further, some of the computational burden can often be removed from the online implementation. For example, parts of the control problem can be formulated as a multi-parametric program with explicit solutions for each feasible system state incorporated in a look-up table [Bem06].

3.2 Alternating direction method of multipliers

In the context of power systems, ADMM solves the power scheduling problem in (3.1) by distributing the optimization problem and hence the computational requirements across the multiple DERs. This casts the optimization problem as a completely decentralized problem whereby each controller computes and exchanges simple messages with only its neighbors in the system in order to reach consensus. The controllers send small quantities of numeric data to neighbors in order to coordinate at each iteration while storing small amounts of local state information and efficiently computing solutions for its own optimization. This method is iterative, passing messages at every iteration before each DER minimizes its own objective function along with a regularization term that depends only on the messages it received in the previous iteration from its neighbors. The only coordination that is required between DERs is synchronizing iterations and the DERs are allowed to model their systems independently and autonomously solve their own local optimization problems with great flexibility. In this way, the optimization problem is solved locally in a parallel fashion and the computational requirements to solve the problem are significantly reduced. Because everything is solved independently, optimization with ADMM is also scalable and can readily incorporate more DERs as they come online. When a change occurs in one part of the system, this is simultaneously felt by DERs throughout the system as messages are passed between them and they respond instantly without the need for a central control system to detect the change. This addresses the problem that as systems become larger and more complex, the behavior of DERs can begin to couple and directly impact one another, making the optimization problem intractable. While we do not explore the rates of convergence and scalability using ADMM in this dissertation, we direct the reader to [BPC+11, PB13, PB14, KCLB14] for an extensive look at the algorithm's ability to operate in real-time on large scale systems with minimal computational requirements.

The exchange of messages can be thought of as an *internal* price signal that all DERs agree on in order to align locally optimized operating policies with common goals that benefit the entire system. At convergence, this optimal *internal* price between participating DERs in the system naturally results from running ADMM and represents the equilibrium price that occurs when the objectives are mutually optimized. Since optimization is independent and allows for autonomous operation with minimal coordination, this bottom-up control approach not only reduces the

communication requirements but also makes it feasible to connect large numbers of these distributed systems to the grid without requiring the implementation of complex top-down control systems that require extensive knowledge of each DER. This paradigm shift allows for a new way to think of operating the grid since ADMM can allow for both efficient energy trade and active flexible control of power flows at the controller level. The cooperative approach between all the DERs searches for an outcome that each finds at least acceptable and for which the total objective function representing the social benefit to all participants is minimized.

The ADMM algorithms we use to solve the optimization problem specified in (3.1) use the methods developed in [BPC+11, KCLB14]. The problem is first rewritten in ADMM form by making copies of the variables:

$$\begin{array}{ll} \text{minimize} & \sum_{i=1}^{N} f_i(p_i) + g(\sum_{i=1}^{N} z_i) \\ \text{subject to} & p_i - z_i = 0 \qquad\qquad i = 1, \ldots, N, \end{array} \tag{3.3}$$

with variables $p_i,\ z_i \in \mathbf{R}^T$. f_i forms the local cost function for controller i, and the shared objective g is a function of the sum of the power variables. By introducing z_i 's into the problem as copies of the power variables, the shared constraint represented by the power flow balance can be moved into g which simply becomes the indicator function of the empty set $\emptyset$, *i.e.*is represented by the sum of the variables. This definition of g leads to a special case of the sharing problem referred to as the optimal exchange problem. In exchange ADMM, we split the additive term into separable objectives which can then be updated separately to drive the variables to consensus. Each controller effectively participates in an internal market with a price adjustment process that is used to attain general market equilibrium. The internal price of power is increased or decreased depending on whether there is an excess demand or excess supply respectively. This internal price will naturally reflect the external price at the PCC but is not necessarily equivalent since it also reflects the individual interests of the independent DER controllers within a system. External price signals can still be used as an input to help the power grid achieve system-wide objectives but this ability to use an internal price to aggregate resources at various levels of the power grid makes ADMM very useful for rolling out a more meshed and interconnected grid. The complete derivation of ADMM for the exchange problem can be found in [KCLB14, BPC+11]. We highlight the main equations from the derivation below and ask the reader to refer to [KCLB14, BPC+11] for more details.

ADMM can be derived directly from the augmented Lagrangian

$$L_\rho(p, z, y) = \sum_{i=1}^{N} \left(f_i(p_i) + g\left(\sum_{i=1}^{N} z_i\right) + y_i^T(p_i - z_i) + (\rho/2)\|p_i - z_i\|_2^2 \right).$$

where $\rho > 0$ is the *penalty parameter*. Minimizing the variables and updating the dual variable independently and iteratively gives the ADMM algorithm

$$p_i^{k+1} := \underset{p_i}{\operatorname{argmin}} \left(f_i(p_i) + (\rho/2)\|p_i - z_i^k + u_i^k\|_2^2 \right) \tag{3.4}$$

$$z^{k+1} := \underset{z}{\operatorname{argmin}} \left(g(\sum_{i=1}^{N} z_i) + (\rho/2) \sum_{i=1}^{N} \|z_i - p_i^{k+1} - u_i^k\|_2^2 \right) \tag{3.5}$$

$$u_i^{k+1} := u_i^k + p_i^{k+1} - z_i^{k+1}. \tag{3.6}$$

with the penalty parameter ρ in the Lagrangian used as the step size in the dual variable update in order to ensure the iterate (p_i^{k+1}, y_i^{k+1}) is dual feasible [BPC+11]. We have combined the linear and quadratic terms in the augmented Lagrangian in order to present ADMM in a more concise and convenient form using a scaled form of the dual variable. The variables p and z are minimized in step (3.4) and step (3.5) respectively, followed by a dual variable update in (3.6). Written in this form, we can also clearly interpret the scaled dual variable as being the running sum of the residuals, We can further simplify the z-update step by taking the Lagrangian and solving the dual problem to determine z analytically [BPC+11]. Substituting the solution in the u-update shows that the dual variables u_i^k are all equal and can be replaced with a single dual variable u representing the scaled price. The ADMM algorithm finally becomes

$$\begin{aligned} p_i^{k+1} &:= \underset{p_i}{\operatorname{argmin}} \left(f_i(p_i) + (\rho/2)\|p_i - p_i^k + \overline{p}^k + u^k\|_2^2 \right) \\ u^{k+1} &:= u^k + \overline{p}^{k+1}. \end{aligned}$$

where $\overline{p} = (1/N)\sum_{i=1}^{N} p_i$ represents the mean of the p_i variables.

In each iteration of the p-update in step (3.7), ADMM augments its own local objective function f_i with a simple quadratic regularization term. The linear parts of the quadratic terms containing the iterative target value are then updated, pulling the variables towards an optimal value and allowing them to converge. This proximal operator contains the scaled price u^k and can be interpreted as a penalty for

p_i^{k+1} deviating from p_i^k projected onto the feasible set, helping pull the variables p_i toward schedules that enable power balancing while still minimizing each local objective subject to system constraints. In other words, it represents each controller's commitment to help reach market equilibrium so that as the power profiles are adjusted and the system converges, the effect of the regularization term vanishes. Since each controller only handles its own objectives and constraints, the p-update can be carried out independently in parallel by all the controllers in a system. ADMM enables decentralization by allowing the regularization term in the augmented Lagrangian to separate so that it can be minimized locally while still maintaining the convergence properties and robustness of the method of multipliers [BPC+11]. This means that the optimization problem with at least as many variables as the number of DERs and loads multiplied by the length of the time horizon is reduced to small local optimization problems with only a few variables for each controller to manage, including local power flows and internal states. The controllers pass their updated power schedules, p_i^{k+1}, to a collector (possibly co-located at the PCC) in the u-update (3.7) which in turn simply gathers the variables and computes the new average power imbalance, $\overline{p}^{k+1}$, in order to update the scaled price u^{k+1}. The computed values are then broadcasted back to the controllers to readjust the proximal operator. In this way, the collection stage projects the power schedules back to feasibility and helps push the system towards equilibrium by adjusting the price up or down depending on whether there is net power demand or generation in the system. The iterative ADMM algorithm converges to optimal price and power schedules by alternating between the controllers and the collector with synchronization (necessary for real-time pricing in any event) being the only coordination that is required between the controllers.

When implementing ADMM, selecting the correct penalty parameter ρ is critically important for convergence rates. The optimal value of ρ will greatly depend on the scheduling problem. While there are heuristic methods to help determine the value of ρ, in many cases it will perform just as well with a fixed value found using a binary search method. Since we are using a scaled form of ADMM, the scaled dual variable $u^k = (1/\rho)y^k$ must also be rescaled after updating ρ. This means that if ρ is halved, u^k should be doubled before computing the ADMM updates.

It is instructive to consider the primal and dual residuals in the ADMM exchange problem since both have useful interpretations. Following from [BPC+11, KCLB14],

the primal and dual residuals for the exchange problem can simply be written as

$$r^k = \overline{p}^{k+1}, \qquad s^k = \rho((p^k - \overline{p}^k) - (p^{k-1} - \overline{p}^{k-1})).$$

The primal residual simply represents the net power imbalance across all controllers and is hence a measure of physical feasibility. The dual residual is equal to the difference between the current and previous iterations of the deviation in power schedules. As the residuals approach 0, the internal price represented by the dual variable y converges to an optimal value. While there can be more than one optimal point, ADMM is guaranteed to converge to an optimal point when the problem is closed, convex and proper [BPC+11]. The residuals also provide a simple criterion for terminating the ADMM algorithm,

$$\|r^k\|_2 = \|\overline{p}^k\|_2 \leq \epsilon^{\text{pri}},$$
$$\|s^k\|_2 = \|\rho((p^k - \overline{p}^k) - (p^{k-1} - \overline{p}^{k-1}))\|_2 \leq \epsilon^{\text{dual}},$$

where ϵ^{pri} and ϵ^{dual} are the primal and dual tolerances respectively.

Discussion. Since ADMM is solving the exact same problem as that specified in (3.1), it is guaranteed to provide an optimal solution if the original problem is feasible. However, as with all optimization methods, it is only as good as its input. Combining ADMM with MPC hence ensures that the control policy is adaptive and robust, and also enables us to take advantage of a symbiotic relationship that enhances the advantages of both. One such mutual benefit is the fact that the computation time per iteration, as well as the data storage and communication requirements, is small and independent of the size of the network when ADMM is solved in a parallel manner. This enables MPC to be used in areas where performance was previously limited due to the size and complexity of the optimization problem and the time required at each computational time step. The computational requirements are effectively reduced by leveraging the local control efforts of each individual controller while respecting the constraints of each DER. And while the complexity of each controller's control effort can remain low, the aggregate intelligence of the coordinated effort is high. The decentralized method of optimization hence addresses the complexity and scale of actively managing a dynamic and rapidly changing power system. But MPC is also a natural extension of ADMM since after one time step is executed, a warm start

can be used to rerun ADMM in the subsequent time steps by taking advantage of the power schedules and price variables computed by each controller in the previous iteration of MPC. The warm start can dramatically speed up computation time to fraction of a second rates [KCLB14]. This is especially true with recent advances in convex optimization which allow relatively inexpensive embedded processors in power conversion devices to efficiently execute ADMM iterations in tens to hundreds of microsecond time scales [MWB10, MB12, MWB11]. It also ensures that ADMM can work with imperfect input and still perform well by adapting to changing system conditions. The combination of a rapid open source convex solver that handles objectives and constraints directly [MB12, MB10] and increasing CPU capabilities enables power schedules to be dynamically computed. Additionally, offline simulations using different control policies can be benchmarked on similar time scales prior to online implementation to enable design and development of grid systems.

Another important characteristic that is shared between ADMM and MPC that enables them to work synergistically is that both algorithms only require each controller to have access to *local* variables and not those concerning other controllers in the system. This allows controllers to operate with their own estimation methods and determine their own control actions without worrying about what other controllers are using since local anomalies are automatically accounted for in the power schedules and messages passed between DERs in ADMM and errors in predictions that may occur at an instance of time are addressed through MPC's iterative process. This resiliency to single point vulnerabilities and unexpected topology changes makes ADMM combined with MPC ideal for rapidly changing and expanding power networks such as microgrids or commercial sites with energy innovations. Because control is distributed and automated across all DERs, failure of any node or line automatically causes them to adapt and reconfigure their power flows. This ability to self-heal means that systems can handle the increase of DERs in the medium and low voltage distribution networks where the proliferation of high numbers of relatively small individual capacities suggests a new way to think of operating the grid through bottom-up control.

3.3 Predictions

To implement MPC, estimates of input variables are required at each time step in order to solve the optimization problem and determine a control policy over the finite time horizon. These estimates can be based on historical data, stochastic models, forecasts, and pricing information. The flexibility of the MPC framework means that it is not tied to any specific forecasting method and can incorporate a range of techniques depending on what information can be accessed. Perfect predictions or a formal statistical model to represent uncertainty are also not required for the method to perform robustly and predictions that capture general trends are sufficient since MPC recalculates power schedules at each time step after executing the first step of the schedule determined by solving the power scheduling problem, dynamically adjusting and self-correcting for any past errors or missing information [WB09].

3.3.1 Power profiles

Within any energy system, predictions are required for unknown independent variables. This includes generation and load power profiles that follow predictable trends. While many methods have been used to predict energy profiles on the power grid, ranging from multiple regression to expert systems [AN02, DKO12, BK01] and total load over a large region can be predicted up to 1% accuracy, predicting generation locally from intermittent renewable resources poses a greater challenge due to nuances that require detailed knowledge of the system environment and geographic diversification cannot smooth out [Ove12]. Since our objective is to only capture general trends and the DERs just needs to forecast their own future power profiles in each time period, we require only local measurements of historical output to get adequate predictions. We illustrate an example of this below using PV data taken from a rooftop array in Northern California to anticipate future generation output.

To predict PV output, we first assume that the historical PV output data $p_{\text{PV}}^{\text{hist}} \in \mathbf{R}_{\text{hist}}^{T}$ has general periodicity over a 24hour period, where T_{hist} is the historical time horizon. This is a good assumption over a period of a few days where the seasonal variation is insignificant since the solar insolation at any geographical location and given time is well determined. Deviations from the expected PV array output are due to weather conditions like cloud cover that result mostly in drops in output with occasional over-irradiance due to cloud enhancement effects. This leads up to propose

an asymmetric least squares fit of the available historical PV output data to provide a periodic baseline that is weighted towards the outer envelope of the observed data. To generate the PV output prediction, we first determine the historical baseline $\hat{p}_{\text{PV}}^{\text{hist}}$ by solving an approximation problem with a smoothing regularization term and a periodicity constraint,

$$\begin{aligned} \underset{\hat{p}_{\text{PV}}^{\text{hist}}}{\text{minimize}} \ \frac{1}{T} \sum_{\tau=t-T_{\text{hist}}+1}^{t-1} & \big((\hat{p}_{\text{PV}}^{\text{hist}}(\tau) - p_{\text{PV}}^{\text{hist}}(\tau))_+^2 + \gamma_{\text{asym}}(\hat{p}_{\text{PV}}^{\text{hist}}(\tau) - p_{\text{PV}}^{\text{hist}}(\tau))_-^2 \\ & + \gamma_{\text{curv}}(\hat{p}_{\text{PV}}^{\text{hist}}(\tau-1) - 2\hat{p}_{\text{PV}}^{\text{hist}}(\tau) + \hat{p}_{\text{PV}}^{\text{hist}}(\tau+1))^2\big) \\ \text{subject to } \hat{p}_{\text{PV}}^{\text{hist}}(\tau) \ &= \ \hat{p}_{\text{PV}}^{\text{hist}}(\tau + T_{\text{period}}) \qquad \tau = t - T_{\text{hist}} + 1, \ldots, t-1, \end{aligned} \tag{3.7}$$

where $(z)_+ = \max(0, z)$ and $(z)_- = \min(0, z)$. γ_{asym} and γ_{curv} represent the weights for the asymmetric and curvature terms respectively and T_{hist} is the amount of time over which $p_{\text{PV}}^{\text{hist}}$ occurs. The first and second term of the objective function represent the positive and negative deviation of the predicted curve from the actual data and the third term smooths the curvature in the predicted curve. The first term is weighted more heavily to push the predicted curve towards the outer envelope of the fluctuating data. The constraint ensures periodicity across $T_{\text{period}} = 24$hrs. Solving this problem de-noises the data to reconstruct a smooth baseline profile [BV09]. The objective function is a weighted sum of squared convex terms and forms a regularized convex problem which trades off an asymmetric least-squares fit against the mean-square curvature of the data. The baseline prediction for the MPC horizon is then defined as $\hat{p}_{\text{PV}} = \hat{p}_{\text{PV}}^{\text{hist}}(t - T_{\text{period}}, \ldots, t + T - T_{\text{period}})$.

Once we determine this baseline prediction, we correct for transient weather phenomena by adjusting the baseline using an error fit with a linear model applied to the residual $r = \hat{p}_{\text{PV}}^{\text{hist}} - p_{\text{PV}}^{\text{hist}}$. The correction is calculated by writing the predicted residual at the time step τ as a weighted sum of the previous residuals, $\hat{r}(\tau) = a_1 r_{\tau-1} + a_2 r_{\tau-2} + \cdots + a_n r_{\tau-n}$ where a is an n-element vector determining what weights to give the n previous residuals. The associated residual prediction error is defined as $e(\tau) = \hat{r}(\tau) - r(\tau)$. To determine a, we minimize the sum of the squared l_2 norms of the prediction error over the entire historical data time horizon $\tau = t - T_{\text{hist}} + 1, \ldots, t-1$. We can rewrite this concisely in matrix form,

$$||e||_2^2 = ||Ma - b||_2^2 \tag{3.8}$$

$$\text{where} \quad b = \begin{bmatrix} r(n+1) \\ r(n+2) \\ \vdots \\ r(T_{\text{hist}}) \end{bmatrix}, \quad M = \begin{bmatrix} r(n) & r(n-1) & \vdots & r(1) \\ r(n+1) & r(n) & \vdots & r(2) \\ \vdots & \vdots & \vdots & \vdots \\ r(T_{\text{hist}}-1) & r(T_{\text{hist}}-2) & \vdots & r(T_{\text{hist}}-n) \end{bmatrix}.$$

This is a least squares problem and has the analytical solution $a = M^{\dagger}b$ where the † symbol denotes the Moore-Penrose pseudoinverse. We assume M is full rank since the residuals are due to random weather patterns and can effectively be treated as independent and identically distributed. The predicted residual corrections across the MPC horizon are then decreased by a factor λ at each future time step, where $0 < \lambda < 1$. This reduces the magnitude of the correction over the MPC horizon moving forward in time so that the prediction reverts back to the baseline. The procedure used to make predictions at each MPC time step is summarized in Algorithm 2.

Algorithm 2 Computing PV predictions from historical data.

1: Compute baseline profile $\hat{p}_{\text{PV}}^{\text{hist}}$ over MPC horizon using $p_{\text{PV}}^{\text{hist}}$ to solve (3.7).
2: Compute residual $r = \hat{p}_{\text{PV}}^{\text{hist}} - p_{\text{PV}}^{\text{hist}}$.
3: Determine the residual weights a by 3.8.
4: Compute predicted residuals over MPC horizon as $\hat{r}(\tau) = \lambda^{\tau-t}a^T[r(\tau-1)\cdots r(\tau-n)]$ for $\tau = t, \ldots, t+T$.
5: Compute prediction as $\hat{p}_{\text{PV}} + \hat{r}$.

We can employ a similar approach for electric load using the historical load profile $p_{\text{load}}^{\text{hist}} \in \mathbf{R}_{\text{hist}}^{T}$. The only difference between the PV and load profiles is that with the load, we weight the first and second term of the objective function equally since we expect the positive and negative deviation of the predicted curve from the actual data to be similar. The adaptability of MPC means that the least squares fit of the load output and the adjusted least squares fit of the PV output data is able to provide sufficiently accurate forecasts to enable dynamic power scheduling. This prediction method is simple to implement using available historical data and requires very little computational effort during real-time implementation.

Examples of PV and load profile predictions over the MPC time horizon at one instance of time are presented in Fig. 3.1. In each plot, the prediction is compared to the actual output. For both PV and load, the predictions are good at capturing the diurnal trends and general shape of the power profiles. Sudden changes in power profiles that occur near the time of measurement where the actual output is known and rapid deviations can be anticipated are also captured by the error correction,

pulling the baseline towards the actual output and ensuring immediate response to sudden changes. Additional measurements and information to supplement the historical power profiles used to make the predictions could help to capture the irregular and intermittent dips and spikes that occur throughout the prediction horizon. This additional information might include cloud cover predictions for PV or anticipating occupancy levels in the case of load. While the flexibility of our prediction method means that it is trivial to incorporate this added data by including an additional weighted term to the asymmetric least squares objective, MPC does not require more than what is shown in Fig. 3.1 in order to perform robustly.

3.3.2 Stochastic parameters

In order to predict variables that are stochastic and discrete in nature, we use maximum likelihood estimates from probability distributions. While we could construct conditional probability distributions that are time dependent using Markov chains that adjust the stationary prior probability distribution, we found through simulations that this method added insignificant or no benefit since MPC already corrects for prediction error and hence already performs near optimal with rough predictions. The fact that the probability distributions are fairly time independent after enough data has been acquired means that this prediction method performs satisfactorily and is preferred for its minimal computational requirements.

The types of variables that would use this approach include the unknown parameters for each EV. We define four stochastic variables in a vector,

$$\theta_{\mathrm{EV},i} = [T_{\mathrm{arr}}, T_{\mathrm{dep}}, q_{\mathrm{init}}, q_{\mathrm{des}}]_{\mathrm{EV},i},$$

for each ith vehicle. These represent the arrival time, departure time, initial charge state and desired charge state respectively. To obtain predictions for $\theta_{\mathrm{EV},i}$, we first consider whether or not a given EV has arrived. In the case where an EV has arrived, we assume $\theta_{\mathrm{EV},i}$ is known. If a vehicle has not arrived, we need to predict $\theta_{\mathrm{EV},i}$ using available data. Ideally, there would be enough data to form approximately stationary prior probability distributions for each of the variables in $\theta_{\mathrm{EV},i}$. In practice, there will initially be insufficient data to make accurate predictions for each vehicle. As time progresses, however, accumulation of data will lead to more stationary distributions and predictions will become increasingly accurate. In fact, for this reason it is common

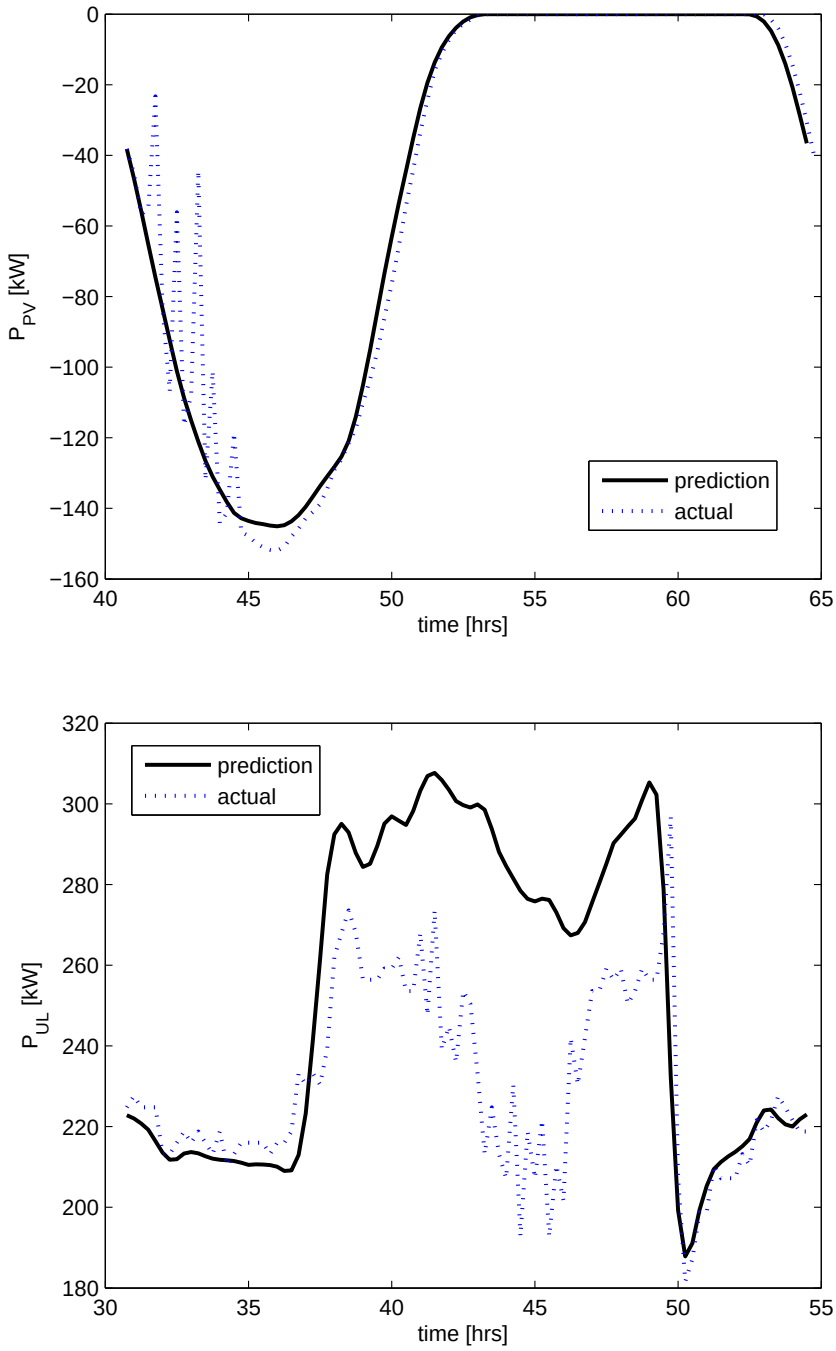

Figure 3.1: Predictions for the PV output (*top*) and electric load profile (*bottom*) over the MPC time horizon at one instance of time.

practice to ignore some number of samples at the beginning before the stationary distribution to be reached. However, for MPC even a general forecast can produce good results as our results will later show.

To predict T_{arr} and T_{dep}, we simply choose maximum likelihood values from their respective distributions. Since there is no reason to assume the state of charge variables are strongly and independently correlated with the time variables, we similarly use stationary distributions to determine arrival and desired departure charge states. For q_{init}, this means computing the maximum likelihood once daily for the vehicles that have not yet arrived. To determine the desired charge state q_{des}, we employ a more conservative approach to determine how much charge the vehicle batteries have to at least have at departure time. Instead of a maximum likelihood, we consider the highest charge state in the distribution since it is assumed that the driver will want to ensure there is enough charge to face the majority of contingency situations. Although this is overly conservative in the case of a plug-in hybrid EV which has the option to use gasoline as backup, gasoline prices will inevitably continue to trend upward and running solely on the electric battery will become increasingly desirable so that the assumptions made serve as a good first approximation for setting a desired charge state. From a carbon standpoint, ensuring the maximum number of electric miles will also have the best possible societal benefit. As more data for each vehicle is accumulated, the model can easily be adapted to the needs of individual vehicle owners in real-time. After a vehicle arrives, the charge states along with the times of arrival and departure are added to the data and the predictions are updated the next day.

Chapter 4

Grid-tied PV storage system

PV systems directly coupled with ES from batteries are uniquely positioned to address the environmental, technological and economic challenges the current power grid faces. When intelligently controlled, such systems can integrate solar resources onto the grid while maintaining and even increasing system reliability, thereby providing means for greater power system flexibility and stability. For example, a grid-connected PV storage system can be leveraged to address supply-demand imbalances by using the battery for continuous bidirectional load balancing while also storing solar power for use when the sun is not shining. While the daily solar cycle and load follow the same general trend, the peak of the latter lags behind the former by several hours. A storage system directly coupled to PV is able to absorb the excess output from local generation and redeliver the stored energy during shortfalls, making curtailment unnecessary and effectively firming up power from the PV array. By firming dispatch, reducing demand peaks and improving grid efficiency, ES displaces the amount of backup generation needed and offsets the need for fossil based spinning reserves and peaking resources in the dispatch stack. The dynamic response of ES located close to the solar resource can act as a buffer to match the availability of generation to the draw from online loads, making solar a much more deployable resource. PV storage systems can exploit the rapid response of ES to handle the intermittency that results from cloud cover. When a cloud casts a shadow on the solar array, the battery units automatically smooth the output of the solar panels by instantaneously dispatching energy to fill in the drop created by the cloud. This increases the reliability of power from solar.

To demonstrate how a grid-connected PV storage system could potentially operate, we model the PNM Prosperity Energy Storage Project and simulate the intelligence and control logic to enable optimization and scheduling of the system. Located near Albuquerque, the PNM system provides 500kW of power from solar panels directly coupled to 1MWh of rated battery capacity and is the first fully grid integrated solar storage facility in the U.S. With one of the largest combinations of PV and battery storage in the nation, the aim of the project is to develop and demonstrate battery-storage technology that will create firm and dispatchable energy derived from a renewable energy source. The battery storage is separated into a high power 500kW battery unit combined with capacitors for rapid short term smoothing and a larger capacity 250kW advanced lead acid battery unit to store and dispatch energy when needed. Currently, the system has demonstrated the ability to shift energy to peak periods and provide energy in rectangular blocks according to predetermined demand schedules. It is simultaneously able to smooth PV output in reaction to recently experienced cloud cover.

The framework we propose for this system optimizes both the smoothing and shifting processes using MPC by framing the problem as a special case of (3.1). This allows the system to operate autonomously and dynamically in response to external price and forecasted power output. The model of the PNM system is capable of integrating various objectives, constraints and forecast methods into a policy that can be configured to the current operating environment. Because control is entirely local in MPC, there is no need for centralized grid management. MPC schedules the PV-storage system given available input data from the PNM system. The input data includes price schedules as well as the operating conditions and constraints defined by the system operator, current line conditions, the storage unit and PV array. Power profiles based on historical information are also considered for the PV array. At each time step, the system is modeled as a convex problem with a convex objective function and linear constraints. This problem is solved dynamically in real-time to enable scheduling of the ES units over the MPC time horizon.

Note that with the PNM system, we employ MPC to enable dynamic control but do not use ADMM for distributed control. As mentioned in Chapter 3, ADMM solves the exact same power scheduling problem as that which is solved centrally. However, it only makes sense to use this when there is coupling between DERs in the system. In the case of the PNM system, the constraints and objectives are not binding and the

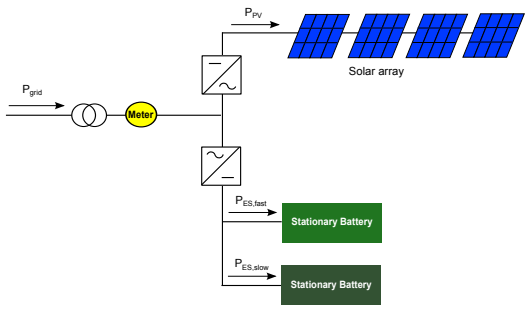

Figure 4.1: Schematic representing grid tied PV storage system.

number of resources is relatively small so it does not warrant the use of a distributed method to implement control.

To demonstrate the benefits of scheduling and dispatching the ES units in the PNM system using MPC, we use our model to determine the ability of the ES units to address intermittent local generation using ramping constraints while simultaneously shaping the power output profile of the system using price schedules. Showing this quantitatively will verifies the system's ability to provide increased system flexibility and responsiveness to both local and system-wide intermittency and unpredictability. This will in turn show the potential of PV-storage to not only help in grid operating efficiency but also provides a significant amount of stability and resiliency to the power grid at times of sudden increase in demand or loss of generation and during times of fluctuating and intermittent solar power. The possibility for undesirable effects on the lifetime and functionality of the ES units is also considered. This is examined by introducing an additional objective that penalizes excessive charging and discharging of the storage units. Finally, we compare these results to the prescient case to demonstrate MPC's ability to operate robustly and perform near optimal.

4.1 Model

4.1.1 System dynamics and constraints

The PV storage system we model and simulate is shown schematically in Fig. 5.1. Within the system, the storage units and PV array each have an associated power schedule and negative power is always defined as power being generated and/or flowing out from the energy resource.

Photovoltaic array

The PV array has a power schedule that is always negative since it only generates and does not consume power. Its output p_{PV} will fluctuate with changing climate and weather patterns and cannot be controlled. In our model, p_{PV} contributes to the system power balance and is predicted using historical profiles to enable scheduling of the storage units.

Battery electric storage

Storage in our system is divided into two separate battery electric storage (BES) units, a large slower moving unit for energy shifting and arbitrage and a small rapid charging unit for smoothing. The larger BES unit holds the generated power from the PV array and uses it to hedge against high prices as well as act as a safeguard against unforeseen events on the power grid occur by dispatching energy when requested. The smaller BES unit acts more like a short-term buffer, rapidly shuttling energy in and out for short term smoothing. Since each unit can charge and discharge, their power schedules can be both positive and negative. The rates of charge and discharge are constrained by capacity limits. This is represented as $-D_{\mathrm{BES}}^{\max} \leq \hat{p}_{\mathrm{BES}} \leq C_{\mathrm{BES}}^{\max}$, where $D_{\mathrm{BES}}^{\max}$ and $C_{\mathrm{BES}}^{\max}$ are the discharging and charging rate limits and p_{BES} is the device power schedule.

The state of charge $q_{\mathrm{BES}}(t) \in \mathbf{R}^n$ of the n BES units is governed by the dynamics equation

$$q_{\mathrm{BES}}(t+1) = \quad \eta_{BES}^{q} q_{\mathrm{BES}}(t) + \eta_{BES}^{p} p_{\mathrm{BES}}(t), \tag{4.1}$$

where $p_{\mathrm{BES}}(t) \in \mathbf{R}^n$ is the charging rate of the BES units at time t. $\eta_{\mathrm{BES}}^{q}, \eta_{\mathrm{BES}}^{p} \in \mathbf{R}^n$ lie in the interval $[0,1]$ and are the storage and charging efficiencies. $qBES(t)$ must remain within the capacity limits of the battery, given as some fraction of the nominal capacity. We specify this as $Q_{\mathrm{BES}}^{\min} Q_{\mathrm{BES}}^{\mathrm{cap}} \leq q_{\mathrm{BES}}(t) \leq Q_{\mathrm{BES}}^{\max} Q_{\mathrm{BES}}^{\mathrm{cap}}$ where $Q_{\mathrm{BES}}^{\min}, Q_{\mathrm{BES}}^{\max} \in \mathbf{R}^n$ lies in the interval $[0,1]$ and $Q_{\mathrm{BES}}^{\mathrm{cap}} \in \mathbf{R}^n$ contains the nominal ratings of the BES units.

Grid connection

The PV storage system is tied to the grid at the PCC where power goes through a step up transformer before moving upstream to be converted from DC to AC. At this

point, the power flow balance is modeled as

$$p_{\text{grid}}(t) = p_{\text{PV}}(t) + \mathbf{1}^T p_{\text{BES}}(t), \tag{4.2}$$

where $p_{grid}(t)$ is the power schedule at the PCC at time t and can take on both positive and negative values. The term $\mathbf{1}^T \hat{p}_{\text{BES}}(t)$ uses the vector of ones, $\mathbf{1}$, to sum the total flow from the BES units. A simple power limiter also caps the amount of power that can be transmitted over the line and we model this as $|p_{grid}(t)| \leq P_{\text{PCC}}$ where P_{PCC} is the power limit at the PCC.

4.1.2 Objective

The system has two primary objectives, namely to provide energy arbitrage and to smooth intermittent output from the PV array. An additional secondary objective is also included to minimize excessive charge-discharge cycles of the BES units. The total cost incurred at each time step is denoted as $f(p_{\text{grid}}(t), p_{\text{BES}}(t))$, where $f : \mathbf{R}^{2n+1} \to \mathbf{R} \cup \{+\infty\}$ is a convex stage function. Because our objectives are distinct and separable, we can consider a stage cost function that is equal to the sum of the system objectives and the individual objectives,

$$f(\hat{p}_{\text{grid}}(t), \hat{p}_{\text{BES}}(t)) = \phi_{\text{shift}}(\hat{p}_{\text{grid}}(t)) + \phi_{\text{smooth}}(\hat{p}_{\text{grid}}(t)) + \phi_{\text{cyc}}(\hat{p}_{\text{BES}}(t)). \tag{4.3}$$

The first term, $\phi_{\text{shift}}(\hat{p}_{\text{grid}}(t))$, represents shifting energy for arbitrage. Because the system we model is a net generator of power, this term is a negative cost *i.e.*a profit. While there are many methods to achieve shifting, the most practical and effective technique is to simply use real-time wholesale pricing since this allows both consumers and providers to participate and benefit from power transactions as the power grid develops into a more dynamic and responsive system [FS09, GHS+04, HGN06, BJR02, SMDP07, NB10, IA09, Bor09]. Although such real-time price schedules are not currently deployed at all levels of the grid, many jurisdictions are moving towards prices that reflect the actual cost of power especially as the technologies for communicating these prices becomes more common and widespread. We thus represent the shifting term as $\phi_{\text{shift}}(\hat{p}_{\text{grid}}(t)) = c(t)\hat{p}_{\text{grid}}(t)$, where $c(t)$ is the energy price at time t. The second term, $\phi_{\text{smooth}}(\hat{p}_{\text{grid}}(t))$, is the cost associated with the smoothness of the grid output. For this term, we consider a weighted sum of three different measures of smoothness to account for the maximum range, slope and curvature of the power

output. This minimizes the variation in magnitude and rate of change of the output to produce a more consistent and smoother power profile.

We represent this term as

$$\begin{aligned}\phi_{\text{smooth}}(\hat{p}_{\text{grid}}(t)) &= \alpha_{\text{range}}(\max \hat{p}_{\text{grid}} - \min \hat{p}_{\text{grid}}) + \alpha_{\text{diff}}|\hat{p}_{\text{grid}}(t) - \hat{p}_{\text{grid}}(t-1)| \\ &+\alpha_{\text{curv}}(\hat{p}_{\text{grid}}(t-1) - 2\hat{p}_{\text{grid}}(t) + \hat{p}_{\text{grid}}(t+1))^2,\end{aligned}$$

where α_{range}, α_{diff} and $\alpha_{\text{curv}}$are the penalty parameters for tuning the range, slope and curvature terms respectively.

The final term, $\phi_{\text{cyc}}(\hat{p}_{\text{BES}}(t))$, adds a penalty for excessive charging and discharging. We represent this term as $\phi_{\text{cyc}}(\hat{p}_{\text{BES}}(t)) = \alpha_{\text{BES,cyc}}|\hat{p}_{\text{BES}}(t) - \hat{p}_{\text{BES}}(t-1)|$, where $\alpha_{\text{BES,cyc}}$ is the cycling penalty parameter. This non-negative cycling cost is similar to the slope term used for the smoothing objective and ensures that system operation costs are balanced against the functionality and lifetime costs of the BES units. Note that $\phi_{\text{cyc}}(\hat{p}_{\text{BES}}(t))$ is shown to depend on all batteries. This is because we give both BES units the freedom to charge and discharge for any objective. Differentiating the BES units for specific objectives is unnecessary since the optimization process chooses the most suited unit to either smooth or shift power [KWAB11].

An important point to emphasize is that the stage cost f is not assumed to be known in advance. Uncertainties in exogenous input and unknown intrinsic system variables are incorporated into f through predictions and forecasts. The overall objective of the system is the average stage cost across all time,

$$J = \lim_{T\to\infty} \frac{1}{T} \sum_{t=0}^{T-1} f(\hat{p}_{\text{grid}}(t), \hat{p}_{\text{BES}}(t)). \tag{4.4}$$

4.1.3 Control policy

The control policy in our PV storage system selects the control variables based on information available at the current time t. Known information includes system device parameters and grid line limits as well as measured quantities. This information along with external wholesale prices and estimates of unknown quantities are then used to calculate the future stage costs and minimize the total cost. Through this optimization process, the policy to determine the power schedules of each BES unit.

4.2 Numerical example

We demonstrate the dynamic and adaptive capabilities of the MPC algorithm on a grid tied PV storage system using data and specifications taken from the PNM Prosperity Energy Storage Project in New Mexico [(PN13]. The PNM system is one of the largest grid tied PV storage systems in the U.S., providing 500kW of power from solar panels which are directly coupled to battery storage capable of providing up to 750kW. The battery storage is separated into a rapid 500kW Ultra™ battery unit for rapid short term smoothing and a 1000kWh, 250kW advanced lead acid battery unit to store and dispatch energy when needed.

The PNM system is connected to the grid at the PCC through a power converter with a bidirectional meter and control system which has access to real-time prices. The connection also has a physical power transfer limit $P_{\text{grid}}^{\max} = 1250\text{kW}$. The system takes measurements and data at 1 second intervals at all points of the system, including the power output of the PV array and the output at the PCC. We incorporate this into the model as historical data and use MPC to enable optimization and scheduling of the BES units. Since the maximum output of the PV array is 500kW, any excess solar insolation falling on the array must be curtailed. For this reason, we use the power output measurements for predictions of future PV output instead of insolation and temperature measurements since they more accurately represent the output to the system.

The 500kW and 250kW BES units have 180kWh and 1000kWh storage capacity respectively. The charging rate $C_{\text{BES}}^{\max}$ and discharging rate $D_{\text{BES}}^{\max}$ are assumed to be equivalent. The efficiencies are based on information provided to us from the battery manufacturers of the system batteries. The large difference in the charge efficiency values is due to the fact that the smaller battery uses a different technology which makes it more efficient for high capacity power transfers but also makes it much more costly per unit of capacity. Since the leakage efficiencies are near unity, we use 1 for both units. We choose the initial state of each BES unit to be 50% of the nominal capacity. The parameters of each unit along with the system specifications and parameters are summarized in table 4.1 below.

The simulations are run using data taken from the time period July $17-24, 2012$. These days were selected to ensure the climate was warm enough to warrant demand shifting and increased prices at peak hours while also incorporating a period of cloud cover where intermittency would provide a need for smoothing PV output. For the

Table 4.1: PNM parameters.

Parameter	Description	Value
$P_{\text{PCC}}^{\max}$	Maximum power output of PNM system	1250kW
$P_{\text{PV}}^{\max}$	Maximum power output of PV array	500kW
$C_{\text{BES},1}^{\max} = D_{\text{BES},1}^{\max}$	Maximum charge and discharge rate of large BES unit	250kW
$C_{\text{BES},2}^{\max} = D_{\text{BES},2}^{\max}$	Maximum charge and discharge rate of small BES unit	500kW
$Q_{\text{BES},1}^{\text{cap}}$	Nominal storage capacity of large BES unit	1000kWh
$Q_{\text{BES},2}^{\text{cap}}$	Nominal storage capacity of small BES unit	180kWh
$\eta_{BES,1}^{p}$	Charge efficiency of large BES unit	0.70
$\eta_{BES,2}^{p}$	Charge efficiency of small BES unit	0.95
$\eta_{BES,1}^{q}$	Leakage efficiency of large BES unit	1.00
$\eta_{BES,2}^{q}$	Leakage efficiency of small BES unit	1.00

simulation period, we use day-ahead hourly wholesale price data published by the CAISO [CAI]. We chose to use the CAISO data since PNM currently does not publish dynamic electricity rates and the climate and load shapes in New Mexico are reasonably close to those found in southern California to warrant similar price schedules.

For each scenario we simulate, we use an MPC time horizon of $T = 180$. This corresponds to 4 minute intervals over a 12 hour period and is a typical time step and horizon for price signals and DR. The time step $t = 1$ corresponds to midnight. The wholesale price and PV output during the simulation time period are added onto the historical data incrementally at each time step and used to update the MPC predictions. The predictions are made using 5 days of historical data to predict the next 12 hours of generation.

4.2.1 Results

To demonstrate how the PV storage system can dynamically and automatically respond to various objectives including smoothing system output and using real-time prices for shifting, we present simulation results for a range of scenarios. Each scenario is evaluated with metrics for cost of shifting, smoothing and lifetime of the BES units. The cost of shifting is simply the area under the $c(t)p_{\text{grid}}(t)$ curve and we use

a straightforward trapezoidal approximation to determine the cost,

$$\Phi_{\text{shift}} = \sum_{t=1}^{T_{\text{f}}-1} \frac{1}{2}(c(t)p_{\text{grid}}(t) + c(t+1)p_{\text{grid}}(t+1))dt,$$

where $T_{\text{f}} = 1800$ is the simulation end time and $dt = 4$minutes. For the smoothing cost, there is as of yet no consensus on how to measure and price intermittency. Since system operators are mainly concerned with rates of change of power, we use a metric that represents the cumulative sum of the absolute values of the curve slope. This is a normalized version of the slope term in our smoothing objective and is similar to what was used in the PNM interim report [Wil12],

$$\Phi_{\text{smooth}} = \frac{\sum_{t=1}^{T_{\text{f}}-1} |p_{\text{grid}}(t+1) - p_{\text{grid}}(t)|}{\sum_{t=1}^{T_{\text{f}}-1} |p_{\text{PV}}(t+1) - p_{\text{PV}}(t)|}.$$

There is also lack of consensus on how to measure the lifetime impact on the BES units, since it is difficult to determine the different effects of deep and shallow cycling. There is general agreement, however, that lifetime degradation is strongly correlated to the amount of energy that has been moved in and out of the battery terminals. This is equivalent to the area under the $|p_{\text{BES}}|$ curve and again we use the trapezoidal approximation to determine the cost,

$$\Phi_{\text{cyc}} = \sum_{t=1}^{T_{\text{f}}-1} \frac{1}{2}|p_{\text{BES}}(t) + p_{\text{BES}}(t+1)|dt.$$

We note here that while there is currently no standard of measurement or established way to measure objectives for the systems coming online since they are the first of their kind and levels of penetration have yet to reach the point where the true impact of intermittency can be determined [DM07a, DM07b, HSJ12, HvVdVT07, Sov09], there is general agreement that smoothing renewable energy is important and lifetime effects on DERs used for managing intermittency are also important. Since there is still no clear idea of how to balance between these as well as other competing objectives, our aim is not to make large assumptions and definitively quantify how one objective will affect the other, but provide the algorithms, benchmarks and platform that will enable system engineers to tune tradeoffs once cost-benefit analyses are conducted and there is a more quantitative examination of the tradeoffs between competing objectives. It will also provide a useful design tool to help nudge such

Table 4.2: Cost metrics for simulated scenarios.

Scenario	Φ_{shift}[\$/kWh]	Φ_{smooth}	Φ_{cyc}[kWh]
base case (PV output)	−4892	1.000	–
smoothing	−4917	0.3920	3044
shifting	−5053	1.740	3971
smoothing and shifting	−5017	0.6674	3635
smoothing and shifting (prescient)	−5031	0.3052	3678
smoothing, shifting and BES cycling	−5040	0.9141	2710

systems to finding acceptable compromises between system objectives.

For each scenario simulated, we provide the cost metrics in table 5.2. The *base case* refers to the scenario where the grid output profile is not modified, *i.e.*is equal to the PV output. We also provide the simulated output and power schedules of the system and BES units for the first 3 days of some of the scenarios. The shifting costs are all negative since energy is being generated so the system is profiting. We present a selection of our results in figures 4.2-4.3, with the first plot in each figure representing the PV and grid output with a green solid line and black dotted line respectively. In the second plot, the large and small battery SOCs are represented by the red solid line and blue dotted line. Note that we have moved p_{grid} to the other side of the power balance constraint in the scheduling problem in order to maintain the convention that consumed power is positive and generated power is negative.

We begin by examining the effect of each system objective independently. The smoothing objective results in p_{grid} being notably smoother and steadier than p_{PV}. This is clearly seen in Fig. 4.2. Contrasting this *smoothing only* scenario with the base case, Φ_{smooth} is reduced by 60.8%. Not surprisingly, most of the high frequency smoothing is allocated to the small BES unit since it is able to respond more rapidly. The larger BES unit primarily handles the lower frequency diurnal swings of the solar resource, using its greater capacity to store the energy and provide it when necessary to help with smoothing the grid output.

Since the main price peak occurs at around 6PM each day and is preceded several hours by the solar peak at around 12PM, the BES units in the *shifting only* scenario are used for arbitrage and shift the PV output towards the peak price times. System output increases rapidly right before each peak price occurs as the storage system

discharges to take advantage of the day ahead price schedule. Similarly, low prices result in the BES units quickly charging using power from the PV array and this results in rapid drops in total system output. The price is effectively shifting the output of the system through economic incentive. In this scenario the batteries have only the price objective to work towards and so as expected both units act similarly in their charging and discharging behavior in order to maximize profit for the system. The slight variations in charging and discharging are due to the differences in efficiency as well as the maximum charging and discharging capacities of the units. Again comparing to the base case, Φ_{shift} in this *shifting only* scenario drops (profit increases) by 3.29%.

We next simulate the *smoothing and shifting* scenario in which the PV storage system simultaneously provides both smoothing and shifting. The α weights on the objectives are adjusted to provide and demonstrate an appropriate trade off between the two objectives. When responding only to price, the BES units tend to discharge and charge rapidly in a knee-jerk fashion in order to maximize profit. We dampen the power swings by increasing the relative weights on the smoothing objectives. The grid output shown in Fig. 4.3 demonstrates this scenario in which the BES units are simultaneously able to shift output to peak price times while smoothing PV output in anticipation of forecasted cloud cover. Comparing the metrics, the decrease in Φ_{smooth} drops by 45.3% relative to the *smoothing only* scenario and the decrease in Φ_{shift} drops by 22.4% when compared to the *shifting only* scenario. But both Φ_{smooth} and Φ_{shift} are still significant improvements from the base case and are able to garner a 33.3% and 2.56% improvement respectively. Note that when the objective is both smoothing and shifting, the optimization algorithm at each MPC time step automatically assigns the small BES unit with a more rapid response to handle most of the high frequency smoothing while using the slower larger BES unit to shift PV output.

We also compare this scenario to the prescient case where instead of using predictions, we assume prescient knowledge of PV output over the MPC horizon. While MPC is only a heuristic policy and hence does not guarantee optimality, it generally demonstrates results that are close to optimal and comparing MPC to the prescient case helps illustrate this. Even with our simple forecasting methods, Φ_{shift} only increases by 11.2% relative to the gain from the prescient case. This performance is not echoed in Φ_{smooth} however. The decrease in this metric approximately doubles the gain of the prescient case. The reason for this discrepancy in performance lies in

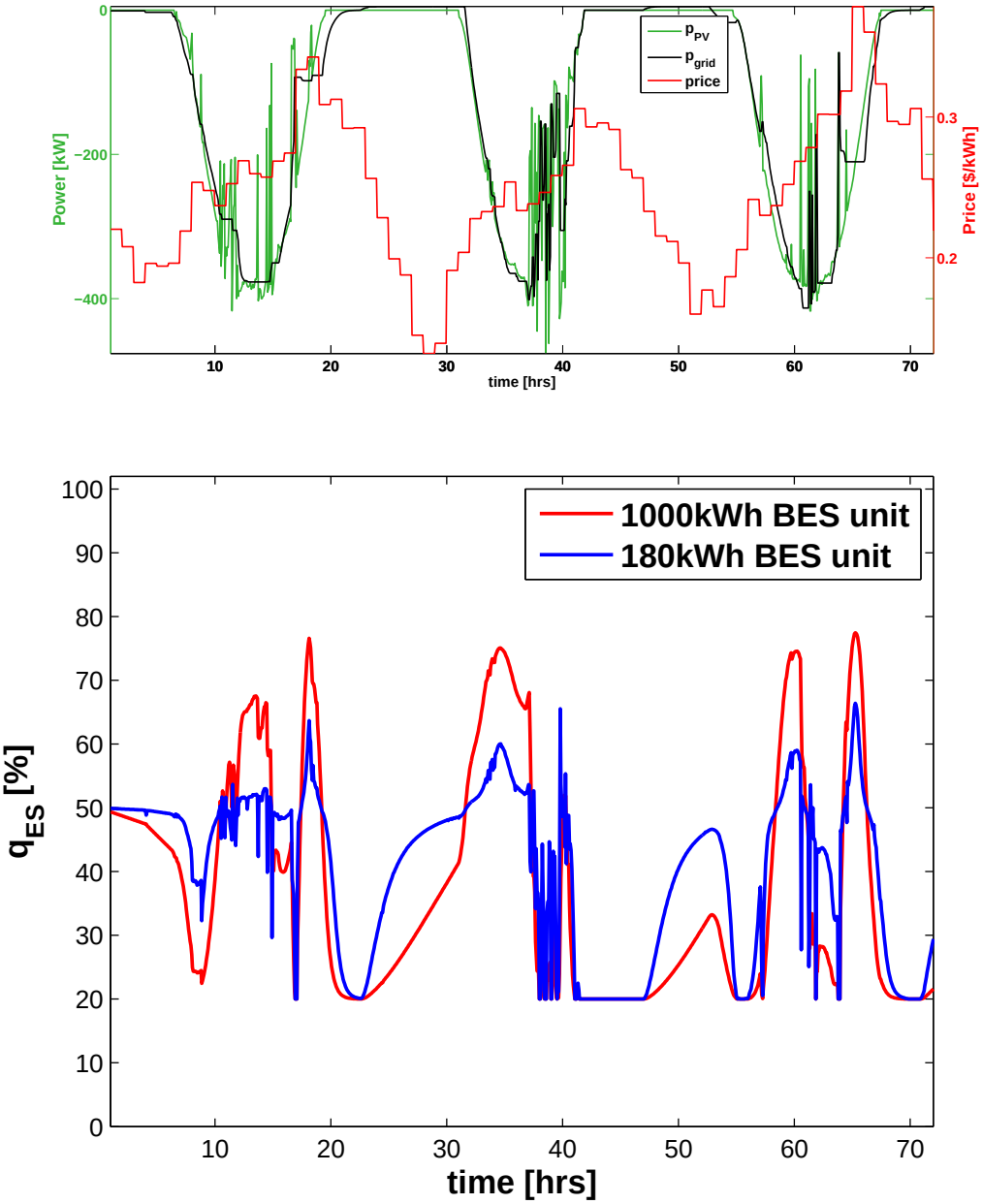

Figure 4.2: Grid output (*top*) and BES units SOC (*bottom*) plotted over the simulation time horizon for the smoothing objective.

how the PV predictions are made. Since we rely only on historical power profiles, our predictions are able to capture periodic trends and peaks very accurately. But without external forecast data, it is not possible to adequately capture transient weather phenomena like cloud cover since historical data does not inherently represent such random chaotic events. Φ_{shift} is able to come much closer to the prescient value than Φ_{smooth} because profits from shifting depend primarily on general periodic trends while the ability to smooth output relies on knowledge of transient weather phenomena to offset rapid unpredictable changes. There is still a significant improvement in Φ_{smooth} from the base case but in agreement with literature [BKP11], the results suggest that smoothing has much more to gain from supplementing historical data with external weather forecasts to make PV output predictions. While we included an error correction to account for sudden changes in PV output and ensure immediate response to sudden changes in weather patterns, performance could be greatly improved if some form of weather prediction was included in the data used to forecast the PV profile. Although MPC does allow for significant cost decrease even with forecasting that only captures general trends, increasing measurements to improve short term forecasts is important for the successful integration of large penetrations of solar generation in order to get smoother power profiles [SHC+12]. PNM is still actively working on acquiring this data but because of the flexibility of our prediction method, it is trivial to incorporate such data by including an additional weighted term to the asymmetric least squares objective once it becomes available and this should greatly improve the smoothing capabilities of the MPC algorithm.

When a BES cycling cost is added to the system objective functions, we see an increase in Φ_{smooth} by about 37.0%. This is in exchange for a reduction in Φ_{cyc} by about 25.4%. A particularly interesting result of introducing the battery cycling cost is that it not only causes Φ_{smooth} to increase in order to decrease Φ_{cyc}, but it also incidentally causes Φ_{shift} to decrease slightly by about 0.46%. This demonstrates how MPC can balance the independent objectives to enable acceptable trade offs to be determined and implemented within a PV storage system when issues like battery lifetime are important and the impact of cycling can be quantified.

Finally, it should be pointed out that the relative improvement in shifting seems relatively small in all scenarios. There are two main reasons for this. First, the batteries were sized primarily for smoothing so their capacity limits the amount of shifting they can do. The second reason is that there was significant overlap between

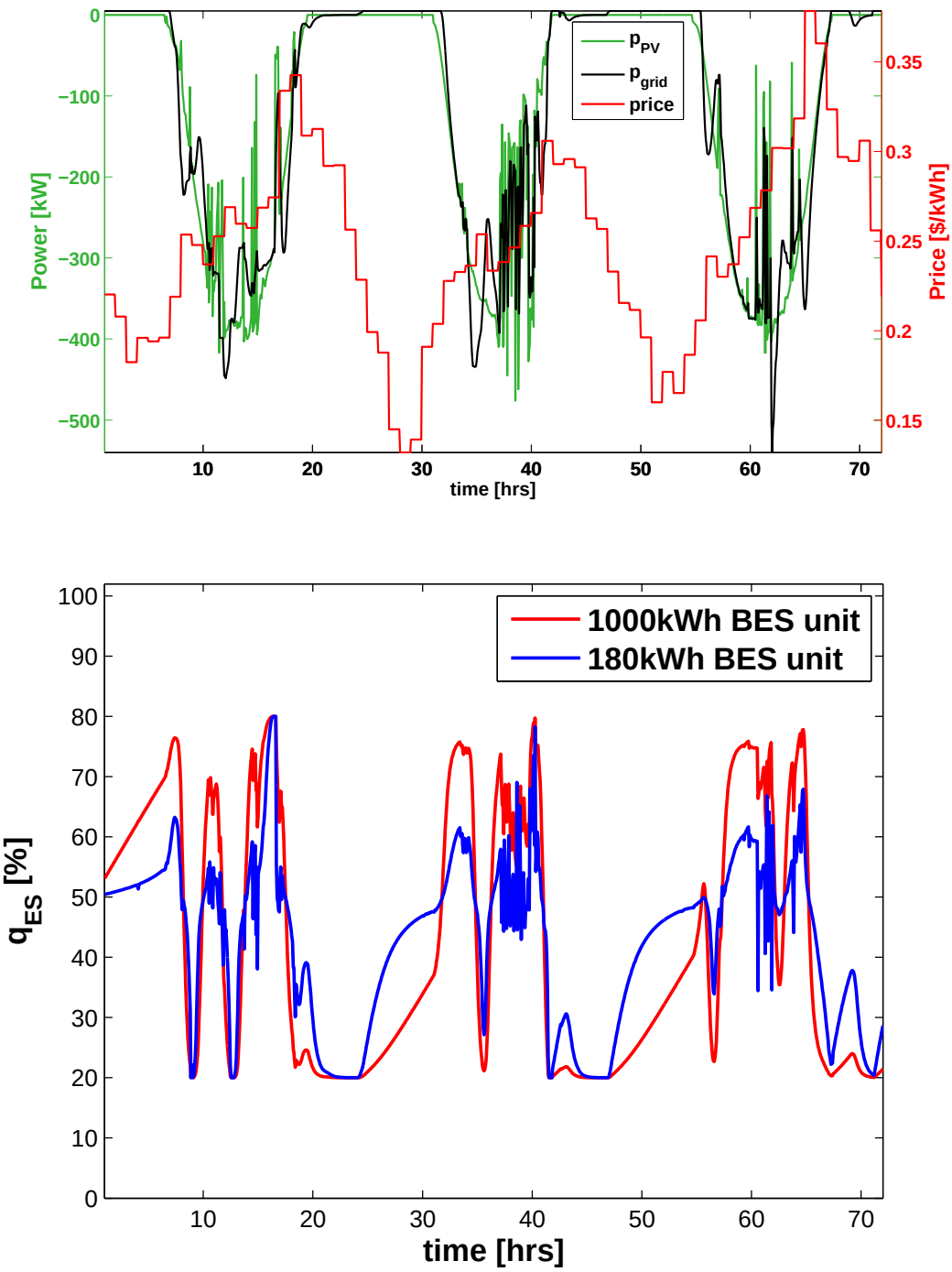

Figure 4.3: Grid output (*top*) and BES units SOC (*bottom*) plotted over the simulation time horizon for simultaneous smoothing and shifting.

the the real-time prices and the PV generation periods so there was less to be gained. This is an important consideration since it could determine the best way to size and deploy a system.

4.2.2 Computation times

The simulation is carried out using CVX [GB08] to solve the optimization at each time step. CVX is a modeling system that solves convex optimization problems by converting them into semidefinite programs (SDPs). Using open-source solvers, it runs in a Matlab environment and is hence ideal for rapid code development and deployment. Since MPC uses the first step of the previous solution to provide a warm start for the next iteration, the optimization problem at each MPC iteration can be solved on an Intel Xeon 3.0GHz dual core processor in approximately 4 seconds. The predictions at each time step are similarly solved using CVX and require an additional 7 seconds. This means that the amount of time required at each MPC step will be significantly less than the amount of time lapsed between MPC time steps, making real-time dynamic solutions entirely feasible. While this demonstrates that our methods are suitable for real-time implementation, we were carrying out simulations using a scripted language as a proof of concept before actual implementation using embedded code. Hence the times reported are conservative approximations only and significantly better performance times can be attained using CVXGEN [MB12] or similar tools to generate custom embedded solvers for deployment on embedded processors.

Chapter 5

Microgrid

Controller optimization naturally addresses whether a mutually beneficial balance can be struck between the power demands of the DERs and their aggregated effect on the power grid. Of particular interest is the fleet of electric vehicles starting to come online at commercial facilities as well as in car sharing programs, public transportation and goods delivery services. The primary reason for electrifying the transportation fleet is to substantially reduce the amount of fossil fuels used for transportation. Since most vehicles only travel short distances on any given day, even relatively short range plug-in hybrid EVs can run primarily on electricity. And as the penetration of solar and wind continues to grow rapidly, charging these vehicles will become increasingly clean. With some storage capacity, the EV has also come to represent one of the DERs with the greatest potential for DR. A study conducted by the Electric Power Research Institute (EPRI) using data from the most recent National Household Travel Survey (NHTS) [AD11] found that across many factors commonly assumed to influence driving behaviors, there is actually little variation in typical driving patterns. This suggests that while EVs will be a significant load on the power grid at the distribution level, they will also be able to provide a consistent resource for energy management services. Plug-in EVs are particularly advantageous since they can be deployed with great flexibility when needed given their relatively long availability window when compared with their actual charge duration. However, as the number of EVs continues to increase, integrating them intelligently will be critical on a power grid that was not inherently designed to handle the additional load. The study concluded that while many of the expected factors like providing high power charging and adding additional charge locations would have little impact on

the magnitude of coincident load reduction, load control through optimized charging may be able to significantly increase and improve energy resource utilization.

In our microgrid example, we propose a method to realize the shift from a centrally run power grid to a decentralized network that will enable real-time management and scheduling of EVs at a commercial site with onsite PV generation, partially curtailable load and a battery storage unit for backup. While the primary charging location for the majority of EVs will be at home, workplace charging will be the second most important charging location since the longest vehicle dwell times during the day occur at work [AD11] . We look at workplace charging because in addition to providing an additional charge location to increase the displacement of gasoline with electricity, it provides useful charging opportunities early in the day when additional load generally has the lowest impact on the grid and it also allows the EVs to directly take advantage of local generation from PV. The regular patterns of arrival and departure at work will also make the vehicles reliable to manage. However, because vehicles arrive at work within a narrower time frame than they arrive at home, there is also potentially a sharp load peak that could occur in the morning at work if the EVs all begin charging immediately. Intelligent load management of the EVs with the help of real-time charge rates or other broadcasted signals could hence be effective if not crucial in flattening load peaks and adding load diversity in addition to allowing the vehicles to take advantage of the PV output. From the utility perspective, this load shifting could have significant benefits for the grid since it would reduce or eliminate the load from vehicle charging coincident with the primary load peak in the evening. The commercial site is connected to the distribution system and operates as a microgrid participating in the wholesale electricity market, with the DERs interacting with each other to arbitrage local generation from the PV array. While each DER is controlled independent of one another, the capacity constraint at the PCC means that in addition to minimizing the cost of power to the microgrid, the controllers must cooperate in such a way as to prevent violation of the physical limitations on the line.

The power converter at the PCC in our model only needs to have access to real-time wholesale prices or know that the grid is in a contingency state and each DER only needs to know its own state and historical power profiles in order to determine control actions for each DER in the microgrid. Using ADMM with MPC to decentralize and distribute the optimization problem, this is sufficient to obtain optimized

power profiles that minimize the total objective while maintaining DER functionality and respecting both individual and system constraints. The distributed algorithms also enable us to explore the full potential of EVs connected to the grid. Many vehicle manufacturers and utilities are already considering vehicle-to-grid operation (V2G) as a way for vehicles to provide additional grid services that they could potentially be remunerated for [KT05a, DEKM10, oE13b, KUH+08, TK07, SES11, CTD+09]. V2G enables bidirectional charging and presents the vehicle batteries with the opportunity to provide various grid services when the vehicles are plugged into the charger. While EVs are unlikely to be able to provide significant base load and peak power services without significant penetration, they have great potential in rapid response short-term power service markets like spinning reserves and regulation [WSE, DEKM10, KUH+08, LLC13, CH10] . Strongly related to this is the ability to help smooth the intermittency from renewable sources of energy [KT05b, DEKM10, LK, SV11, PLAS]. With a bi-directional power converter, plug-in EVs can also provide backup power in situations where the microgrid needs to be isolated from the utility grid, either intentionally due to a fault or other abnormal grid conditions. In this stand-alone mode, the vehicles allow the microgrid to continue operating without power from the grid [CTD+09, PLAS, KT05b]. However, while there are many efforts looking at how to use EVs to provide additional grid services and make the vehicles more economical and profitable, there is still much uncertainty as to whether the profit provided to each vehicle will be sufficient incentive to the vehicle owner who must give up utility, long-term functionality and privacy of the vehicle in return [DEKM10, LPMC10, SES11, PAW, HH13, KT05a, KUH+08]. The algorithms we propose address each of these issues, allowing the vehicle owners to determine their own utility functions, balancing economic benefits against vehicle utility as well as the lifetime degradation from additional cycling on the vehicle. Privacy is also retained since no external control is required to schedule the vehicles. While this means that the system operator cannot know with complete certainty what actions each EV will take, the optimality of the ADMM algorithm does ensure that the aggregate behavior of the microgrid can be aligned with common goals including operational efficiency, flexibility, stability and resiliency. We note here that we deliberately remain agnostic about the viability of V2G: the EVs are allowed to both charge and discharge in our model in order to allow the algorithms to determine the tradeoff between competing objectives within each vehicle and from other DERs in the microgrid. The decision

as to whether or not it is economical for the EVs to participate in the market for V2G will naturally fall out of the solution to the optimization problem.

By using dynamic and distributed algorithms in this example, the microgrid we explore effectively becomes a self-correcting system and the flexibility in the framework allows DERs to plug into the power network with their own configurations, objectives and constraints specified simply and directly without any reconfiguration. This is particularly useful at the distribution level where both individual utility and social good will be specific to the particulars of a given operating environment. Each independent DER will be able to act autonomously in a way that enables it to participate responsibly on the grid while still maximizing benefit to itself. In this way, collaboration and cooperation through distributed optimization enables sustainable operation of the microgrid. DERs can take advantage of dynamic situations where they can work together in order to increase net benefits, ranging from minimizing system costs due to real-time rates to maintaining power quality and preventing catastrophic failure on the grid. The complexity of managing a decentralized network is divided up among all the controllers to make control of the microgrid tractable and reflective of local needs while leveraging local contribution of distribution-level resources including DG from renewable resources and other controllable DERs. From the perspective of the system operator, aggregation of all DERs behind the PCC allows it to treat the microgrid as a well behaved single entity that is flexible and responsive to system needs. The algorithms can also easily be scaled up with each DER or aggregate of DERs participating at the power system level and helping convert the grid into a flexible and simultaneously more resilient system. In each scenario, we compare avoided costs against potential tradeoffs. We also compare these results to the prescient case for 24 hour look ahead horizons to demonstrate that even with simple forecasting models, we are able to operate robustly and very near optimal using ADMM with MPC. The aim is to show that decentralized, distributed control can increase flexibility and responsiveness to both local and system-wide contingencies. In addition to demonstrating that DER integration leads to load balancing and rapid response times, we also show that functionality is maintained with each resources and the DERs all operate within their constraints.

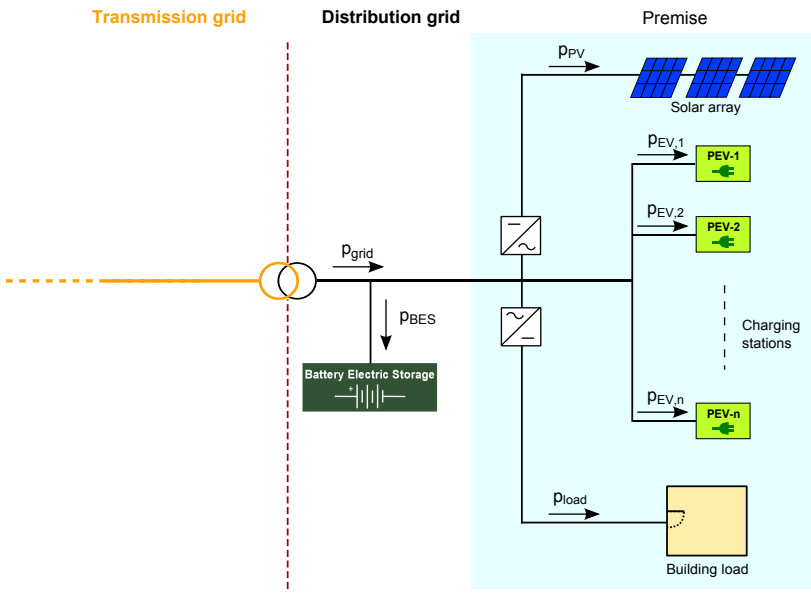

Figure 5.1: Schematic representing microgrid with electric load, PV array, BES and EV charge stations connected to power system. Arrows indicate positive direction of power flow.

5.1 Model

5.1.1 System dynamics and constraints

The microgrid modeled and simulated is shown schematically in Fig. 5.1. In addition to the electric load on-site and local generation from an installed PV array, there are N_{EV} EVs on-site and storage from a BES unit. Collectively, we refer to the PV array, the charging EVs and the BES unit as the DERs. Each DER and the electric load has an associated power schedule across the simulation time horizon T, with negative power always defined as being power being generated and/or flowing out from a point in the microgrid. Note that we have also treated the connection at the PCC as an effective DER representing the grid with its own objectives and constraints.

For simplicity, we consider only DC power without constraints on the phase schedule since smaller microgrids operating at the distribution level will likely be confined to a single phase [Bal08]. The results can easily be extended to an AC network, however, by imposing an additional phase schedule constraint for each DER modeled [KCLB14]. We also note here that while all our DERs have been modeled using objective functions that are convex, we do not require either finiteness or strict convexity of any objective function.

The models associated with the microgrid, electric load, and DERs are described in detail below, including the individual objectives and constraints.

Electric load

The electric load is represented with a consumption profile, $p_{\text{load}} \in \mathbf{R}^T$. This consumption profile has a diurnal cycle and must be predicted and anticipated using historical load data. We represent the predicted profile as $\hat{p}_{\text{load}} \in \mathbf{R}^T$. When grid-connected or during a contingency situation, the load is curtailable but a minimum time-dependent base load must be met. This is modeled as $\beta\hat{p}_{\text{load}} \leq p_{\text{load}} \leq \hat{p}_{\text{load}}$ where $\beta \in [0, 1]$ is the minimal load fraction that must be met. The value of β will depend on the whether the microgrid is operating in a normally functioning power system or if it is facing a contingency situation.

In order to ensure the amount of load that is curtailed on the system is based on the utility factor of that load, a curtailment cost is also added:

$$\alpha_{\text{load}} \|\hat{p}_{\text{load}} - p_{\text{load}}\|_2^2 \tag{5.1}$$

where $\alpha_{\text{load}} \geq 0$ is the load curtailment penalty parameter.

Photovoltaic array

A PV array also follows a diurnal cycle. The power schedule, $p_{\text{PV}} \in \mathbf{R}^T$, is always negative since it only generates power and the peak is offset from the load peak since it occurs when the sun is at its highest. As with the load, the power profile of the PV array is predicted and anticipated using measured historical data. We represent this as $\hat{p}_{\text{PV}}$. The array output will fluctuate with cloud cover and contributes to the power balance on the microgrid but the PV inverter has the option to curtail power if it cannot be pushed back out onto the grid or stored. We model this as $p_{\text{PV}} \leq \hat{p}_{\text{PV}}$.

Since generated power should not be curtailed unnecessarily, a curtailment cost is also included:

$$\alpha_{\text{PV}} \|\hat{p}_{\text{PV}} - p_{\text{PV}}\|_2^2 \tag{5.2}$$

where $\alpha_{\text{PV}} \geq 0$ is the PV curtailment penalty parameter.

Battery electric storage

While the inverter of the PV array can do little more than curtail output, a BES unit can store power from the PV array and other DERs and use this to hedge against high prices as well as act as a buffer for requests or unforeseen events on the power grid. Since the BES unit is able to both charge and discharge, its power schedule can be both positive and negative. The rates of charge and discharge are constrained by capacity limits. This is represented as $-D_{\text{BES}}^{\max} \leq p_{\text{BES}} \leq C_{\text{BES}}^{\max}$, where $D_{\text{BES}}^{\max}$ and $C_{\text{BES}}^{\max}$ are the discharging and charging rate limits and $p_{\text{BES}} \in \mathbf{R}^T$ is the BES power schedule.

The dynamics equation governing the state of charge of the BES unit over the time interval $t = 1, \ldots, T$ is given by

$$q_{\text{BES}}(t+1) = \quad \eta_{BES}^{q} q_{\text{BES}}(t) + \eta_{BES}^{p} p_{\text{BES}}(t), \tag{5.3}$$

where $\eta_{\text{BES}}^{q}, \eta_{\text{BES}}^{p}$ lie in the interval $[0, 1]$ and represent the storage and charging efficiencies respectively. The components of $qBES \in \mathbf{R}^{T+1}$ are given as some fraction of the nominal capacity and must remain within the capacity limits of the battery. We specify this as $Q_{\text{BES}}^{\min} Q_{\text{BES}}^{\text{cap}} \leq q_{\text{BES}} \leq Q_{\text{BES}}^{\max} Q_{\text{BES}}^{\text{cap}}$ where $Q_{\text{BES}}^{\min}, Q_{\text{BES}}^{\max}$ lies in the interval $[0, 1]$ and $Q_{\text{BES}}^{\text{cap}}$ represents the nominal rating of the unit. The BES unit can also have associated costs, including a cycling cost to penalize excessive charge-discharge cycles,

$$\alpha_{\text{cyc}} \textstyle\sum_{t=1}^{T-1} |p_{\text{BES}}(t+1) - p_{\text{BES}}(t)|, \tag{5.4}$$

where $\alpha_{\text{cyc}} \geq 0$ is the cycling penalty parameter.

A terminal constraint can additionally be added to the BES unit to ensure that the storage system is not depleted at the end of the time horizon, with a common choice of q_{final} being $0.5Q_{cap}$ or the initial charge state of the battery.

Electric vehicle

An EV is a flexible load with storage capabilities that allows charging to be deferred as well as discharging when economical. It differs from the BES unit in that it has additional constraints in both availability and required charge capacity. The microgrid has N_{EV} EVs on-site and each EV has an associated charging schedule $p_{\text{EV},i} \in \mathbf{R}^T$ and charge state $q_{\text{EV},i} \in \mathbf{R}^{T+1}$ where $i = 1, \ldots, N_{\text{EV}}$. When the vehicle is not plugged in, $p_{\text{EV},i}(t) = 0.$ and $t = T_{\text{dep}}(i) + 1, \ldots, T$. The EVs are also associated with

four stochastic variables, namely the arrival time, departure time, initial charge state and desired charge state. We write this as the four-component vector defined in the predictions sections in the previous chapter: $\theta_{\mathrm{EV},i} = [T_{\mathrm{arr}}, T_{\mathrm{dep}}, q_{\mathrm{init}}, q_{\mathrm{des}}]_{\mathrm{EV},i}$. For each EV, the components of $\theta_{\mathrm{EV},i}$ need to be predicted using probability distributions based on historical data prior to the arrival time. After arrival, the vector is completely defined.

The vehicles also have charge constraints $-C^{\max}_{\mathrm{EV},i} \leq p_{\mathrm{EV},i} \leq C^{\max}_{\mathrm{EV},i}$ where $C^{\max}_{\mathrm{EV},i}$ is the maximum charge/discharge rate of the ith EV.

$$q_{\mathrm{EV},i}(t+1) = \eta^{q}_{\mathrm{EV},i} q_{\mathrm{EV},i}(t) + \eta^{p}_{\mathrm{EV},i} p_{\mathrm{EV},i}(t), \tag{5.5}$$

where $\eta^{q}_{\mathrm{EV},i}, \eta^{p}_{\mathrm{EV},i}$ lie in the interval $[0,1]$ and are the storage and charging efficiencies respectively. $q_{\mathrm{EV},i}$ is additionally constrained by the capacity of the battery. This is specified as $Q^{\min}_{\mathrm{EV},i} Q^{\mathrm{cap}}_{\mathrm{EV},i} \leq q_{\mathrm{EV},i}(t) \leq Q^{\max}_{\mathrm{EV},i} Q^{\mathrm{cap}}_{\mathrm{EV},i}$ where $Q^{\min}_{\mathrm{EV},i}, Q^{\max}_{\mathrm{EV},i}$ lie between $[0,1]$ and are the minimum and maximum charge levels of the batteries. $Q^{\mathrm{cap}}_{\mathrm{EV},i}$ is the nominal rating of the ith EV battery.

The desired state of charge for each vehicle $q_{\mathrm{des},i}$ is also used to represent the utility of each vehicle through the constraint

$$q_{\mathrm{EV},i}(T_{\mathrm{dep},i}) \leq \alpha_{\mathrm{des},i} q_{\mathrm{des},i}, \tag{5.6}$$

where a higher $q_{\mathrm{des},i}$ indicates a greater demand for power and $\alpha_{\mathrm{des},i}$ represents the vehicle owner's flexibility for not meeting the desired departure state. The utility functions of each vehicle can be learned and/or specified, enabling vehicle owners to assign values to desired charging services and then scheduling their vehicles to maximize net benefits based on these assignments. While methods of determining the utility functions of individual vehicles can readily be incorporated into our model, for simplicity we assume that the utility functions have already been determined and focus on showing that the distributed algorithms can handle varying utility functions for the vehicles simultaneously.

A penalty for excessive cycling of the vehicle battery can also be included for each vehicle,

$$\alpha_{\mathrm{cyc},i} \textstyle\sum_{t=1}^{T-1} |p_{\mathrm{EV},i}(t+1) - p_{\mathrm{EV},i}(t)|, \tag{5.7}$$

where $\alpha_{\mathrm{cyc},i} \geq 0$ is a penalty parameter that weights the excessive cycling cost against the utility function of the vehicle.

Grid connection

The microgrid is connected to the power grid at the PCC where power is stepped up or down in voltage and the wholesale rate for power is applied. From the perspective of the power controller that sits behind the meter, this connection acts as a power limiter that caps the amount of power that can be transmitted over the line. We model this as

$$|p_{\text{grid}}| \leq P_{\text{PCC}}, \tag{5.8}$$

where $p_{\text{grid}} \in \mathbf{R}^T$ is the power schedule at the PCC and P_{PCC} is the power limit at the PCC. p_{grid} can take on both positive and negative values depending on whether the power is being taken from the grid or put back onto the grid. At the PCC, the cost function minimizes energy cost as well as the cost of regulation,

$$\textstyle\sum_{t=1}^{T} c(t)p_{\text{grid}}(t) + f_{\text{smooth}}(p_{\text{grid}}(t)). \tag{5.9}$$

In the first term, c is the real-time wholesale price schedule for power. The microgrid receives this price schedule from the independent system operator (ISO) and uses it in the first term to determine what power profile will minimize the cost of power drawn from the grid over the price time horizon. When the term is negative, the microgrid is selling power to the power grid and making a profit. As with the PNM example, we use real-time wholesale pricing since they are best positioned to allow both consumers and providers to participate and benefit from power transactions as the power grid develops into a more dynamic and responsive system [FS09, GHS$^+$04, HGN06, BJR02, SMDP07, NB10, IA09, Bor09]. In a microgrid scenario, this is a very practical and effective technique to achieve energy arbitrage and shift demand at the distribution level. While such dynamic price schedules are not currently deployed at all levels of the grid, many jurisdictions are moving towards models that pass on prices to consumers reflecting the actual cost of power, especially as the technologies for communicating these prices becomes more common and widespread. The second term is the cost associated with the smoothness of the output at the PCC. It ensures a smooth power profile that does not sharply increase or decrease due to PV output or sudden load changes in the microgrid. This measure of power quality can in some cases be remunerated for by the ISO and in other cases is required in order to connect to the power grid and avoid penalties. For this term, we consider a weighted sum of three different measures of smoothness to account for the maximum range, slope and

curvature of the power output. This minimizes the variation in magnitude and rate of change of the output to produce a more consistent and smoother power profile. We represent this term as

$$\begin{aligned} f_{\text{smooth}}(p_{\text{grid}}(t)) &= \alpha_{\text{range}}(\max_t p_{\text{grid}}(t) - \min_t p_{\text{grid}}(t)) \\ &+\alpha_{\text{diff}} \textstyle\sum_{t=1}^{T-1} |p_{\text{grid}}(t+1) - p_{\text{grid}}(t)| \\ &+\alpha_{\text{curv}} \textstyle\sum_{t=1}^{T-2} (p_{\text{grid}}(t) - 2p_{\text{grid}}(t+1) + p_{\text{grid}}(t+2))^2, \end{aligned} \tag{5.10}$$

where α_{range}, α_{diff} and α_{curv} are the penalty parameters for tuning the range, slope and curvature terms respectively.

5.1.2 Objective

Because our objectives are distinct and separable, we can consider a cost function that is equal to the sum of the individual objectives. This objective is minimized subject to the power flow balance constraint,

$$\begin{array}{ll} \text{minimize} & \sum_{i=1}^{N} f_i(p_i) \\ \text{subject to} & \sum_{i=1}^{N} p_i = 0, \end{array} \tag{5.11}$$

where $N = 4 + N_{\text{EV}}$ is the total number of independent controllers on-site (*i.e.*load, PV, BES, PCC, EVs) and f_i represents the objective and constraints of each controller. We set $f_i(p_i) = \infty$ to represent infeasibility of the power schedule of the ith controller. When $f_i(p_i) < \infty$ for $i = 1, \ldots, N$ and the constraint ensuring power balance is satisfied, the solution is feasible and p_i represents a realizable schedule with $f_i(p_i)$ being the associated cost of that schedule. It is important to note that only the controller at the PCC is provided with the external price schedule and it determines its own power profile. As was explained in Chapter 3, even though it is the interface between the power grid and the other DERs, it does not need to know anything about them or control their behavior. The power balance constraint is a physical constraint that ensures all controllers in the microgrid coordinate to achieve a feasible solution.

5.1.3 Control policy

The control policy for each DER in our microgrid selects the control variables based on information available at the current time. Known information includes DER parameters, grid line limits and conditions at the PCC, and measured states of the DERs

and loads. This information along with external wholesale prices and estimates of unknown quantities are then used to calculate and minimize the total cost. Through this optimization process, the control policy determines the power schedules of each controller on-site.

5.2 Numerical example

To demonstrate the dynamic capabilities of the ADMM algorithm used in conjunction with MPC, we look specifically at a commercial site operating as a microgrid equipped with the set of DERs described in our model. The DERs each have access to their own historical data to make estimates and predictions. Simulations are run using load and generation data and price information taken from May $18 - 25, 2013$. For the simulation period, we use day-ahead hourly wholesale price data published by the CAISO [CAI] when the microgrid is grid-connected. For each scenario we simulated, we selected an MPC time horizon of $T = 96$. This corresponds to 15 minute intervals over a 24 hour period and is a typical horizon and time step for schedule updates. The time step $\tau = 1$ corresponds to midnight.

The microgrid is connected to the distribution system through a bidirectional meter which has access to the real-time wholesale prices. The connection also has a physical transfer constraint $P_{\text{PCC}} = 200\text{kW}$. The power profile for the electric load is taken from measurements at a typical commercial site and the PV output is taken from the output of a rooftop PV array, both geographically co-located in Northern California. We size the array to 1200 kW and scale the output accordingly so that it is able to meet all of the local energy demands of the load over a diurnal cycle when islanded. This data is incorporated into the model both as simulation data and as historical data to help make predictions and schedule the DERs. As time proceeds, the simulation data is added onto the historical data incrementally at each time step and used to update the predictions. The predictions are made using 5days of historical data to predict the next 24hours of generation. While it may seem that P_{PCC} is overly limiting, the aim is to demonstrate that the microgrid can run flexibly and reliably on a grid of limited capacity using dynamic algorithms to handle the coupling constraints.

The on-site BES unit is sized to provide sufficient capacity to arbitrage energy for islanding situations while maintaining at least 50% baseload. This capacity is

Parameter	Value
$C_{\text{BES}}^{\max} = D_{\text{BES}}^{\max}$	500kW
$\eta_{BES,c} = \eta_{BES,d}$	0.85
$\eta_{BES,q}$	0.90
$Q_{\text{BES}}^{\min}$	0.20
$Q_{\text{BES}}^{\max}$	0.90
$Q_{\text{BES}}^{\text{cap}}$	3000kWh

Table 5.1: BES parameters.

commensurate with typical BES unit sizes located on microgrid sites [Pow13, Ger12, CDC+11, CGW12, KK13]. The charging rate $C_{\text{BES}}^{\max}$ and discharging rate $D_{\text{BES}}^{\max}$ are assumed to be equivalent. These rates as well as the other BES parameters are provided in table 5.1.

Within the microgrid, a fleet of 20 EVs are available and capable of level 2 charging at 7.2kW without requiring a dedicated circuit [oE13a, Wik13]. As with the BES unit, the charging rate $C^{\max}$ and discharging rate $D^{\max}$ are considered to be equal. Battery efficiency values $\eta_{\text{EV},i}^{q}, \eta_{\text{EV},i}^{p}$ are both set at 90% [MKD, CNHD10, PSM10, GME+, CKY+, KD12, MEED12]. The battery capacity for each vehicle is selected based on driver need. In other words, as an appropriate first approximation it is assumed that vehicles with longer commute distances will have vehicle owners who desire larger batteries. Using individual vehicle data taken over several hundred days in a study conducted by EPRI [Ale12], we consider the longest trip distance and use a 2× buffer along with a typical mileage conversion rate of 0.311kWh/mile [AD11, IA09, DKA07] to determine the battery capacity. The minimum and maximum charge states of the batteries are selected to be 30% and 90% respectively to avoid deep cycling of the battery [SBD+07, WNs+09, CNHD10, DS06].

The arrival times T_{arr} and departure times T_{dep} for each vehicle are chosen from distributions constructed using the vehicle data. Since this data only provides travel distances for each vehicle going between work and home, we determine the initial charge state q_{init} by assuming that the EVs have had enough time to charge to at least 50% at home overnight and look at a distribution of how far a vehicle is driven to get to work in order to determine how much the vehicle battery has been depleted from the journey to work. The miles are then converted to kilowatt-hours before subtracting this from 50% of the battery capacity. The predicted value of q_{init} is then determined using the maximum likelihood method described in Chapter 3.3 while the actual value of q_{init} is determined using a Monte Carlo method to select from the

Table 5.2: Cost metrics for simulated scenarios.

Method	External cost [$]	Smoothing cost [kW]	PV curtailed [kWh]	Load curtailed [kWh]	EV shortfall [kWh]
centralized (prescient)	−1682.59	14.99	902.63	122.05	0
centralized	−1680.62	16.36	918.48	114.94	0
ADMM decentralized	−1674.16	16.67	919.76	119.95	0

distribution of initial charge states. For the desired charge state q_{des}, we similarly look at how far the vehicles have to drive to get home but instead of using a Monte Carlo method to determine how much charge the vehicle batteries have to at least have at departure time, we use the conservative approach described in Chapter 3.3 . Since we do not have charge distributions for each vehicle and only have distances traveled, we consider the longest distance traveled and ensure that each vehicle has at least the kWh equivalent of this distance in addition to the depleted charge from driving to work added to the initial charge state. This forms the minimum a vehicle must have before departing and ensures that most contingency situations can be handled. As with the prediction method, this approach gives a good first approximation for how much a driver will want before departing but as more data is accumulated for each vehicle and the preferences of the driver are specified, the value of q_{des} can be fine tuned.

5.2.1 Results

The results of our simulations over a 3 day period are provided below in Table 5.2. In addition to carrying out the simulation using the ADMM method to distribute the optimization calculations among the controllers and determine the power profiles at each point in the microgrid, we compare this case against a microgrid carrying out the same calculations through centralized optimization as well as a centralized case with prescient knowledge in order to provide a benchmark for how ADMM with MPC performs even with simple prediction methods. In the prescient case, instead of using predictions we assume PV and load schedules as well as EV parameters are fully known over the MPC horizon. For each scenario simulated, we calculate the cost of energy at the PCC as well as for each DER. We also determine the ramping cost at the PCC represented by (5.10), the power curtailed by the PV array and load, and the total energy shortfall when charging the EVs to their desired departure state of charge. Note that the external energy costs are negative since energy is being generated so the system is profiting.

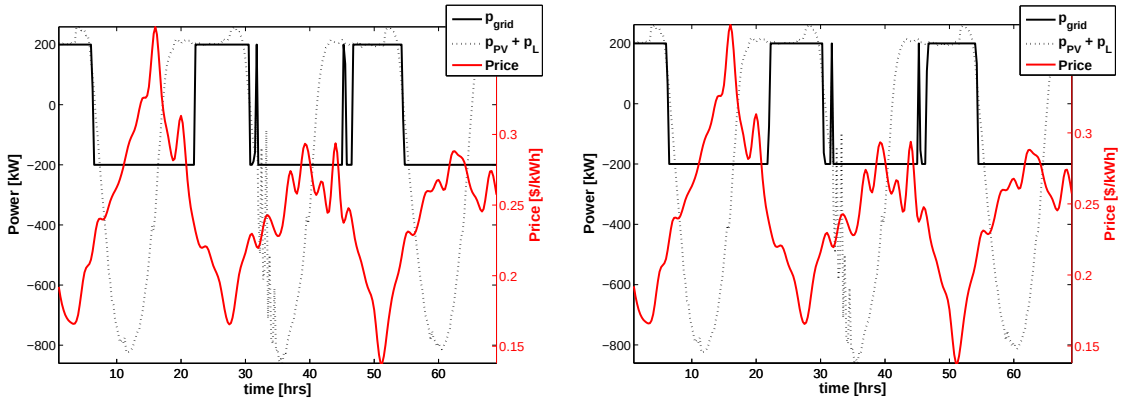

Figure 5.2: Power profile of microgrid at PCC over the simulation time horizon plotted against price and the unoptimized net power profile of the PV array and load. The *left* plot represents the ADMM case and the *right* plot represents the centralized case.

As the table shows, ADMM performs similarly to the centralized method and both come very close to the performance of the centralized prescient method. Comparing the ADMM and the centralized case, there is less than a 2.1% difference between the total system costs. Contrasting both of them against the ideal centralized prescient case, there is less than a 3.9% difference between the total system costs. Below, plots of the ADMM and the centralized cases for the simulation period are provided. The net power profile of the microgrid at the PCC is shown first in Fig. 5.2, plotted against the net power profile that results from PV and load without any optimization to compare the performance of the microgrid against the base case. Note that we have moved p_{grid} to the other side of the power balance constraint in the scheduling problem in order to maintain the convention that consumed power is positive and generated power is negative.This plot is followed by plots of the PV array output in Fig. 5.3 and the load power profile in Fig. 5.4 after curtailing. The power profile and state of charge of the BES unit and the EVs are provided in Fig. 5.5 and Fig. 5.6 respectively. The state of charge in the latter case is plotted as a percentage of the total capacity of all the vehicles beginning each day at the time when the first vehicle arrives and ending at when the last vehicle departs. From the plots comparing ADMM with the centralized case, it can be seen that the results are very similar and this corroborates the quantitative results from Table 5.2.

Since the capacity limit of the line is binding, the DERs use their resources to ensure the microgrid can still operate within these limits, absorbing excess generation from the PV array to sustain the load when PV generation drops off. This can be seen

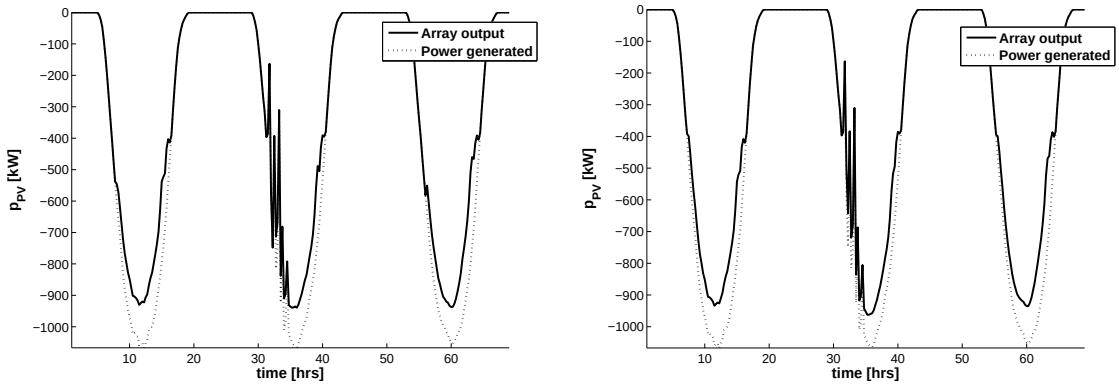

Figure 5.3: Power output of the PV array over the simulation time horizon plotted against the actual power produced by the array before curtailing. The *left* plot represents the ADMM case and the *right* plot represents the centralized case.

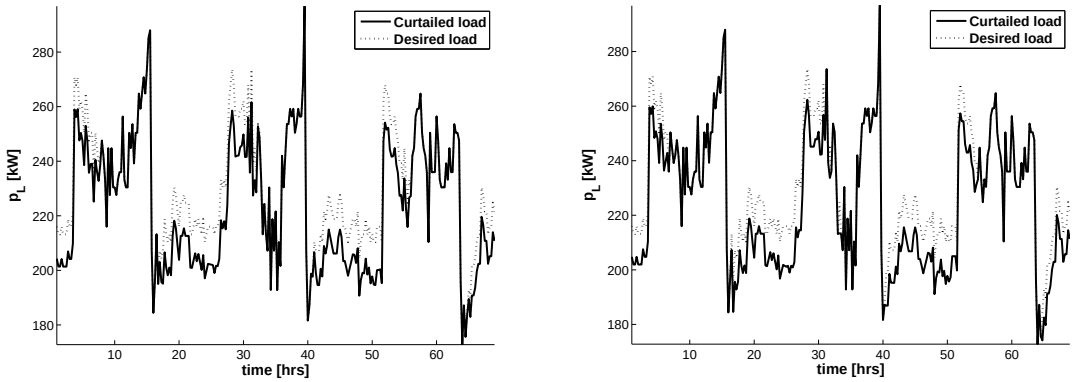

Figure 5.4: Power profile of the curtailed building load over the simulation time horizon plotted against the desired power prior to curtailing. The *left* plot represents the ADMM case and the *right* plot represents the centralized case.

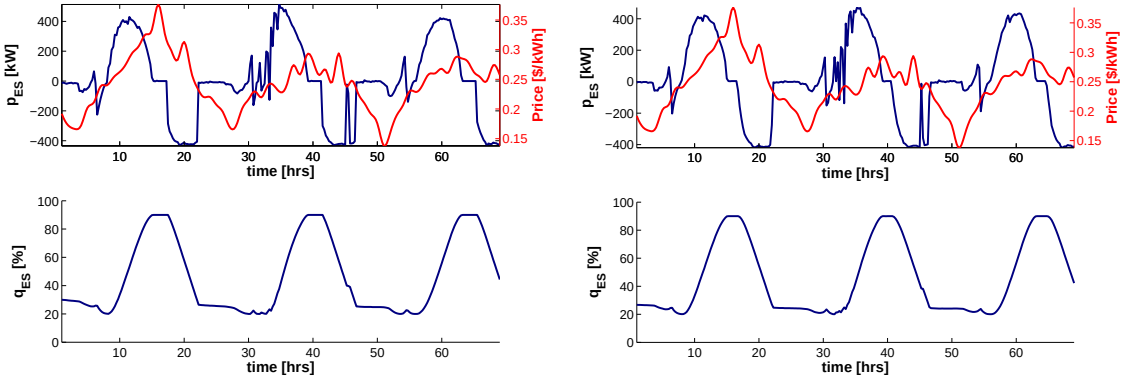

Figure 5.5: The plots on the *left* represent the ADMM case and the plots on the *right* represent the centralized case. In both cases, the *top* plot shows the power profile of the BES unit against price. In the *bottom* plots, the state of charge of the BES unit is plotted as a percentage of the total battery capacity.

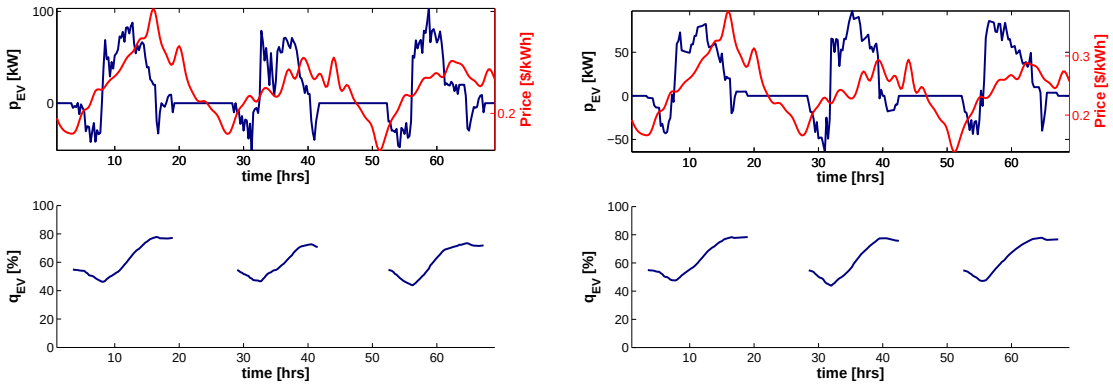

Figure 5.6: The plots on the *left* represent the ADMM case and the plots on the *right* represent the centralized case. In both cases, the *top* plots show the net power profile of the EVs against price. In the *bottom* plot, the state of charge of the EVs is plotted as a percentage of the total vehicle battery capacity.

in Fig. 5.2, where some smoothing in the power profile is noted with fewer intermittent spikes due to either load or generation but the PCC profile is shaped primarily by the capacity limit. The generated output is relatively flat throughout the peak price hours before dropping off each evening as the price begins to decrease. There is then a constant load for the remaining hours until the PV output begins to increase again along with the price. The PV output is initially insufficient to meet the load which begins to go up earlier so both the BES unit and EVs discharge slightly to help with the load in the face of increasing prices. There is still some flexibility with the storage unit and EVs after the PV output falls off each day, allowing the DERs to use their capacity for energy arbitrage and respond to the still high prices by shifting the power generated by the PV array from its peak time at 12PM to 5PM when the peak price occurs. This completely offsets the load at the peak price time in order to increase net profit.

From Fig. 5.3, there is evidence of some curtailing of generated PV power during peak generation hours when there is not enough capacity both on-site and through the grid connection to absorb that energy. The power is curtailed around the peak generation hour when the price for power is still relatively low. In Fig. 5.4, a small amount of load curtailment also occurs during the evening hours when no solar energy is being generated and the line capacity has been reached in terms of how much power can be imported. In both cases, however, the curtailing is not substantial since both the BES unit and the EVs use their storage capabilities to minimize these effects. The BES unit undergoes a deep discharge daily as shown in Fig. 5.5 during peak prices when there is no PV output and charges again to full capacity when prices drop and PV output increases. The EVs behave similarly in Fig. 5.6, with the aggregate of vehicles charging as the price generally decreases and discharge excess capacity when the price generally increases. The reason for the smooth charging profile of the vehicles is that while individual charging can be more intermittent when responding to real-time pricing and each vehicle has arrival and departure constraints that limit their flexibility, the aggregate effect of a diverse population can actually produce a smoother net power profile with some degree of coordination without having to impose a smoothing constraint or centralized top-down control [OPMO13, WGM$^+$13, Eto07]. This is helped by the fact that when optimizing power schedules through ADMM, the line limit forces each EV to account for the implications of simultaneously drawing power at start up when voltages are low and this presents a natural way to reintroduce

diversity by automatically giving priority to the vehicles most willing to accept the high costs without any subjective ordering. Hence the vehicles can be viewed within the microgrid as one single entity with predictable emergent characteristics that can be incentivized to charge in a way which helps to meet system wide objectives even though each vehicle is controlled independently in a decentralized fashion and does not need to over-cycle significantly on an individual basis. This is a very notable finding since there is currently a large amount of concern over how the proliferation of EVs may tax an already strained grid with their relatively substantial charging requirements at the distribution level [Fai09, Net13, Bul13]. Both the predictability of the net power profile and its flexibility to charge at specific hours shows in fact that the EVs can play a very useful role in DR even when control is distributed to each vehicle. Responding in this way through the demand side can provide a potentially more economical way to automate and dynamically respond to changing conditions on the power grid compared to transformers, load tap changers and capacitor banks. Note also that while only the controller at the PCC is aware of the wholesale price schedule and the grid connection limit, all DERs on-site operating independently can help shape P_{grid} through ADMM and ensure the limiting constraints are not violated.

While the primary aim of the simulations was to demonstrate the ability of dynamic distributed algorithms to schedule DERs cooperating when faced with a coupling constraint, this example also shows that a microgrid using such algorithms can produce very reliable and consistent power profiles when provided with the right incentives even if its only generation source is intermittent solar. This means that power can be provided with higher power quality and a higher load factor in addition to responding to real-time prices. Such services are critical in a more distributed and decentralized grid where contingency events can lead to catastrophic circumstances as the grid becomes increasingly less foreseeable and controllable. The ability to shape p_{grid} using the capacities and demands of each DER benefits both the system operator and the microgrid since it means the system can be run more efficiently at a lower cost and the impacts of tiered pricing can also be avoided.

5.2.2 Computation times

The simulation is carried out using CVXGEN [MB12] to solve the optimization at each time step. Using open-source solvers, it runs in a Matlab environment and is hence ideal for rapid code development and deployment. Since MPC uses the

first step of the previous solution to provide a warm start for ADMM in the next iteration, the optimization problem at each MPC iteration can be solved on an Intel Xeon 3.0GHz dual core processor in approximately $1 - 4$ seconds for each DER. The predictions at each time step are similarly solved and require an additional $1 - 4$ seconds. This means that the amount of time required at each time step will be significantly less than the amount of time lapsed between time steps, making real-time dynamic solutions entirely feasible. While this demonstrates that our methods are suitable for real-time implementation, it should be noted that we were carrying out simulations using Matlab's interpreted language as a proof of concept before actual implementation using embedded compiled code. Hence the times reported are conservative approximations only and significantly better performance times can be attained in the milliseconds time frame [KCLB14].

Chapter 6

Conclusions and extensions

In this dissertation, we have modeled and simulated two real world systems using an integrated dynamic optimization and control framework through which DERs can be scheduled and controlled. The simulations from each example demonstrate that even with limited or incomplete information, the algorithms used to control the DERs perform close to optimal. Additionally, no assumptions need to be made about the distributions of the data and this is particularly advantageous when having to deal with real data for which such assumptions do not always hold.

In the first example, we looked at a grid-tied PV storage system operated by PNM. Here we showed how BES units could be controlled and coordinated in a PV storage system for smoothing the intermittency of the solar resource and shifting output to more closely match the load profile. The PV storage system is modeled as a convex optimization problem and an MPC framework solves the optimization problem using external price signals and predictions of PV output at each time step. Our predictions consist of an asymmetric least squares fit of the PV output data with a linear error correction to handle uncertainties and provide forecasts to enable dynamic power scheduling. This method is sufficiently accurate given the adaptability and self-correcting nature of the MPC algorithm. Using real-time data taken from the PNM system we simulated and compared a number of scenarios. We showed that using only locally available information, MPC could schedule the BES units to accommodate various objectives and help to balance trade-offs between smoothing intermittent local generation, shifting output in response to price signals and maintaining the functional requirements of the BES units and power converters. Extensive cost savings were attainable with all cost metrics and the performance was shown to come very close

to the prescient scenario.

For the second example, we modeled a distributed microgrid with onsite PV generation, battery storage, curtailable load and a fleet of DR-enabled EVs. A framework using MPC and ADMM was employed to control and optimize the microgrid in a dynamic and distributed manner. When used together, the algorithms only require local information and simple prediction methods to work while retaining the ability to incorporate any additionally available information into the objectives and constraints of both the entire system and individual DERs as time progresses. By distributing control through ADMM, the problem of managing the microgrid is made more tractable by enabling a cooperative approach between the resources while still respecting coupling constraints due to capacity limited lines. MPC ensures the executed power schedules adapt and keep the system both flexible and resilient when there is imperfect information or sudden unexpected changes to the system. Using data taken from the the EPRI campus in Northern California and transportation data taken from a national survey, we simulated and compared the performance of the algorithms. Our simulations demonstrated that each device was able to retain functionality while allowing the microgrid to respond to external price signals and physical power line limits as well as contingency events. With minimal information sharing between devices, we obtained performance results that are comparable to those obtained when the optimization problem is solved centrally with prescient knowledge.

With both examples, we have demonstrated that the dynamic and distributed algorithms are able to optimize the power schedules in such a way as to provide more flexible and responsive systems as seen by the power grid. Each DER cooperates to ensure the systems behave in a more predictable manner even while maintaining their autonomy. This allows more intermittent renewable energy resources to be integrated onto the grid without requiring more operational reserves or upgrade in line capacities to handle the variable and unpredictable generation. It moves us away from a centrally controlled power system by enabling control of power flow down to the distribution level and building collaborative networks of DERs that can communicate information on and respect individual operating states and costs while collecting external information on prices and grid conditions to optimize system behavior. This helps pave the way for a more participatory infrastructure is which all DERs are incentivized to take part in operating the grid more flexible, efficiently and robustly. Increasing responsiveness in the power grid in this way also has significant

implications for reducing costs associated with system upgrades and procuring more operating reserves.

Extensions of this work will explore interactions between DERs in scenarios where grid constraints and different shifting and smoothing requirements create additional demands on the system. It will also be of interest to see how the algorithms handle larger scale power systems with line data and other objectives such as frequency and voltage control. Since the scheduling methods could be used as both a method for real-time control and as a tool for planning and designing future power grids, another natural extension would be to take the next step and go from simulations to testing on actual hardware. Finally, since forecasting will foreseeably play a more critical role as the amount of data available on the power grid increases, the tools we developed would be useful for exploring different forecasting techniques.

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